

Photon Voice

v2.64

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# Chapter 1

## Photon Voice

Photon Voice 2 has three key classes:

- `Photon.Voice.Unity.VoiceConnection` (extends `Photon.Realtime.ConnectionHandler`)
- `Photon.Voice.Unity.Recorder`
- `Photon.Voice.Unity.Speaker`

If you also use the integration with PUN 2, we added two components for ease-of-use and more convenience:

- `Photon.Voice.PUN.PhotonVoiceNetwork`
- `Photon.Voice.PUN.PhotonVoiceView`

Photon Voice 2 also comes with a WebRTC based DSP (`Photon.Voice.Unity.WebRtcAudioDsp` using `Photon.Voice.WebRTCAudioProcessor`).

Read more in the official documentation [here](#).

You can download Photon Voice 2 [here](#).



## Chapter 2

# Namespace Documentation

## 2.1 Photon Namespace Reference

## 2.2 Photon.Voice Namespace Reference

### Classes

- class [ArrayPool< T >](#)  
*Pool of Arrays with components of type T, with ObjectPool info being the array's size.*
- class [ArrayPoolSet< T >](#)  
*ArrayPool set of limited size.*
- class [ArraySegmentPool< T >](#)  
*Set of ArrayPool's of N<sup>2</sup>-sized arrays used to produce an ArraySegment with the requested size.*
- class [AudioDesc](#)
- class [AudioInChangeNotifierNotSupported](#)
- class **AudioInEnumeratorNotSupported**
- class [AudioOutDelayControl](#)
- class [AudioOutDelayControl< T >](#)
- class [AudioOutDummy< T >](#)
- class [AudioReceptionSettings](#)
- class [AudioSenderSettings](#)
- class [AudioStreamReceiver](#)  
*Basic audio stream receiver Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)*
- class [AudioStreamSender](#)  
*Basic audio stream sender Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)*
- class [AudioSyncBuffer< T >](#)
- class [AudioUtil](#)  
*Collection of Audio [Utility](#) functions and classes.*
- class [BufferReaderPushAdapterAsyncPool< T >](#)  
*BufferReaderPushAdapter<T> implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#).*
- class [BufferReaderPushAdapterAsyncPoolFloatToShort](#)

*BufferedReaderPushAdapter<T> implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#), converting float samples to short.*

- class [BufferedReaderPushAdapterAsyncPoolShortToFloat](#)

*BufferedReaderPushAdapter<T> implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#), converting short samples to float.*

- class [BufferedReaderPushAdapterBase< T >](#)

*Adapter base reading data from [IDataReader<T>.Read](#) and pushing it to [LocalVoice](#).*

- class [ByteStreamDecoder](#)
- class [ByteStreamEncoder](#)
- class [DecoderConfigFrame](#)

*Stores the config frame and prevents other frames decoding until the decoder is ready.*

- class [DeviceEnumeratorBase](#)
- class [DeviceEnumeratorNotSupported](#)
- class [DeviceEnumeratorSingleDevice](#)
- class [DeviceFeatures](#)
- struct [DeviceInfo](#)
- struct [Flip](#)
- struct [FrameBuffer](#)
- class [FrameOut< T >](#)
- class [Framer< T >](#)

*Utility class to re-frame packets.*

- class [FramerResampler< T >](#)
- interface [IAudioDesc](#)

*Audio Source interface.*

- interface [IAudioInChangeNotifier](#)
- interface [IAudioOut< T >](#)
- interface [IAudioPusher< T >](#)

*Audio Pusher interface.*

- interface [IAudioReader< T >](#)

*Audio Reader interface.*

- interface [IDataReader< T >](#)

*Interface for pulling data, in case this is more appropriate than pushing it.*

- interface [IDecoder](#)

*Generic decoder interface.*

- interface [IDecoderDirect< B >](#)

*Interface for an decoder which outputs data via explicit call.*

- interface [IDeviceEnumerator](#)
- interface [IEncoder](#)

*Generic encoder interface.*

- interface [IEncoderDirect< B >](#)

*Interface for an encoder which consumes input data via explicit call.*

- interface [IEncoderDirectImage](#)

*Interface for an encoder which consumes images via explicit call.*

- interface [ILocalVoiceAudio](#)

*Interface for an outgoing audio stream.*

- interface [ILogger](#)
- struct [ImageBufferInfo](#)
- class [ImageBufferNative](#)
- class [ImageBufferNativeAlloc](#)
- class [ImageBufferNativeGCHandleBytes](#)
- class [ImageBufferNativeGCHandleSinglePlane](#)
- class [ImageBufferNativePool< T >](#)
- interface [IPreviewManager](#)

- interface [IProcessor< T >](#)  
*Processor interface.*
- interface [IResettable](#)
- interface [IServiceable](#)  
*Interface for classes that want their [Service\(\)](#) function to be called regularly in the context of a [LocalVoice](#).*
- interface [IStreamReceiver](#)  
*Interface for component relying on a [LoadBalancingTransport](#) to receive a stream of data (audio, video, ...)*
- interface [IStreamSender](#)  
*Interface for component relying on a [LoadBalancingTransport](#) to send a stream of data (audio, video, ...)*
- interface [IStreamTransportManager](#)  
*Interface for components capable of providing a [LoadBalancingTransport](#) for an [IStreamSender](#) and/or [IStreamReceiver](#) Those components should forward the stream callbacks to their registered listeners:*
- interface [IStreamTransportManagerListener](#)
- interface [IVideoPlayer](#)
- interface [IVideoPreview](#)
- interface [IVideoRecorder](#)
- interface [IVideoRecorderPusher](#)
- interface [IVideoSink](#)
- interface [IVoiceTransport](#)
- class [LoadBalancingTransport](#)  
*Extends [LoadBalancingClient](#) with media streaming functionality.*
- class [LoadBalancingTransport2](#)  
*Variant of [LoadBalancingTransport](#). Aims to be non-alloc at the cost of breaking compatibility with older clients.*
- class [LocalVoice](#)  
*Represents outgoing data stream.*
- class [LocalVoiceAudio< T >](#)  
*Outgoing audio stream.*
- class [LocalVoiceAudioDummy](#)  
*Dummy [LocalVoiceAudio](#).*
- class [LocalVoiceAudioFloat](#)  
*Specialization of [LocalVoiceAudio< T >](#) for float audio.*
- class [LocalVoiceAudioShort](#)  
*Specialization of [LocalVoiceAudio< T >](#) for short audio.*
- class [LocalVoiceFramed< T >](#)  
*Typed re-framing [LocalVoice](#).*
- class [LocalVoiceVideo](#)
- class [MonoPInvokeCallbackAttribute](#)
- interface [ObjectFactory< TType, TInfo >](#)  
*Object factory with optional info useful in combination with [ObjectPool< TType, TInfo >](#).*
- class [ObjectPool< TType, TInfo >](#)  
*Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).*
- class [OpusCodec](#)
- class [PermissionsRequester](#)  
*Launch successively permission requests (Android only for now)*
- class [PermissionWaiter](#)  
*Wait for a permission to be granted.*
- class [PhotonAppSettings](#)  
*Collection of connection-relevant settings, used internally by [PhotonNetwork.ConnectUsingSettings](#).*
- class [PhotonTransportProtocol](#)
- class [Platform](#)
- class [PreviewManager](#)

- class [RawCodec](#)
- class **RemoteVoice**
- class [RemoteVoiceInfo](#)
  - Information about a remote voice (incoming stream).*
- struct [RemoteVoiceOptions](#)
  - Event Actions and other options for a remote voice (incoming stream).*
- struct [SendFrameParams](#)
- class **SpacingProfile**
- class [StreamMatchmakingSettings](#)
- class [StreamTransportClient](#)
  - Basic implementation of a [IRegisterableStreamTransportManager](#), initiating a [LoadBalancingTransport](#) and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses (that manage recorders, offer a deeper app life cycle management, ...)*
- class [StreamTransportClientSettings](#)
- class [StreamTransportManagerListener](#)
  - Base class for stream senders/receivers which rely on a [IRegisterableStreamTransportManager](#) to provide the underlying [LoadBalancingTransport](#) upon which the stream run Automatically tries to find [\\_streamTransportManager](#) [IRegisterableStreamTransportManager](#), through [StreamTransportObserver.FindStreamTransportManager](#).*
- class [StreamTransportObserver](#)
  - Wrapper handling [LoadBalancingTransport](#) callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the [IRegisterableStreamTransportManager](#) interface.*
- struct [StringSessionProperty](#)
- class [UnsupportedCodecException](#)
  - Exception thrown if an unsupported codec is encountered.*
- class [UnsupportedPlatformException](#)
  - Exception thrown if an unsupported platform is encountered.*
- class [UnsupportedSampleTypeException](#)
  - Exception thrown if an unsupported audio sample type is encountered.*
- class **Util**
- class [Utility](#)
- class **VideoInEnumeratorNotSupported**
- class [VideoPlayer](#)
- class [VideoSinkArray](#)
- struct [VideoSourceSizeMode](#)
  - Defines how the video source is resized before passing to the encoder. These options are available depending on the video source.*
- class [VoiceClient](#)
  - [Voice](#) client interact with other clients on network via [IVoiceTransport](#).*
- class [VoiceConnectionMatchmaking](#)
  - Use matchmaking to connect and join a [VoiceConnection](#).*
- class [VoiceConnectionPermissionWaiter](#)
  - Wait for Microphone access permissions to be available before letting [VoiceConnection](#) connect Note: [recordWhen](#)↔[Joined](#) should be disabled on the recorder.*
- struct [VoiceCreateOptions](#)
  - Used to initialize optional properties of the [LocalVoice](#) instance at creation time.*
- class [VoiceEvent](#)
- class [VoiceFollowClient](#)
  - This class can be used to automatically sync client states between Leader and [Voice](#) clients.*
- struct [VoiceInfo](#)
  - Describes stream properties.*
- class [VPxCodec](#)
- class [WebRTCAudioLib](#)
- class [WebRTCAudioProcessor](#)



## Enumerations

- enum **WebcamSelectionMode**
- enum **CameraFacing**
- enum [AudioSampleType](#)  
*The type of samples used for audio processing.*
- enum **LogLevel**
- enum **FrameFlags** : byte
- enum [Codec](#)  
*Enum for Media Codecs supported by PhotonVoice.*
- enum **ImageFormat**
- enum **Rotation**

## 2.2.1 Enumeration Type Documentation

### 2.2.1.1 Codec

enum [Codec](#)

Enum for Media Codecs supported by PhotonVoice.

Transmitted in [VoiceInfo](#). Do not change the values of this Enum!

#### Enumerator

|  |           |             |
|--|-----------|-------------|
|  | AudioOpus | OPUS audio. |
|--|-----------|-------------|

## 2.3 Photon.Voice.FMOD Namespace Reference

### Classes

- class [AudioInEnumerator](#)
- class [AudioInReader< T >](#)
- class [AudioOut< T >](#)
- class [AudioOutEvent< T >](#)

## 2.4 Photon.Voice.Fusion Namespace Reference

### Classes

- class [FusionVoiceClient](#)
- class [VoiceNetworkObject](#)

### Typedefs

- using **PhotonAppSettings** = global::Fusion.Photon.Realtime.PhotonAppSettings
- using **LogLevel** = Photon.Voice.LogLevel

## 2.5 Photon.Voice.IOS Namespace Reference

### Classes

- class [AudioInChangeNotifier](#)
- class [AudioInPusher](#)
- class [AudioInReader](#)
- struct [AudioSessionParameters](#)
- class [AudioSessionParametersPresets](#)
- class [PreviewManagerLayer](#)
- class [VideoCapture](#)
- class [VideoCaptureLayer](#)
- class [VideoCaptureUnityTexture](#)
- class [VideoDecoder](#)
- class [VideoDecoderLayer](#)
- class [VideoDecoderUnityTexture](#)
- class [VideoEncoder](#)
- class [VideoEncoderFromCapture](#)
- class [VideoEncoderUnityTextureInput](#)
- class [VideoInEnumerator](#)
  - Enumerates cameras available on device.*
- class [VideoPlayerUnityTexture](#)
- class [VideoRecorderLayer](#)
- class [VideoRecorderUnityTexture](#)
- class [VideoRecorderUnityTextureInput](#)

### Enumerations

- enum [AudioSessionCategory](#)
- enum [AudioSessionMode](#)
- enum [AudioSessionCategoryOption](#)

### 2.5.1 Enumeration Type Documentation

#### 2.5.1.1 AudioSessionCategory

enum [AudioSessionCategory](#)

##### Enumerator

|  |                 |                                                                                                              |
|--|-----------------|--------------------------------------------------------------------------------------------------------------|
|  | Ambient         |                                                                                                              |
|  | SoloAmbient     |                                                                                                              |
|  | Playback        |                                                                                                              |
|  | Record          |                                                                                                              |
|  | PlayAndRecord   |                                                                                                              |
|  | AudioProcessing |                                                                                                              |
|  | MultiRoute      | Use this category to customize the usage of available audio accessories and built-in audio hardware. For exa |

#### 2.5.1.2 AudioSessionCategoryOption

enum [AudioSessionCategoryOption](#)

## Enumerator

|  |                  |                                                                                                                |
|--|------------------|----------------------------------------------------------------------------------------------------------------|
|  | MixWithOthers    | This allows an application to set whether or not other active audio apps will be interrupted or mixed with whe |
|  | DuckOthers       |                                                                                                                |
|  | AllowBluetooth   | This allows an application to change the default behavior                                                      |
|  | DefaultToSpeaker |                                                                                                                |

## 2.5.1.3 AudioSessionMode

enum [AudioSessionMode](#)

## Enumerator

|  |                |                                                                                                                    |
|--|----------------|--------------------------------------------------------------------------------------------------------------------|
|  | Default        |                                                                                                                    |
|  | VoiceChat      | Only valid with AVAudioSessionCategoryPlayAndRecord. Appropriate for <a href="#">Voice</a> over IP (VoIP) applicat |
|  | VideoRecording |                                                                                                                    |
|  | Measurement    |                                                                                                                    |
|  | MoviePlayback  |                                                                                                                    |
|  | VideoChat      | Only valid with kAudioSessionCategory_PlayAndRecord. Reduces the number of allowable audio routes to be            |

## 2.6 Photon.Voice.MacOS Namespace Reference

## Classes

- class [AudioInChangeNotifier](#)
- class [AudioInEnumerator](#)
  - Enumerates microphones available on device.*
- class [AudioInPusher](#)
- class [AudioInReader](#)
- class [VideoDecoder](#)
- class [VideoEncoder](#)
- class [VideoEncoderUnityTextureInput](#)
- class [VideoRecorderUnityTextureInput](#)

## 2.7 Photon.Voice.PUN Namespace Reference

## Classes

- class [PhotonVoiceView](#)
  - Component that should be attached to a networked [PUN](#) prefab that has [PhotonView](#). It will bind remote [Recorder](#) with local [Speaker](#) of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.*
- class [PunVoiceClient](#)
  - This class can be used to automatically sync client states between [PUN](#) and [Voice](#). It also finds the [Speaker](#) component for a character's voice. For this to work attach a [PhotonVoiceView](#) next to the [PhotonView](#) of your player's prefab.*

## 2.8 Photon.Voice.PUN.UtilityScripts Namespace Reference

### Classes

- class [VoiceDebugScript](#)

*Utility script to be attached next to [PhotonVoiceView](#) & [PhotonView](#) on the player prefab to be network instantiated. Call `voiceDebugScript.CantHearYou()` on the networked object of the remote (or local) player if you can't hear the corresponding player.*

## 2.9 Photon.Voice.Unity Namespace Reference

### Classes

- class [AndroidAudioInAEC](#)
- struct [AndroidAudioInParameters](#)
- class [AndroidCamera](#)
- class [AndroidCameraVideoEncoder](#)
- class [AndroidCameraVideoEncoderSurfaceView](#)
- class [AndroidCameraVideoEncoderTexture](#)
- class [AndroidNDKCamera](#)
- class [AndroidNDKVideoEncoder](#)
- class [AndroidNDKVideoRecorderSurfaceView](#)
- class [AndroidNDKVideoRecorderUnityTexture](#)
- class [AndroidPreviewManagerSurfaceView](#)
- class [AndroidSurfaceView](#)
- class [AndroidTextureVideoEncoder](#)
- class [AndroidTextureView](#)
- class [AndroidVideoDecoder](#)
- class [AndroidVideoDecoderSurfaceView](#)
- class [AndroidVideoDecoderTexture](#)
- class [AndroidVideoEncoder](#)
- class [AndroidVideoGkPluginEventIssuer](#)
- class [AndroidVideoGkPluginEventIssuerMB](#)
- class [AndroidVideoInEnumerator](#)

*Enumerates cameras available on device.*

- class [AndroidVideoPlayerUnityTexture](#)
- class [AndroidVideoRecorderSurfaceView](#)
- class [AndroidVideoRecorderTextureInput](#)
- class [AndroidVideoRecorderUnityTexture](#)
- class [AudioChangesHandler](#)

*This component is useful to handle audio device and config changes.*

- class [AudioClipWrapper](#)
- class [AudioInEnumerator](#)
- class [AudioOutCapture](#)
- class [Logger](#)
- class [MicWrapper](#)
- class [MicWrapperPusher](#)
- class [MicWrapperPusherOnAudioFilterRead](#)
- class [PhotonVoiceCreatedParams](#)
- class [PreviewManagerScreenQuadTexture](#)
- class [PreviewManagerUnityGUI](#)
- class [Recorder](#)

*Component representing outgoing audio stream in scene.*

- class [RecorderPreset](#)
- class [RemoteVoiceLink](#)
- class [ScreenRecorder](#)
- class [Speaker](#)
- class [SpeakerAudioFilterRead](#)
- class [TimerWorker](#)
- class [UnityAudioOut](#)
- class [UnityLogger](#)
- class [UnityMicrophone](#)

*A wrapper around [UnityEngine.Microphone](#) to be able to safely use [Microphone](#) and compile for WebGL.*

- class [UnityVoiceClient](#)

*Component that represents a [Voice](#) client and manages a simple [Unity](#) integration: a single [Recorder](#) and multiple remote speakers.*

- class [VideoInEnumerator](#)
- class [VideoPlayerUnity](#)
- class [VideoPlayerUnityMB](#)
- class [VideoRecorderUnity](#)
- class [VideoRecorderUnityMB](#)
- class [VideoTexture](#)
- class [VideoUtil](#)
- class [VoiceComponent](#)
- class [VoiceComponentImpl](#)
- class [VoiceConnection](#)

*Component that represents a [Voice](#) client.*

- class [VoiceLogger](#)
- class [WebAudioAudioOut](#)
- class [WebAudioInEnumerator](#)
- class [WebAudioMicIn](#)
- class [WebCodecsCamera](#)
- class [WebCodecsCameraRecorderUnityTexture](#)
- class [WebCodecsScreenRecorderUnityTexture](#)
- class [WebCodecsScreenShare](#)
- class [WebCodecsVideoCapture](#)
- class [WebCodecsVideoDecoderUnityTexture](#)
- class [WebCodecsVideoEncoder](#)
- class [WebCodecsVideoEncoderDirectImage](#)
- class [WebCodecsVideoPlayerUnityTexture](#)
- class [WebCodecsVideoRecorderUnityTexture](#)
- class [WebDeviceEnumerator](#)
- class [WebRtcAudioDsp](#)
- class [WebVideoInEnumerator](#)

## 2.10 Photon.Voice.Unity.FMOD Namespace Reference

### Classes

- class [FMODRecorderSetup](#)
- class [SpeakerFMOD](#)

## 2.11 Photon.Voice.Unity.UtilityScripts Namespace Reference

### Classes

- class [ConnectAndJoin](#)
- class [MicAmplifier](#)
- class [MicAmplifierFloat](#)
- class [MicAmplifierShort](#)
- class [MicrophonePermission](#)
  - Helper to request Microphone permission on Android or iOS.*
- class [PhotonVoiceLagSimulationGui](#)
- class [PhotonVoiceStatsGui](#)
  - Basic GUI to show traffic and health statistics of the connection to [Photon](#), toggled by shift+tab.*
- class [SaveIncomingStreamToFile](#)
- class [SaveOutgoingStreamToFile](#)
- class [TestTone](#)
- class [WaveWriter](#)

## 2.12 Photon.Voice.UWP Namespace Reference

### Classes

- class [AudioInEnumerator](#)
- class [AudioInPusher](#)
- class [CaptureDevice](#)
- class [DeviceEnumerator](#)
- class [PreviewManagerMediaPlayerElement](#)
- class [UnitySharedVideoTexture](#)
- class [VideoDecoder](#)
- class [VideoDecoderMediaPlayerElement](#)
- class [VideoDecoderUnityTexture](#)
- class [VideoEncoder](#)
- class [VideoEncoderMediaPlayerElement](#)
- class [VideoEncoderUnityTexture](#)
- class [VideoInEnumerator](#)
- class [VideoPlayerUnityTexture](#)
- class [VideoRecorderMediaPlayerElement](#)
- class [VideoRecorderUnityTexture](#)

### Functions

- delegate void [MediaCaptureInitCompleted](#) (MediaCapture mediaCpture, bool ok)

## 2.13 Photon.Voice.UWP.PlatformAPI Namespace Reference

### Classes

- interface [IMediaCapture](#)
- interface [IMixedRealityCapture](#)

## 2.14 Photon.Voice.Windows Namespace Reference

### Classes

- class [AudioInEnumerator](#)  
*Enumerates microphones available on device.*
- class [MFTCodec](#)
- class [WindowsAudioInPusher](#)





## Chapter 3

# Class Documentation

### 3.1 AndroidAudioInAEC Class Reference

Inherits [IAudioPusher< T >](#), and [IResettable](#).

#### Public Member Functions

- **AndroidAudioInAEC** (Voice.ILogger logger, bool enableAEC=false, bool enableAGC=false, bool enableNS=false)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimalFrameSize)
- void [Reset](#) ()
- void **Dispose** ()

#### Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

#### Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

#### 3.1.1 Member Function Documentation

##### 3.1.1.1 Reset()

```
void Reset ()
```

Implements [IResettable](#).

### 3.1.2 Property Documentation

#### 3.1.2.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.1.2.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

#### 3.1.2.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.2 AndroidAudioInParameters Struct Reference

### Public Attributes

- `bool EnableAEC`
- `bool EnableAGC`
- `bool EnableNS`

### Static Public Attributes

- `static AndroidAudioInParameters Default = new AndroidAudioInParameters() { EnableAEC = true, EnableAGC = true, EnableNS = true }`

## 3.3 AndroidCamera Class Reference

Inherits `IDisposable`.

### Classes

- `class OnReadyCallback`

**Public Member Functions**

- **AndroidCamera** ([ILogger](#) logger, [VoicelInfo](#) info, string deviceId)
- void **AddSurface** (AndroidJavaObject surface)
- void **Start** ()
- void **Dispose** ()

**Properties**

- int **Width** [get]
- int **Height** [get]

## 3.4 AndroidCameraVideoEncoder Class Reference

Inherits [AndroidVideoEncoder](#).

Inherited by [AndroidCameraVideoEncoderSurfaceView](#), and [AndroidCameraVideoEncoderTexture](#).

**Classes**

- class [OnReadyCallback](#)

**Public Member Functions**

- **AndroidCameraVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- override void [Dispose](#) ()

**Public Member Functions inherited from [AndroidVideoEncoder](#)**

- **AndroidVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- void **Start** (int width, int height)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*

**Protected Member Functions**

- void **startEncoderAndCamera** (AndroidJavaObject camera)

**Additional Inherited Members****Protected Attributes inherited from [AndroidVideoEncoder](#)**

- [ILogger](#) logger
- [VoicelInfo](#) info
- AndroidJavaObject activity

## Properties inherited from [AndroidVideoEncoder](#)

- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]
- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### 3.4.1 Member Function Documentation

#### 3.4.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [AndroidVideoEncoder](#).

## 3.5 AndroidCameraVideoEncoderSurfaceView Class Reference

Inherits [AndroidCameraVideoEncoder](#).

### Public Member Functions

- **AndroidCameraVideoEncoderSurfaceView** ([ILogger](#) logger, [VoicelInfo](#) info, string cameraID)

### Public Member Functions inherited from [AndroidCameraVideoEncoder](#)

- **AndroidCameraVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- override void **Dispose** ()

### Public Member Functions inherited from [AndroidVideoEncoder](#)

- **AndroidVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- void **Start** (int width, int height)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I **GetPlatformAPI**< I > ()  
*Returns an platform-specific interface.*

### Properties

- AndroidJavaObject **Preview** [get]

### Properties inherited from [AndroidVideoEncoder](#)

- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]
- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### Additional Inherited Members

### Protected Member Functions inherited from [AndroidCameraVideoEncoder](#)

- void **startEncoderAndCamera** (AndroidJavaObject camera)

### Protected Attributes inherited from [AndroidVideoEncoder](#)

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- AndroidJavaObject **activity**

## 3.6 AndroidCameraVideoEncoderTexture Class Reference

Inherits [AndroidCameraVideoEncoder](#).

### Public Member Functions

- **AndroidCameraVideoEncoderTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string cameraID, Action< [AndroidCameraVideoEncoderTexture](#) > onReady)

### Public Member Functions inherited from [AndroidCameraVideoEncoder](#)

- **AndroidCameraVideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- override void **Dispose** ()

### Public Member Functions inherited from [AndroidVideoEncoder](#)

- **AndroidVideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Start** (int width, int height)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I **GetPlatformAPI**< I > ()  
*Returns an platform-specific interface.*

## Properties

- Texture **Preview** [get]

## Properties inherited from [AndroidVideoEncoder](#)

- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]
- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## Additional Inherited Members

## Protected Member Functions inherited from [AndroidCameraVideoEncoder](#)

- void **startEncoderAndCamera** (AndroidJavaObject camera)

## Protected Attributes inherited from [AndroidVideoEncoder](#)

- [ILogger](#) **logger**
- [VoicelInfo](#) **info**
- AndroidJavaObject **activity**

## 3.7 AndroidNDKCamera Class Reference

Inherits IDisposable.

### Public Member Functions

- **AndroidNDKCamera** ([ILogger](#) logger, [VoicelInfo](#) info, string deviceId)
- void **AddSurface** (IntPtr surface)
- void **Start** ()
- void **Dispose** ()

### Protected Member Functions

- static IntPtr **create\_camera\_manager** (string deviceId, int width, int height)
- static IntPtr **start\_camera\_session** (IntPtr camera)
- static void **stop\_camera\_session** (IntPtr camera)
- static void **is\_camera\_active** (IntPtr camera)
- static void **get\_camera\_error** (IntPtr camera)
- static void **add\_surface\_to\_camera** (IntPtr camera, IntPtr surface)
- static int **get\_camera\_width** (IntPtr camera)
- static int **get\_camera\_height** (IntPtr camera)

**Static Protected Attributes**

- static Dictionary< IntPtr, [AndroidNDKVideoEncoder](#) > **instancePerHandle** = new Dictionary<IntPtr, [AndroidNDKVideoEncoder](#)>()

**Properties**

- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]

## 3.8 AndroidNDKVideoEncoder Class Reference

Inherits [IEncoder](#).

**Public Member Functions**

- **AndroidNDKVideoEncoder** (ILogger logger, int width, int height, [VoicelInfo](#) info)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I **GetPlatformAPI**< I > ()  
*Returns an platform-specific interface.*
- virtual void **Dispose** ()

**Protected Member Functions**

- static IntPtr **get\_encoder\_surface** (IntPtr Encoder)

**Static Protected Attributes**

- static Dictionary< IntPtr, [AndroidNDKVideoEncoder](#) > **instancePerHandle** = new Dictionary<IntPtr, [AndroidNDKVideoEncoder](#)>()

**Properties**

- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

### 3.8.1 Member Function Documentation

#### 3.8.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

#### 3.8.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

#### 3.8.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

### 3.8.2 Property Documentation

#### 3.8.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

#### 3.8.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).



## 3.9 AndroidNDKVideoRecorderSurfaceView Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **AndroidNDKVideoRecorderSurfaceView** ([ILogger](#) logger, [VoiceInfo](#) info, string cameraID, Action<[IVideoRecorder](#)> onReady)
- void **Dispose** ()

### Properties

- [IEncoder](#) **Encoder** [get]
- object [PlatformView](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

### 3.9.1 Property Documentation

#### 3.9.1.1 Encoder

[IEncoder](#) **Encoder** [get]

Implements [IVideoRecorder](#).

#### 3.9.1.2 Error

string **Error** [get]

Implements [IVideoRecorder](#).

#### 3.9.1.3 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).

#### 3.9.1.4 Height

int **Height** [get]

Implements [IVideoPreview](#).

### 3.9.1.5 PlatformView

object PlatformView [get]

Implements [IVideoPreview](#).

### 3.9.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.9.1.7 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.10 AndroidNDKVideoRecorderUnityTexture Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **AndroidNDKVideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string cameraID, Action<[IVideoRecorder](#)> onReady)
- void **Dispose** ()

### Properties

- [IEncoder](#) Encoder [get]
- object [PlatformView](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) Flip [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

## 3.10.1 Property Documentation

### 3.10.1.1 Encoder

[IEncoder](#) Encoder [get]

Implements [IVideoRecorder](#).

### 3.10.1.2 Error

`string Error [get]`

Implements [IVideoRecorder](#).

### 3.10.1.3 Flip

`Flip Flip [get]`

Implements [IVideoPreview](#).

### 3.10.1.4 Height

`int Height [get]`

Implements [IVideoPreview](#).

### 3.10.1.5 PlatformView

`object PlatformView [get]`

Implements [IVideoPreview](#).

### 3.10.1.6 Rotation

`Rotation Rotation [get]`

Implements [IVideoPreview](#).

### 3.10.1.7 Width

`int Width [get]`

Implements [IVideoPreview](#).

## 3.11 AndroidPreviewManagerSurfaceView Class Reference

Inherits [PreviewManager](#).

### Public Member Functions

- **AndroidPreviewManagerSurfaceView** ([ILogger](#) logger)
- override void [AddView](#) (object id, [IVideoPreview](#) view)
- override void [RemoveView](#) (object id)

## Public Member Functions inherited from [PreviewManager](#)

- virtual bool [Has](#) (object id)
- virtual void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), Rotation rot=Rotation.Rotate0)  
*Positions the preview of the view / image. Not optimized to be called every frame.*

## Protected Member Functions

- override void [Apply](#) ([ViewState](#) v)

## Protected Member Functions inherited from [PreviewManager](#)

- void [Apply](#) ([ViewState](#) v)

## Properties

- AndroidJavaObject [ViewManager](#) [get]

## Additional Inherited Members

## Protected Attributes inherited from [PreviewManager](#)

- Dictionary< object, [ViewState](#) > **views** = new Dictionary<object, [ViewState](#)>()

## 3.11.1 Member Function Documentation

### 3.11.1.1 AddView()

```
override void AddView (
    object id,
    IVideoPreview view) [virtual]
```

Reimplemented from [PreviewManager](#).

### 3.11.1.2 RemoveView()

```
override void RemoveView (
    object id) [virtual]
```

Reimplemented from [PreviewManager](#).

## 3.12 AndroidSurfaceView Class Reference

Inherits [IDisposable](#).

## Classes

- class [OnReadyCallback](#)

## Public Member Functions

- **AndroidSurfaceView** ([ILogger](#) logger, [VoicelInfo](#) info, Action< AndroidJavaObject, IntPtr > onSurfaceReady)
- void **Dispose** ()

## Properties

- string **Error** [get, protected set]
- AndroidJavaObject **View** [get]

# 3.13 AndroidTextureVideoEncoder Class Reference

Inherits [AndroidVideoEncoder](#).

## Public Member Functions

- **AndroidTextureVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- void [encodeTexture](#) (Texture texture)
  - Encode a [Unity](#) texture (works with both OpenGL and Vulkan)*
- void [encodeTexture](#) (Texture texture, long timestampUs)
  - Encode a [Unity](#) texture with explicit timestamp.*
- override void [Dispose](#) ()

## Public Member Functions inherited from [AndroidVideoEncoder](#)

- **AndroidVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- void **Start** (int width, int height)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)
  - Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()
  - Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()
  - Returns an platform-specific interface.*

## Additional Inherited Members

## Protected Attributes inherited from [AndroidVideoEncoder](#)

- [ILogger](#) logger
- [VoicelInfo](#) info
- AndroidJavaObject **activity**

## Properties inherited from [AndroidVideoEncoder](#)

- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]
- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### 3.13.1 Member Function Documentation

#### 3.13.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [AndroidVideoEncoder](#).

#### 3.13.1.2 encodeTexture() [1/2]

```
void encodeTexture (
    Texture texture)
```

Encode a [Unity](#) texture (works with both OpenGL and Vulkan)

##### Parameters

|                         |                               |
|-------------------------|-------------------------------|
| <a href="#">texture</a> | <a href="#">Unity</a> texture |
|-------------------------|-------------------------------|

#### 3.13.1.3 encodeTexture() [2/2]

```
void encodeTexture (
    Texture texture,
    long timestampUs)
```

Encode a [Unity](#) texture with explicit timestamp.

##### Parameters

|                             |                                        |
|-----------------------------|----------------------------------------|
| <a href="#">texture</a>     | <a href="#">Unity</a> texture          |
| <a href="#">timestampUs</a> | Presentation timestamp in microseconds |

## 3.14 AndroidTextureView Class Reference

Inherits [IDisposable](#).

**Public Member Functions**

- **AndroidTextureView** ([ILogger](#) logger, [VoicelInfo](#) info)
- void **Dispose** ()

**Properties**

- string **Error** [get, protected set]
- Texture **Texture** [get]
- AndroidJavaObject **Surface** [get]
- IntPtr **SurfaceNative** [get]

## 3.15 AndroidVideoDecoder Class Reference

Inherits [IDecoder](#).

Inherited by [AndroidVideoDecoderSurfaceView](#), and [AndroidVideoDecoderTexture](#).

**Public Member Functions**

- **AndroidVideoDecoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- void **Open** ([VoicelInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

**Protected Attributes**

- [ILogger](#) **logger**
- [VoicelInfo](#) **info**
- AndroidJavaObject **decoder**

**Properties**

- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

### 3.15.1 Member Function Documentation

#### 3.15.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

### 3.15.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|                      |                                          |
|----------------------|------------------------------------------|
| <a href="#">info</a> | Properties of the data stream to decode. |
|----------------------|------------------------------------------|

Implements [IDecoder](#).

## 3.15.2 Property Documentation

### 3.15.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.16 AndroidVideoDecoderSurfaceView Class Reference

Inherits [AndroidVideoDecoder](#).

### Public Member Functions

- **AndroidVideoDecoderSurfaceView** ([ILogger](#) logger, [VoiceInfo](#) info)

### Public Member Functions inherited from [AndroidVideoDecoder](#)

- **AndroidVideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

### Properties

- object **Preview** [get]



### Properties inherited from [AndroidVideoDecoder](#)

- int **Width** [get]
- int **Height** [get]
- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*

### Additional Inherited Members

### Protected Attributes inherited from [AndroidVideoDecoder](#)

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- AndroidJavaObject **decoder**

## 3.17 AndroidVideoDecoderTexture Class Reference

Inherits [AndroidVideoDecoder](#).

### Public Member Functions

- **AndroidVideoDecoderTexture** ([ILogger](#) logger, [VoiceInfo](#) info)

### Public Member Functions inherited from [AndroidVideoDecoder](#)

- **AndroidVideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

### Properties

- Texture **Preview** [get]

### Properties inherited from [AndroidVideoDecoder](#)

- int **Width** [get]
- int **Height** [get]
- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*

## Additional Inherited Members

## Protected Attributes inherited from [AndroidVideoDecoder](#)

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- [AndroidJavaObject](#) **decoder**

## 3.18 AndroidVideoEncoder Class Reference

Inherits [IEncoder](#).

Inherited by [AndroidCameraVideoEncoder](#), and [AndroidTextureVideoEncoder](#).

## Public Member Functions

- **AndroidVideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Start** (int width, int height)
- [ArraySegment](#)< byte > **DequeueOutput** (out [FrameFlags](#) flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- [I](#) **GetPlatformAPI**< [I](#) > ()  
*Returns an platform-specific interface.*
- virtual void **Dispose** ()

## Protected Attributes

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- [AndroidJavaObject](#) **activity**

## Properties

- [AndroidJavaObject](#) **Surface** [get]
- [IntPtr](#) **SurfaceNative** [get]
- int **Width** [get]
- int **Height** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- [Action](#)< [ArraySegment](#)< byte >, [FrameFlags](#) > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.18.1 Member Function Documentation

### 3.18.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.18.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.18.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

## 3.18.2 Property Documentation

### 3.18.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.18.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.19 AndroidVideoGkPluginEventIssuer Class Reference

#### Static Public Member Functions

- static void **Start** ([ILogger](#) logger\_)
- static void **Stop** ()

### 3.20 AndroidVideoGkPluginEventIssuerMB Class Reference

Inherits [MonoBehaviour](#).

#### Public Member Functions

- void **SetPluginEventPtr** (IntPtr pluginEventPtr, int eventID)
- void **Update** ()

### 3.21 AndroidVideoInEnumerator Class Reference

Enumerates cameras available on device.

Inherits [DeviceEnumeratorBase](#).

#### Public Member Functions

- **AndroidVideoInEnumerator** ([ILogger](#) logger)
- override void **Refresh** ()  
*Refreshes the microphones list.*
- override void **Dispose** ()

#### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerator< [DeviceInfo](#) > **GetEnumerator** ()
- void **Refresh** ()
- void **Dispose** ()

#### Properties

- override bool **IsSupported** [get]

#### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool **IsSupported** [get]
- virtual string **Error** [get, protected set]
- Action **OnReady** [protected get, set]

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

#### 3.21.1 Detailed Description

Enumerates cameras available on device.

#### 3.21.2 Member Function Documentation

##### 3.21.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements [IDeviceEnumerator](#).

#### 3.21.3 Property Documentation

##### 3.21.3.1 IsSupported

```
override bool IsSupported [get]
```

Implements [IDeviceEnumerator](#).

## 3.22 AndroidVideoPlayerUnityTexture Class Reference

Inherits [IVideoPlayer](#).

### Public Member Functions

- **AndroidVideoPlayerUnityTexture** ([ILogger](#) logger, [VoicelInfo](#) info, Action< [IVideoPlayer](#) > onReady)
- void **Dispose** ()

### Properties

- [IDecoder](#) **Decoder** [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]

### 3.22.1 Property Documentation

#### 3.22.1.1 Decoder

`IDecoder` `Decoder` [get]

Implements [IVideoPlayer](#).

#### 3.22.1.2 Flip

`Flip` `Flip` [get]

Implements [IVideoPreview](#).

#### 3.22.1.3 Height

`int` `Height` [get]

Implements [IVideoPreview](#).

#### 3.22.1.4 PlatformView

`object` `PlatformView` [get]

Implements [IVideoPreview](#).

#### 3.22.1.5 Rotation

`Rotation` `Rotation` [get]

Implements [IVideoPreview](#).

#### 3.22.1.6 Width

`int` `Width` [get]

Implements [IVideoPreview](#).

## 3.23 AndroidVideoRecorderSurfaceView Class Reference

Inherits [IVideoRecorder](#).

## Public Member Functions

- **AndroidVideoRecorderSurfaceView** ([ILogger](#) logger, [VoiceInfo](#) info, string cameraID, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

## Properties

- [IEncoder](#) **Encoder** [get]
- object [PlatformView](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

## 3.23.1 Property Documentation

### 3.23.1.1 Encoder

[IEncoder](#) Encoder [get]

Implements [IVideoRecorder](#).

### 3.23.1.2 Error

string Error [get]

Implements [IVideoRecorder](#).

### 3.23.1.3 Flip

[Flip](#) Flip [get]

Implements [IVideoPreview](#).

### 3.23.1.4 Height

int Height [get]

Implements [IVideoPreview](#).

### 3.23.1.5 PlatformView

object PlatformView [get]

Implements [IVideoPreview](#).

### 3.23.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.23.1.7 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.24 AndroidVideoRecorderTextureInput Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **AndroidVideoRecorderTextureInput** ([ILogger](#) logger, [VoicelInfo](#) info, Action< [IVideoRecorder](#) > onReady)
- int **EncodeFrame** ()
- void **Dispose** ()

### Properties

- [IEncoder](#) **Encoder** [get, protected set]
- UnityEngine.RenderTexture **InputTexture** [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- **Flip Flip** [get]
- object **PlatformView** [get]
- string **Error** [get]

## 3.24.1 Property Documentation

### 3.24.1.1 Encoder

[IEncoder](#) Encoder [get], [protected set]

Implements [IVideoRecorder](#).

### 3.24.1.2 Error

string Error [get]

Implements [IVideoRecorder](#).



### 3.24.1.3 Flip

`Flip Flip` [get]

Implements [IVideoPreview](#).

### 3.24.1.4 Height

`int Height` [get]

Implements [IVideoPreview](#).

### 3.24.1.5 PlatformView

`object PlatformView` [get]

Implements [IVideoPreview](#).

### 3.24.1.6 Rotation

`Rotation Rotation` [get]

Implements [IVideoPreview](#).

### 3.24.1.7 Width

`int Width` [get]

Implements [IVideoPreview](#).

## 3.25 AndroidVideoRecorderUnityTexture Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **AndroidVideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string cameraID, Action<[IVideoRecorder](#)> onReady)
- void **Dispose** ()

## Properties

- [IEncoder Encoder](#) [get, protected set]
- Rotation [Rotation](#) [get]
- [Flip Flip](#) [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

## 3.25.1 Property Documentation

### 3.25.1.1 Encoder

[IEncoder Encoder](#) [get], [protected set]

Implements [IVideoRecorder](#).

### 3.25.1.2 Error

string [Error](#) [get]

Implements [IVideoRecorder](#).

### 3.25.1.3 Flip

[Flip Flip](#) [get]

Implements [IVideoPreview](#).

### 3.25.1.4 Height

int [Height](#) [get]

Implements [IVideoPreview](#).

### 3.25.1.5 PlatformView

object [PlatformView](#) [get]

Implements [IVideoPreview](#).

### 3.25.1.6 Rotation

Rotation [Rotation](#) [get]

Implements [IVideoPreview](#).

## 3.25.1.7 Width

```
int Width [get]
```

Implements [IVideoPreview](#).

## 3.26 ArrayPool&lt; T &gt; Class Template Reference

Pool of Arrays with components of type T, with ObjectPool info being the array's size.

Inherits [ObjectPool< TType, TInfo >](#).

## Public Member Functions

- **ArrayPool** (int capacity, string name, int info)

Public Member Functions inherited from [ObjectPool< TType, TInfo >](#)

- [ObjectPool](#) (int capacity, string name, TInfo info)  
*Create a new ObjectPool instance with the given info structure.*
- TType [New](#) ()  
*Acquire an existing object, or create a new one if none are available.*
- TType [New](#) (TInfo info)  
*Acquire an existing object (if info matches), or create a new one from the passed info.*
- virtual bool [Free](#) (TType obj, TInfo objInfo)  
*Returns object to pool.*
- virtual bool [Free](#) (TType obj)  
*Returns object to pool, or destroys it if the pool is full.*
- void **Dispose** ()  
*Free resources associated with this ObjectPool.*

## Protected Member Functions

- override T[] **createObject** (int info)
- override void **destroyObject** (T[] obj)
- override bool **infosMatch** (int i0, int i1)

Protected Member Functions inherited from [ObjectPool< TType, TInfo >](#)

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

### Additional Inherited Members

### Protected Attributes inherited from [ObjectPool< TType, TInfo >](#)

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

### Properties inherited from [ObjectPool< TType, TInfo >](#)

- TInfo **Info** [get]  
*The property (info) that objects in this Pool must match.*

### 3.26.1 Detailed Description

Pool of Arrays with components of type T, with ObjectPool info being the array's size.

#### Template Parameters

|          |                     |
|----------|---------------------|
| <b>T</b> | Array element type. |
|----------|---------------------|

## 3.27 [ArrayPoolSet< T >](#) Class Template Reference

ArrayPool set of limited size.

Inherits [ObjectFactory< TType, TInfo >](#).

### Public Member Functions

- **ArrayPoolSet** (int capacity, string name, int defaultInfo, int setSize)
- T[] **New** ()
- bool **Free** (T[] obj)
- T[] **New** (int info)
- bool **Free** (T[] obj, int info)
- void **Dispose** ()

### Public Member Functions inherited from [ObjectFactory< TType, TInfo >](#)

- TType **New** (TInfo info)
- bool **Free** (TType obj)
- bool **Free** (TType obj, TInfo info)

### 3.27.1 Detailed Description

ArrayPool set of limited size.

## Template Parameters

|          |                     |
|----------|---------------------|
| <b>T</b> | Array element type. |
|----------|---------------------|

### 3.27.2 Member Function Documentation

#### 3.27.2.1 New()

`T [ ] New ()`

Implements [ObjectFactory< TType, TInfo >](#).

## 3.28 ArraySegmentPool< T > Class Template Reference

Set of ArrayPool's of  $N^2$ -sized arrays used to produce an ArraySegment with the requested size.

Inherits [ObjectFactory< TType, TInfo >](#).

### Public Member Functions

- **ArraySegmentPool** (int capacity, string name, int defaultInfo)
- ArraySegment< T > [New](#) ()
- ArraySegment< T > **New** (int info)
- bool **Free** (ArraySegment< T > obj, int info)
- bool **Free** (ArraySegment< T > obj)
- void **Dispose** ()

### Public Member Functions inherited from [ObjectFactory< TType, TInfo >](#)

- TType [New](#) (TInfo info)
- bool [Free](#) (TType obj)
- bool [Free](#) (TType obj, TInfo info)

### Static Public Member Functions

- static void **test** ()

### 3.28.1 Detailed Description

Set of ArrayPool's of  $N^2$ -sized arrays used to produce an ArraySegment with the requested size.

## Template Parameters

|          |                     |
|----------|---------------------|
| <b>T</b> | Array element type. |
|----------|---------------------|

### 3.28.2 Member Function Documentation

#### 3.28.2.1 New()

```
ArraySegment< T > New ()
```

Implements [ObjectFactory< TType, TInfo >](#).

## 3.29 AudioChangesHandler Class Reference

This component is useful to handle audio device and config changes.

Inherits [VoiceComponent](#).

### Public Attributes

- bool **HandleDeviceChange** = true  
*Try to react to device change notification when [Recorder](#) is started.*
- bool **HandleDeviceChangeIOS**  
*iOS: Try to react to device change notification when [Recorder](#) is started.*
- bool **HandleDeviceChangeAndroid**  
*Android: Try to react to device change notification when [Recorder](#) is started.*

### Protected Member Functions

- override void [Awake](#) ()

### Additional Inherited Members

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

#### 3.29.1 Detailed Description

This component is useful to handle audio device and config changes.

### 3.29.2 Member Function Documentation

#### 3.29.2.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

## 3.30 AudioClipWrapper Class Reference

Inherits [IAudioReader< T >](#).

### Public Member Functions

- **AudioClipWrapper** (AudioClip audioClip)
- bool **Read** (float[] buffer)
- void **Dispose** ()

### Public Member Functions inherited from [IDataReader< T >](#)

- bool [Read](#) (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Properties

- bool **Loop** [get, set]
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.30.1 Property Documentation

#### 3.30.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.30.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.30.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.31 AudioDesc Class Reference

Inherits [IAudioDesc](#).

### Public Member Functions

- **AudioDesc** (int samplingRate, int channels, string error)
- void **Dispose** ()

### Properties

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

## 3.31.1 Property Documentation

### 3.31.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.31.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).



### 3.31.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.32 AudioInChangeNotifier Class Reference

Inherits [IAudioInChangeNotifier](#).

### Public Member Functions

- **AudioInChangeNotifier** (Action callback, [ILogger](#) logger)
- void **Dispose** ()  
*Disposes enumerator. Call it to free native resources.*

### Properties

- bool [IsSupported](#) [get]
- string [Error](#) [get]  
*If not null, the enumerator is in invalid state.*

## 3.32.1 Property Documentation

### 3.32.1.1 Error

```
string Error [get]
```

If not null, the enumerator is in invalid state.

Implements [IAudioInChangeNotifier](#).

### 3.32.1.2 IsSupported

```
bool IsSupported [get]
```

Implements [IAudioInChangeNotifier](#).

## 3.33 AudioInChangeNotifier Class Reference

Inherits [IAudioInChangeNotifier](#).

## Public Member Functions

- **AudioInChangeNotifier** (Action callback, [ILogger](#) logger)
- void **Dispose** ()

*Disposes enumerator. Call it to free native resources.*

## Properties

- bool [IsSupported](#) [get]
- string [Error](#) [get]

*If not null, the enumerator is in invalid state.*

## 3.33.1 Property Documentation

### 3.33.1.1 Error

string [Error](#) [get]

If not null, the enumerator is in invalid state.

Implements [IAudioInChangeNotifier](#).

### 3.33.1.2 IsSupported

bool [IsSupported](#) [get]

Implements [IAudioInChangeNotifier](#).

## 3.34 AudioInChangeNotifierNotSupported Class Reference

Inherits [IAudioInChangeNotifier](#).

## Public Member Functions

- **AudioInChangeNotifierNotSupported** (Action callback, [ILogger](#) logger)
- void **Dispose** ()

## Properties

- bool [IsSupported](#) [get]
- string [Error](#) [get]

### 3.34.1 Property Documentation

#### 3.34.1.1 Error

string Error [get]

Implements [IAudioInChangeNotifier](#).

#### 3.34.1.2 IsSupported

bool IsSupported [get]

Implements [IAudioInChangeNotifier](#).

## 3.35 AudioInEnumerator Class Reference

Inherits [DeviceEnumeratorBase](#).

### Public Member Functions

- **AudioInEnumerator** (FMODLib.System coreSystem, [ILogger](#) logger)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

### 3.35.1 Member Function Documentation

#### 3.35.1.1 Refresh()

```
override void Refresh ()
```

Implements [IDeviceEnumerator](#).

## 3.36 AudioInEnumerator Class Reference

Enumerates microphones available on device.

Inherits [DeviceEnumeratorBase](#).

### Public Member Functions

- **AudioInEnumerator** ([ILogger](#) logger)
- override void [Refresh](#) ()  
*Refreshes the microphones list.*
- override void **Dispose** ()  
*Disposes enumerator. Call it to free native resources.*

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- [IEnumerator](#)< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- [List](#)< [DeviceInfo](#) > **devices** = new [List](#)<[DeviceInfo](#)>()
- [ILogger](#) logger

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

#### 3.36.1 Detailed Description

Enumerates microphones available on device.

### 3.36.2 Member Function Documentation

#### 3.36.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements [IDeviceEnumerator](#).

## 3.37 AudioInEnumerator Class Reference

Inherits [DeviceEnumeratorBase](#).

### Public Member Functions

- **AudioInEnumerator** ([ILogger](#) logger)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- [IEnumerator](#)< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Properties

- override bool [IsSupported](#) [get]
- override string [Error](#) [get]

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### 3.37.1 Member Function Documentation

#### 3.37.1.1 Refresh()

```
override void Refresh ()
```

Implements [IDeviceEnumerator](#).

### 3.37.2 Property Documentation

#### 3.37.2.1 Error

```
override string Error [get]
```

Implements [IDeviceEnumerator](#).

#### 3.37.2.2 IsSupported

```
override bool IsSupported [get]
```

Implements [IDeviceEnumerator](#).

## 3.38 AudioInEnumerator Class Reference

Inherits [DeviceEnumerator](#).

### Public Member Functions

- **AudioInEnumerator** ([ILogger](#) logger)

### Public Member Functions inherited from [DeviceEnumerator](#)

- **DeviceEnumerator** ([ILogger](#) logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

#### Additional Inherited Members

#### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

#### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## 3.39 AudioInEnumerator Class Reference

Enumerates microphones available on device.

Inherits [DeviceEnumeratorBase](#).

#### Public Member Functions

- **AudioInEnumerator** ([ILogger](#) logger)
- override void [Refresh](#) ()  
*Refreshes the microphones list.*
- override void **Dispose** ()  
*Disposes enumerator. Call it to free native resources.*

#### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

#### Additional Inherited Members

#### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

#### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

### 3.39.1 Detailed Description

Enumerates microphones available on device.

### 3.39.2 Member Function Documentation

#### 3.39.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements [IDeviceEnumerator](#).

## 3.40 AudioInPusher Class Reference

Inherits [IAudioPusher< T >](#), and [IResettable](#).

### Public Member Functions

- **AudioInPusher** ([AudioSessionParameters](#) sessParam, [ILogger](#) logger)
- void **SetCallback** (Action< float[ ]> callback, ObjectFactory< float[ ], int > bufferFactory, int optimalFrameSize)
- void [Reset](#) ()
- void **Dispose** ()

### Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[ ]> callback, ObjectFactory< T[ ], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

### Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.40.1 Member Function Documentation

#### 3.40.1.1 Reset()

```
void Reset ()
```

Implements [IResettable](#).



## 3.40.2 Property Documentation

### 3.40.2.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.40.2.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.40.2.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.41 AudioInPusher Class Reference

Inherits [IAudioPusher< T >](#).

### Public Member Functions

- **AudioInPusher** (int deviceId, [ILogger](#) logger)
- void **SetCallback** (Action< float[ ]> callback, ObjectFactory< float[ ], int > bufferFactory, int optimalFrameSize)
- void **Dispose** ()

### Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[ ]> callback, ObjectFactory< T[ ], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

## Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

## 3.41.1 Property Documentation

### 3.41.1.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.41.1.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.41.1.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.42 AudiInPusher Class Reference

Inherits [IAudioPusher< T >](#).

## Public Member Functions

- **AudiInPusher** ([ILogger](#) logger, int samplingRate, int channels, string deviceId)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimalFrameSize)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)
- void **EndOfStream** ()
- I [GetPlatformAPI](#)< I > ()
- void **Dispose** ()

## Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

## Properties

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*
- bool [ErrorAccess](#) [get]

## 3.42.1 Member Function Documentation

### 3.42.1.1 [GetPlatformAPI< I >\(\)](#)

`I GetPlatformAPI< I > ()`

#### Type Constraints

***I : class***

## 3.42.2 Property Documentation

### 3.42.2.1 [Channels](#)

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.42.2.2 [Error](#)

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.42.2.3 SamplingRate

int SamplingRate [get]

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.43 AudioInReader Class Reference

Inherits [IAudioReader< T >](#), and [IResettable](#).

### Public Member Functions

- **AudioInReader** ([AudioSessionParameters](#) sessParam, [ILogger](#) logger)
- void **Reset** ()
- void **Dispose** ()
- bool **Read** (float[] buf)

### Public Member Functions inherited from [IDataReader< T >](#)

- bool **Read** (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Properties

- int **Channels** [get]  
*Number of channels in the audio signal.*
- int **SamplingRate** [get]  
*Sampling rate of the audio signal (in Hz).*
- string **Error** [get]  
*If not null, audio object is in invalid state.*

## 3.43.1 Member Function Documentation

### 3.43.1.1 Reset()

void Reset ()

Implements [IResettable](#).

### 3.43.2 Property Documentation

#### 3.43.2.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.43.2.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

#### 3.43.2.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.44 AudioInReader Class Reference

Inherits [IAudioReader< T >](#).

### Public Member Functions

- **AudioInReader** (int deviceId, [ILogger](#) logger)
- void **Dispose** ()
- bool **Read** (float[] buf)

### Public Member Functions inherited from [IDataReader< T >](#)

- bool [Read](#) (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.44.1 Property Documentation

#### 3.44.1.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.44.1.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

#### 3.44.1.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.45 AudioInReader< T > Class Template Reference

Inherits [IAudioReader< T >](#).

### Public Member Functions

- **AudioInReader** (FMODLib.System coreSystem, int device, int suggestedFrequency, [ILogger](#) logger)
- void **Dispose** ()
- bool **Read** (T[] readBuf)

*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Public Attributes

- bool **isRecording**

### Properties

- int **SamplingRate** [get]  
*Sampling rate of the audio signal (in Hz).*
- int **Channels** [get]  
*Number of channels in the audio signal.*
- string **Error** [get]  
*If not null, audio object is in invalid state.*

### 3.45.1 Member Function Documentation

#### 3.45.1.1 Read()

`bool Read (  
    T[] buffer)`

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

## Parameters

|                     |                 |
|---------------------|-----------------|
| <code>buffer</code> | Buffer to fill. |
|---------------------|-----------------|

## Returns

True if buffer was filled successfully, false otherwise.

Implements [IDataReader< T >](#).

## 3.45.2 Property Documentation

### 3.45.2.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.45.2.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.45.2.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.46 AudioOut< T > Class Template Reference

Inherits [AudioOutDelayControl< T >](#).

Inherited by [AudioOutEvent< T >](#).

### Public Member Functions

- **AudioOut** (FMODLib.System coreSystem, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int samplingRate, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[] frame, int offsetSamples)
- override void [Stop](#) ()

## Public Member Functions inherited from [AudioOutDelayControl< T >](#)

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void [Start](#) (int frequency, int channels, int frameSamples)
- void [Service](#) ()
- void [Push](#) (T[] frame)
- void [Flush](#) ()

## Protected Attributes

- int **channels**
- int **frequency**

## Protected Attributes inherited from [AudioOutDelayControl< T >](#)

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

## Properties

- FMODLib.Sound **Sound** [get]
- FMODLib.Channel **Channel** [get]
- override long **OutPos** [get]
- string **Error** [get]

## Properties inherited from [AudioOutDelayControl< T >](#)

- long **OutPos** [get]
- int [Lag](#) [get]
- bool **IsFlushed** [get]
- bool [IsPlaying](#) [get]

## Additional Inherited Members

## Static Public Attributes inherited from [AudioOutDelayControl< T >](#)

- const int **FRAME\_POOL\_CAPACITY** = 50



### 3.46.1 Member Function Documentation

#### 3.46.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from [AudioOutDelayControl< T >](#).

## 3.47 AudioOutCapture Class Reference

Inherits MonoBehaviour.

### Events

- Action< float[], int > **OnAudioFrame**

## 3.48 AudioOutDelayControl Class Reference

Inherited by [AudioOutDelayControl< T >](#).

### Classes

- struct [PlayDelayConfig](#)

## 3.49 AudioOutDelayControl< T > Class Template Reference

Inherits [AudioOutDelayControl](#), and [IAudioOut< T >](#).

Inherited by [AudioSyncBuffer< T >](#), [AudioOut< T >](#), [UnityAudioOut](#), and [WebAudioAudioOut](#).

### Public Member Functions

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void **Start** (int frequency, int channels, int frameSamples)
- void **Service** ()
- void **Push** (T[] frame)
- void **Flush** ()
- virtual void **Stop** ()

### Static Public Attributes

- const int **FRAME\_POOL\_CAPACITY** = 50

### Protected Attributes

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

### Properties

- long **OutPos** [get]
- int [Lag](#) [get]
- bool **IsFlushed** [get]
- bool [IsPlaying](#) [get]

## 3.49.1 Member Function Documentation

### 3.49.1.1 Flush()

```
void Flush ()
```

Implements [IAudioOut< T >](#).

### 3.49.1.2 Push()

```
void Push (  
    T[] frame)
```

Implements [IAudioOut< T >](#).

### 3.49.1.3 Service()

```
void Service ()
```

Implements [IAudioOut< T >](#).

### 3.49.1.4 Start()

```
void Start (  
    int frequency,  
    int channels,  
    int frameSamples)
```

Implements [IAudioOut< T >](#).

#### 3.49.1.5 Stop()

```
virtual void Stop () [virtual]
```

Implements [IAudioOut< T >](#).

### 3.49.2 Property Documentation

#### 3.49.2.1 IsPlaying

```
bool IsPlaying [get]
```

Implements [IAudioOut< T >](#).

#### 3.49.2.2 Lag

```
int Lag [get]
```

Implements [IAudioOut< T >](#).

## 3.50 AudioOutDummy< T > Class Template Reference

Inherits [IAudioOut< T >](#).

### Public Member Functions

- void [Flush](#) ()
- void [Push](#) (T[] frame)
- void [Service](#) ()
- void [Start](#) (int frequency, int channels, int frameSamplesPerChannel)
- void [Stop](#) ()

### Properties

- bool [IsPlaying](#) [get]
- int [Lag](#) [get]

### 3.50.1 Member Function Documentation

#### 3.50.1.1 Flush()

```
void Flush ()
```

Implements [IAudioOut< T >](#).

### 3.50.1.2 Push()

```
void Push (  
    T[] frame)
```

Implements [IAudioOut< T >](#).

### 3.50.1.3 Service()

```
void Service ()
```

Implements [IAudioOut< T >](#).

### 3.50.1.4 Start()

```
void Start (  
    int frequency,  
    int channels,  
    int frameSamplesPerChannel)
```

Implements [IAudioOut< T >](#).

### 3.50.1.5 Stop()

```
void Stop ()
```

Implements [IAudioOut< T >](#).

## 3.50.2 Property Documentation

### 3.50.2.1 IsPlaying

```
bool IsPlaying [get]
```

Implements [IAudioOut< T >](#).

### 3.50.2.2 Lag

```
int Lag [get]
```

Implements [IAudioOut< T >](#).

## 3.51 AudioOutEvent< T > Class Template Reference

Inherits [AudioOut< T >](#).

**Public Member Functions**

- **AudioOutEvent** (FMODLib.System coreSystem, FMODLib.Studio.EventInstance fmodEvent, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutStart** ()
- override void **Stop** ()

**Public Member Functions inherited from [AudioOut< T >](#)**

- **AudioOut** (FMODLib.System coreSystem, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int samplingRate, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[] frame, int offsetSamples)

**Public Member Functions inherited from [AudioOutDelayControl< T >](#)**

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void **Start** (int frequency, int channels, int frameSamples)
- void **Service** ()
- void **Push** (T[] frame)
- void **Flush** ()

**Properties**

- override long **OutPos** [get]

**Properties inherited from [AudioOut< T >](#)**

- FMODLib.Sound **Sound** [get]
- FMODLib.Channel **Channel** [get]
- override long **OutPos** [get]
- string **Error** [get]

**Properties inherited from [AudioOutDelayControl< T >](#)**

- long **OutPos** [get]
- int **Lag** [get]
- bool **IsFlushed** [get]
- bool **IsPlaying** [get]

**Additional Inherited Members****Static Public Attributes inherited from [AudioOutDelayControl< T >](#)**

- const int **FRAME\_POOL\_CAPACITY** = 50

### Protected Attributes inherited from [AudioOut< T >](#)

- int **channels**
- int **frequency**

### Protected Attributes inherited from [AudioOutDelayControl< T >](#)

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

## 3.51.1 Member Function Documentation

### 3.51.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from [AudioOut< T >](#).

## 3.52 AudioReceptionSettings Class Reference

### Public Attributes

- int **AudioJitterBufferMs** = 200
- int **AudioJitterBufferMsHigh** = 400
- int **AudioJitterBufferMsMax** = 1000

## 3.53 AudioSenderSettings Class Reference

### Public Attributes

- int **AudioBitrate** = 30000
- bool **AudioEcho** = false

### Static Public Attributes

- const [POpusCodec.Enums.SamplingRate](#) **AudioEncoderSamplingRate** = [POpusCodec.Enums.SamplingRate.Sampling24000](#)
- const int **AudioMicSamplingRate** = 24000

## 3.54 AudioSessionParameters Struct Reference

### Public Member Functions

- int **CategoryOptionsToInt** ()
- override string **ToString** ()

**Public Attributes**

- [AudioSessionCategory](#) **Category**
- [AudioSessionMode](#) **Mode**
- [AudioSessionCategoryOption](#)[] **CategoryOptions**

## 3.55 AudioSessionParametersPresets Class Reference

**Static Public Attributes**

- static [AudioSessionParameters](#) **Game**
- static [AudioSessionParameters](#) **VoIP**

### 3.55.1 Member Data Documentation

#### 3.55.1.1 Game

[AudioSessionParameters](#) **Game** [static]

**Initial value:**

```
= new AudioSessionParameters()
{
    Category = AudioSessionCategory.PlayAndRecord,
    Mode = AudioSessionMode.VoiceChat,
    CategoryOptions = new AudioSessionCategoryOption[] {
        AudioSessionCategoryOption.DefaultToSpeaker, AudioSessionCategoryOption.AllowBluetooth }
}
```

#### 3.55.1.2 VoIP

[AudioSessionParameters](#) **VoIP** [static]

**Initial value:**

```
= new AudioSessionParameters()
{
    Category = AudioSessionCategory.PlayAndRecord,
    Mode = AudioSessionMode.VoiceChat,

    CategoryOptions = new AudioSessionCategoryOption[] { AudioSessionCategoryOption.AllowBluetooth }
}
```

## 3.56 AudioStreamReceiver Class Reference

Basic audio stream receiver Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

Inherits [StreamTransportManagerListener](#), and [IStreamReceiver](#).

## Public Member Functions

- virtual void [OnRemoteVoiceAdd](#) (int channelId, int playerId, byte voiceId, [VoiceInfo](#) i, ref [RemoteVoiceOptions](#) options)  
*Called when a remote stream (voice) has been detected.*

## Public Attributes

- [AudioReceptionSettings](#) **audioReceptionSettings** = new [AudioReceptionSettings](#)()

## Public Attributes inherited from [StreamTransportManagerListener](#)

- GameObject **streamTransportGameObject** = null

## Protected Member Functions

- override void **OnStreamTransportManagerFound** ()
- virtual void **Update** ()
- virtual void **StreamService** ()

## Protected Member Functions inherited from [StreamTransportManagerListener](#)

- virtual void **Awake** ()
- virtual void **OnDestroy** ()
- virtual IEnumerator [FindStreamTransportManagerCoroutine](#) ()  
*Try to find a IRegisterableStreamTransportManager, using [StreamTransportObserver.FindStreamTransportManager](#), that will fill the `_streamTransportManager` field if found Retries a few times (around 10s) if not found, before giving up.*
- void **OnStreamTransportManagerFound** ()  
*Called when a `_streamTransportManager` is set, and so a trnasport can be used.*

## Protected Attributes

- List< [IAudioOut](#)< float > > **audioPlayers** = new List<[IAudioOut](#)<float>>()

## Protected Attributes inherited from [StreamTransportManagerListener](#)

- [IStreamTransportManager](#) **\_streamTransportManager** = null
- [Photon.Voice.Unity.Logger](#) **\_logger** = new [Photon.Voice.Unity.Logger](#)(LogLevel.Info)

## Properties

- IEnumerable< [IAudioOut](#)< float > > **AudioPlayers** [get]

## Properties inherited from [StreamTransportManagerListener](#)

- GameObject [StreamTransportGameObject](#) [get, set]
- [IStreamTransportManager](#) [StreamTransportManager](#) [get, set]
- StreamTransport **Transport** [get]



## Properties inherited from [IStreamTransportManagerListener](#)

- `GameObject` **gameObject** [get]

### 3.56.1 Detailed Description

Basic audio stream receiver Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

### 3.56.2 Member Function Documentation

#### 3.56.2.1 OnRemoteVoiceAdd()

```
virtual void OnRemoteVoiceAdd (
    int channelId,
    int playerId,
    byte voiceId,
    VoiceInfo i,
    ref RemoteVoiceOptions options) [virtual]
```

Called when a remote stream (voice) has been detected.

Implements [IStreamReceiver](#).

## 3.57 AudioStreamSender Class Reference

Basic audio stream sender Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

Inherits [StreamTransportManagerListener](#), and [IStreamSender](#).

### Public Types

- enum **Status**

### Public Member Functions

- virtual bool **IsRecordingPossible** ()
- virtual void **StartSending** ()
- virtual void **StopSending** ()
- virtual void **ToggleSending** ()
- virtual void [OnRoomJoined](#) ()
 

*Called when the [Photon Voice](#) room is joined, and creating streams (voice) is possible.*
- virtual void [OnLeftRoom](#) ()

## Public Attributes

- [AudioSenderSettings](#) **audioSenderSettings** = new [AudioSenderSettings](#)()
- bool **sendOnJoinRoom** = true
- Status **status** = Status.NotSending

## Public Attributes inherited from [StreamTransportManagerListener](#)

- GameObject **streamTransportGameObject** = null

## Protected Member Functions

- override void **OnStreamTransportManagerFound** ()
- virtual void **Update** ()
- override void **OnDestroy** ()
- void **CreateLocalStreamAudio** ()
- void **RemoveLocalStreamAudio** ()
- void **CreateStream** ()
- virtual IEnumerator **CreateStreamCoroutine** ()
 

*Check if collecting is possible (by testing [IsRecordingPossible](#)()) before starting the recording. If the recording is not possible for [MaxWaitTimeforCollectingAvailability](#) seconds, displays an error and gives up.*
- virtual IEnumerator **WaitForRecordingBeingPossible** ()
- virtual void **WillTrytoRecord** ()
- virtual void **AskForPermissions** ()
- virtual void **OnPermissionGranted** (bool granted)

## Protected Member Functions inherited from [StreamTransportManagerListener](#)

- virtual void **Awake** ()
- virtual IEnumerator **FindStreamTransportManagerCoroutine** ()
 

*Try to find a [IRegisterableStreamTransportManager](#), using [StreamTransportObserver.FindStreamTransportManager](#), that will fill the `_streamTransportManager` field if found. Retries a few times (around 10s) if not found, before giving up.*
- void **OnStreamTransportManagerFound** ()
 

*Called when a `_streamTransportManager` is set, and so a trnasport can be used.*

## Protected Attributes

- bool **\_shouldSend** = false
- bool **\_roomJoined** = false
- [Photon.Voice.DeviceInfo](#) **\_micDevice** = [Photon.Voice.DeviceInfo.Default](#)
- [IAudioDesc](#) **\_audioSource**
- bool **\_permissionGranted** = false

## Protected Attributes inherited from [StreamTransportManagerListener](#)

- [IStreamTransportManager](#) **\_streamTransportManager** = null
- [Photon.Voice.Unity.Logger](#) **\_logger** = new [Photon.Voice.Unity.Logger](#)(LogLevel.Info)

## Properties

- virtual int **AudioChannels** [get]
- int **AudioTransportChannel** [get]
- virtual float **MaxWaitTimeforCollectingAvailability** [get]

## Properties inherited from [StreamTransportManagerListener](#)

- GameObject [StreamTransportGameObject](#) [get, set]
- [IStreamTransportManager](#) [StreamTransportManager](#) [get, set]
- StreamTransport **Transport** [get]

## Properties inherited from [IStreamTransportManagerListener](#)

- GameObject **gameObject** [get]

### 3.57.1 Detailed Description

Basic audio stream sender Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

### 3.57.2 Member Function Documentation

#### 3.57.2.1 OnDestroy()

```
override void OnDestroy () [protected], [virtual]
```

Reimplemented from [StreamTransportManagerListener](#).

#### 3.57.2.2 OnLeftRoom()

```
virtual void OnLeftRoom () [virtual]
```

Implements [IStreamSender](#).

#### 3.57.2.3 OnRoomJoined()

```
virtual void OnRoomJoined () [virtual]
```

Called when the [Photon Voice](#) room is joined, and creating streams (voice) is possible.

Implements [IStreamSender](#).

### 3.58 AudioSyncBuffer< T > Class Template Reference

Inherits [AudioOutDelayControl< T >](#).

#### Public Member Functions

- **AudioSyncBuffer** ([PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[] data, int offsetSamples)
- override void [Stop](#) ()
- void **Read** (T[] outBuf, int outChannels, int outSampleRate)

#### Public Member Functions inherited from [AudioOutDelayControl< T >](#)

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void [Start](#) (int frequency, int channels, int frameSamples)
- void [Service](#) ()
- void [Push](#) (T[] frame)
- void [Flush](#) ()

#### Properties

- override long **OutPos** [get]

#### Properties inherited from [AudioOutDelayControl< T >](#)

- long **OutPos** [get]
- int [Lag](#) [get]
- bool **IsFlushed** [get]
- bool [IsPlaying](#) [get]

#### Additional Inherited Members

#### Static Public Attributes inherited from [AudioOutDelayControl< T >](#)

- const int **FRAME\_POOL\_CAPACITY** = 50

#### Protected Attributes inherited from [AudioOutDelayControl< T >](#)

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

### 3.58.1 Member Function Documentation

#### 3.58.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from [AudioOutDelayControl< T >](#).

## 3.59 AudioUtil Class Reference

Collection of Audio [Utility](#) functions and classes.

### Classes

- class [GeneratorPusher< T >](#)  
*IAudioPusher that provides a constant tone signal.*
- class [GeneratorReader< T >](#)
- interface [ILevelMeter](#)  
*Audio Level Metering interface.*
- interface [IVoiceDetector](#)  
*Voice Activity Detector interface.*
- class [LevelMeter< T >](#)  
*Audio Level Meter.*
- class [LevelMeterDummy](#)  
*Dummy Audio Level Meter that doesn't actually do anything.*
- class [LevelMeterFloat](#)  
*LevelMeter specialization for float audio.*
- class [LevelMeterShort](#)  
*LevelMeter specialization for short audio.*
- class [Resampler< T >](#)  
*Sample-rate conversion Audio Processor.*
- class [TempoUp< T >](#)
- class [ToneAudioPusher< T >](#)  
*IAudioPusher that provides a constant tone signal.*
- class [ToneAudioReader< T >](#)  
*IAudioReader that provides a constant tone signal.*
- class [VoiceDetector< T >](#)  
*Simple voice activity detector triggered by signal level.*
- class [VoiceDetectorCalibration< T >](#)  
*Calibration Utility for Voice Detector.*
- class [VoiceDetectorDummy](#)  
*Dummy VoiceDetector that doesn't actually do anything.*
- class [VoiceDetectorFloat](#)  
*VoiceDetector specialization for float audio.*
- class [VoiceDetectorShort](#)  
*VoiceDetector specialization for short audio.*
- class [VoiceLevelDetectCalibrate< T >](#)  
*Utility Audio Processor Voice Detection Calibration.*
- class [WaveformAudioPusher< T >](#)  
*IAudioPusher that provides the given waveform.*
- class [WaveformAudioReader< T >](#)  
*IAudioReader that provides the given waveform.*

## Static Public Member Functions

- static int **ToneToBuf**< T > (T[] buf, long timeSamples, int channels, double amp, double k, double phase↔Mod=0)
- static int **ToneToBuf**< T > (T[] buf, int offset, int length, long timeSamples, int channels, double amp, double k, double phaseMod=0)
- static int **WaveformToBuf**< T > (T[] buf, T[] waveform, long timePos)
- static void **Resample**< T > (T[] src, T[] dst, int dstCount, int channels)  
*Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer.*
- static void **Resample**< T > (T[] src, int srcOffset, int srcCount, T[] dst, int dstOffset, int dstCount, int channels)
- static void **Resample**< T > (T[] src, int srcOffset, int srcCount, int srcChannels, T[] dst, int dstOffset, int dstCount, int dstChannels)
- static void **ResampleAndConvert** (short[] src, float[] dst, int dstCount, int channels)  
*Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert short to float samples along the way.*
- static void **ResampleAndConvert** (float[] src, short[] dst, int dstCount, int channels)  
*Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert float to short samples along the way.*
- static void **Convert** (float[] src, short[] dst, int dstCount)  
*Convert audio buffer from float to short samples.*
- static void **Convert** (short[] src, float[] dst, int dstCount)  
*Convert audio buffer from short to float samples.*
- static void **ForceToStereo**< T > (T[] src, T[] dst, int srcChannels)  
*Convert audio buffer with arbitrary number of channels to stereo.*

### 3.59.1 Detailed Description

Collection of Audio [Utility](#) functions and classes.

### 3.59.2 Member Function Documentation

#### 3.59.2.1 Convert() [1/2]

```
void Convert (
    float[] src,
    short[] dst,
    int dstCount) [static]
```

Convert audio buffer from float to short samples.

##### Parameters

|                 |                                                                                                |                     |
|-----------------|------------------------------------------------------------------------------------------------|---------------------|
|                 | <i>src</i>                                                                                     | Source buffer.      |
|                 | <i>dst</i>                                                                                     | Destination buffer. |
| <i>dstCount</i> | Size of destination buffer (in total samples), source buffer must be of same length or longer. |                     |

#### 3.59.2.2 Convert() [2/2]

```
void Convert (
    short[] src,
    float[] dst,
    int dstCount) [static]
```

Convert audio buffer from short to float samples.

## Parameters

|                 |                                                                                                |                     |
|-----------------|------------------------------------------------------------------------------------------------|---------------------|
|                 | <i>src</i>                                                                                     | Source buffer.      |
|                 | <i>dst</i>                                                                                     | Destination buffer. |
| <i>dstCount</i> | Size of destination buffer (in total samples), source buffer must be of same length or longer. |                     |

**3.59.2.3 ForceToStereo< T >()**

```
void ForceToStereo< T > (
    T[] src,
    T[] dst,
    int srcChannels) [static]
```

Convert audio buffer with arbitrary number of channels to stereo.

For mono sources (*srcChannels*==1), the signal will be copied to both Left and Right stereo channels. For all others, the first two available channels will be used, any other channels will be discarded.

## Parameters

|                    |                                                  |                     |
|--------------------|--------------------------------------------------|---------------------|
|                    | <i>src</i>                                       | Source buffer.      |
|                    | <i>dst</i>                                       | Destination buffer. |
| <i>srcChannels</i> | Number of (interleaved) channels in <i>src</i> . |                     |

**3.59.2.4 Resample< T >()**

```
void Resample< T > (
    T[] src,
    T[] dst,
    int dstCount,
    int channels) [static]
```

Resample audio data so that the complete *src* buffer fits into *dstCount* samples in the *dst* buffer.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

## Parameters

|                 |                                                                   |                     |
|-----------------|-------------------------------------------------------------------|---------------------|
|                 | <i>src</i>                                                        | Source buffer.      |
|                 | <i>dst</i>                                                        | Destination buffer. |
| <i>dstCount</i> | Target size of destination buffer (in samples per channel).       |                     |
| <i>channels</i> | Number of channels in the signal (1=mono, 2=stereo). Must be > 0. |                     |

**3.59.2.5 ResampleAndConvert() [1/2]**

```
void ResampleAndConvert (
    float[] src,
    short[] dst,
    int dstCount,
    int channels) [static]
```

Resample audio data so that the complete *src* buffer fits into *dstCount* samples in the *dst* buffer, and convert float to short samples along the way.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

## Parameters

|                 |                                                                   |                     |
|-----------------|-------------------------------------------------------------------|---------------------|
|                 | <i>src</i>                                                        | Source buffer.      |
|                 | <i>dst</i>                                                        | Destination buffer. |
| <i>dstCount</i> | Target size of destination buffer (in samples per channel).       |                     |
| <i>channels</i> | Number of channels in the signal (1=mono, 2=stereo). Must be > 0. |                     |

**3.59.2.6 ResampleAndConvert()** [2/2]

```
void ResampleAndConvert (
    short[] src,
    float[] dst,
    int dstCount,
    int channels) [static]
```

Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert short to float samples along the way.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

## Parameters

|                 |                                                                   |                     |
|-----------------|-------------------------------------------------------------------|---------------------|
|                 | <i>src</i>                                                        | Source buffer.      |
|                 | <i>dst</i>                                                        | Destination buffer. |
| <i>dstCount</i> | Target size of destination buffer (in samples per channel).       |                     |
| <i>channels</i> | Number of channels in the signal (1=mono, 2=stereo). Must be > 0. |                     |

**3.60 BufferReaderPushAdapterAsyncPool< T > Class Template Reference**

BufferReaderPushAdapter<T> implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#).

Inherits [BufferReaderPushAdapterBase< T >](#).

**Public Member Functions**

- [BufferReaderPushAdapterAsyncPool](#) (IDataReader< T > reader)  
*Create a new BufferReaderPushAdapter instance.*
- override void [Service](#) ([LocalVoice](#) localVoice)  
*Do the actual data read/push.*

**Public Member Functions inherited from [BufferReaderPushAdapterBase< T >](#)**

- void [Service](#) ([LocalVoice](#) localVoice)  
*Do the actual data read/push.*
- [BufferReaderPushAdapterBase](#) (IDataReader< T > reader)  
*Create a new BufferReaderPushAdapterBase instance.*
- void **Dispose** ()  
*Release resources associated with this instance.*



### Additional Inherited Members

### Protected Attributes inherited from [BufferReaderPushAdapterBase< T >](#)

- `IDataReader< T > reader`

#### 3.60.1 Detailed Description

`BufferReaderPushAdapter<T>` implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#).

Acquires a buffer from pool before each Read, releases buffer after last Read (brings Acquire/Release overhead). Expects localVoice to be a [LocalVoiceFramed<T>](#) of same T.

#### 3.60.2 Member Function Documentation

##### 3.60.2.1 BufferReaderPushAdapterAsyncPool()

```
BufferReaderPushAdapterAsyncPool (
    IDataReader< T > reader)
```

Create a new `BufferReaderPushAdapter` instance.

###### Parameters

|                     |                                        |
|---------------------|----------------------------------------|
| <code>reader</code> | <code>IDataReader</code> to read from. |
|---------------------|----------------------------------------|

##### 3.60.2.2 Service()

```
override void Service (
    LocalVoice localVoice)
```

Do the actual data read/push.

###### Parameters

|                         |                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>localVoice</code> | <code>LocalVoice</code> instance to push data to. Must be a <a href="#">LocalVoiceFramed&lt;T&gt;</a> of same T. |
|-------------------------|------------------------------------------------------------------------------------------------------------------|

Implements [IServiceable](#).

## 3.61 BufferReaderPushAdapterAsyncPoolFloatToShort Class Reference

`BufferReaderPushAdapter<T>` implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#), converting float samples to short.

Inherits [BufferReaderPushAdapterBase< T >](#).

## Public Member Functions

- [BufferReaderPushAdapterAsyncPoolFloatToShort](#) (IDataReader< float > reader)  
*Create a new BufferReaderPushAdapter instance.*
- override void [Service](#) ([LocalVoice](#) localVoice)  
*Do the actual data read/push.*

## Public Member Functions inherited from [BufferReaderPushAdapterBase< T >](#)

- void [Service](#) ([LocalVoice](#) localVoice)  
*Do the actual data read/push.*
- [BufferReaderPushAdapterBase](#) (IDataReader< T > reader)  
*Create a new BufferReaderPushAdapterBase instance.*
- void **Dispose** ()  
*Release resources associated with this instance.*

## Additional Inherited Members

## Protected Attributes inherited from [BufferReaderPushAdapterBase< T >](#)

- IDataReader< T > **reader**

### 3.61.1 Detailed Description

BufferReaderPushAdapter<T> implementation using asynchronous [LocalVoiceFramed<T>.PushDataAsync](#), converting float samples to short.

This adapter works exactly like [BufferReaderPushAdapterAsyncPool<T>](#), but it converts float samples to short. Acquires a buffer from pool before each Read, releases buffer after last Read.

Expects localVoice to be a [LocalVoiceFramed<T>](#) of same T.

### 3.61.2 Constructor & Destructor Documentation

#### 3.61.2.1 BufferReaderPushAdapterAsyncPoolFloatToShort()

```
BufferReaderPushAdapterAsyncPoolFloatToShort (
    IDataReader< float > reader)
```

Create a new BufferReaderPushAdapter instance.

#### Parameters

|                                        |                          |
|----------------------------------------|--------------------------|
| <input type="checkbox"/> <i>reader</i> | DataReader to read from. |
|----------------------------------------|--------------------------|

### 3.61.3 Member Function Documentation

#### 3.61.3.1 Service()

```
override void Service (
    LocalVoice localVoice)
```

Do the actual data read/push.

## Parameters

|                         |                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>localVoice</code> | <code>LocalVoice</code> instance to push data to. Must be a <code>LocalVoiceFramed&lt;T&gt;</code> of same T. |
|-------------------------|---------------------------------------------------------------------------------------------------------------|

Implements [IServiceable](#).

## 3.62 BufferReaderPushAdapterAsyncPoolShortToFloat Class Reference

`BufferReaderPushAdapter<T>` implementation using asynchronous `LocalVoiceFramed<T>.PushDataAsync`, converting short samples to float.

Inherits [BufferReaderPushAdapterBase< T >](#).

### Public Member Functions

- [BufferReaderPushAdapterAsyncPoolShortToFloat](#) (`IDataReader< short > reader`)  
*Create a new BufferReaderPushAdapter instance.*
- override void [Service](#) (`LocalVoice localVoice`)  
*Do the actual data read/push.*

### Public Member Functions inherited from [BufferReaderPushAdapterBase< T >](#)

- void [Service](#) (`LocalVoice localVoice`)  
*Do the actual data read/push.*
- [BufferReaderPushAdapterBase](#) (`IDataReader< T > reader`)  
*Create a new BufferReaderPushAdapterBase instance.*
- void **Dispose** ()  
*Release resources associated with this instance.*

### Additional Inherited Members

### Protected Attributes inherited from [BufferReaderPushAdapterBase< T >](#)

- `IDataReader< T > reader`

### 3.62.1 Detailed Description

`BufferReaderPushAdapter<T>` implementation using asynchronous `LocalVoiceFramed<T>.PushDataAsync`, converting short samples to float.

This adapter works exactly like [BufferReaderPushAdapterAsyncPool<T>](#), but it converts short samples to float. Acquires a buffer from pool before each Read, releases buffer after last Read.

Expects localVoice to be a [LocalVoiceFramed<T>](#) of same T.

### 3.62.2 Constructor & Destructor Documentation

#### 3.62.2.1 BufferReaderPushAdapterAsyncPoolShortToFloat()

```
BufferReaderPushAdapterAsyncPoolShortToFloat (
    IDataReader< short > reader)
```

Create a new BufferReaderPushAdapter instance.

## Parameters

|                     |                          |
|---------------------|--------------------------|
| <code>reader</code> | DataReader to read from. |
|---------------------|--------------------------|

### 3.62.3 Member Function Documentation

#### 3.62.3.1 Service()

```
override void Service (
    LocalVoice localVoice)
```

Do the actual data read/push.

## Parameters

|                         |                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>localVoice</code> | <a href="#">LocalVoice</a> instance to push data to. Must be a <a href="#">LocalVoiceFramed&lt;T&gt;</a> of same T. |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|

Implements [IServiceable](#).

## 3.63 BufferReaderPushAdapterBase< T > Class Template Reference

Adapter base reading data from [IDataReader<T>.Read](#) and pushing it to [LocalVoice](#).

Inherits [IServiceable](#).

Inherited by [BufferReaderPushAdapterAsyncPool< T >](#), [BufferReaderPushAdapterAsyncPoolFloatToShort](#), and [BufferReaderPushAdapterAsyncPoolShortToFloat](#).

#### Public Member Functions

- void [Service](#) ([LocalVoice](#) localVoice)  
*Do the actual data read/push.*
- [BufferReaderPushAdapterBase](#) (IDataReader< T > reader)  
*Create a new BufferReaderPushAdapterBase instance.*
- void **Dispose** ()  
*Release resources associated with this instance.*

#### Protected Attributes

- IDataReader< T > **reader**

#### 3.63.1 Detailed Description

Adapter base reading data from [IDataReader<T>.Read](#) and pushing it to [LocalVoice](#).

Use this with a [LocalVoice](#) of same T type.

### 3.63.2 Member Function Documentation

#### 3.63.2.1 BufferReaderPushAdapterBase()

```
BufferReaderPushAdapterBase (
    IDataReader< T > reader)
```

Create a new BufferReaderPushAdapterBase instance.

## Parameters

|                     |                          |
|---------------------|--------------------------|
| <code>reader</code> | DataReader to read from. |
|---------------------|--------------------------|

## 3.63.2.2 Service()

```
void Service (
    LocalVoice localVoice) [abstract]
```

Do the actual data read/push.

## Parameters

|                         |                                      |
|-------------------------|--------------------------------------|
| <code>localVoice</code> | LocalVoice instance to push data to. |
|-------------------------|--------------------------------------|

Implements [IServiceable](#).

## 3.64 ByteStreamDecoder Class Reference

Inherits [IDecoder](#).

## Public Member Functions

- delegate void **OutputDelegate** (ref [FrameBuffer](#) buf)
- **ByteStreamDecoder** (OutputDelegate output, Action onMissingFrame)
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Dispose** ()

## Properties

- string **Error** [get]  
*If not null, the object is in invalid state.*

## 3.64.1 Member Function Documentation

## 3.64.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

## 3.64.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

## Parameters

|                   |                                          |
|-------------------|------------------------------------------|
| <code>info</code> | Properties of the data stream to decode. |
|-------------------|------------------------------------------|

Implements [IDecoder](#).

### 3.64.2 Property Documentation

#### 3.64.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.65 ByteStreamEncoder Class Reference

Inherits [IEncoderDirect< B >](#).

### Public Member Functions

- void **Input** (byte[] buf)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect< B >](#)

- void [Input](#) (B buf)  
*Consumes the given raw data.*

### Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.65.1 Member Function Documentation

### 3.65.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.65.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.65.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

## 3.65.2 Property Documentation

### 3.65.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.65.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.66 CaptureDevice Class Reference

### Public Types

- enum **Media**

### Public Member Functions

- **CaptureDevice** (ILogger logger, Media media, string deviceId)
- void **Initialize** ()
- void **InitializeAsync** ()
- async Task **CleanUpAsync** ()  
*Asynchronous method cleaning up resources and stopping recording if necessary.*
- async Task< IMediaEncodingProperties > **SelectPreferredCameraStreamSettingAsync** (MediaStreamType mediaStreamType, Func< IMediaEncodingProperties, bool > filterSettings)  
*Allow selection of camera settings.*
- async Task **StartRecordingAsync** (MediaEncodingProfile encodingProfile, Action< byte[], FrameFlags > encoderCallback)  
*Starts media recording asynchronously.*
- async Task **StopRecordingAsync** ()  
*Stops recording asynchronously.*

### Static Public Member Functions

- static async Task< bool > **CheckForRecordingDeviceAsync** ()

### Properties

- MediaCapture **CaptureSource** [get]  
*Creates url object from MediaCapture.*

### Events

- EventHandler< MediaCaptureFailedEventArgs > **CaptureFailed**

### 3.66.1 Member Function Documentation

#### 3.66.1.1 SelectPreferredCameraStreamSettingAsync()

```
async Task< IMediaEncodingProperties > SelectPreferredCameraStreamSettingAsync (
    MediaStreamType mediaStreamType,
    Func< IMediaEncodingProperties, bool > filterSettings)
```

Allow selection of camera settings.



## Parameters

|                              |                                                                            |
|------------------------------|----------------------------------------------------------------------------|
| <code>mediaStreamType</code> | Type of a the media stream.                                                |
| <code>filterSettings</code>  | A predicate function, which will be called to filter the correct settings. |

## 3.66.1.2 StartRecordingAsync()

```
async Task StartRecordingAsync (
    MediaEncodingProfile encodingProfile,
    Action< byte[], FrameFlags > encoderCallback)
```

Starts media recording asynchronously.

## Parameters

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| <code>encodingProfile</code> | Encoding profile used for the recording session |
|------------------------------|-------------------------------------------------|

## 3.67 ConnectAndJoin Class Reference

Inherits MonoBehaviour, IConnectionCallbacks, and IMatchmakingCallbacks.

## Public Member Functions

- void **ConnectNow** ()
- void **OnCreatedRoom** ()
- void **OnCreateRoomFailed** (short returnCode, string message)
- void **OnFriendListUpdate** (List< FriendInfo > friendList)
- void **OnJoinedRoom** ()
- void **OnJoinRandomFailed** (short returnCode, string message)
- void **OnJoinRoomFailed** (short returnCode, string message)
- void **OnLeftRoom** ()
- void **OnConnected** ()
- void **OnConnectedToMaster** ()
- void **OnDisconnected** (DisconnectCause cause)
- void **OnRegionListReceived** (RegionHandler regionHandler)
- void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- void **OnCustomAuthenticationFailed** (string debugMessage)

## Public Attributes

- bool **RandomRoom** = true
- string **RoomName**

## Properties

- bool **IsConnected** [get]

## 3.68 VoiceClient.CreateOptions Struct Reference

### Public Attributes

- byte **VoiceIDMin**
- byte **VoiceIDMax**

### Static Public Attributes

- static [CreateOptions Default](#)

### 3.68.1 Member Data Documentation

#### 3.68.1.1 Default

[CreateOptions Default](#) [static]

#### Initial value:

```
= new CreateOptions()  
{  
    VoiceIDMin = 1,  
    VoiceIDMax = 15  
}
```

## 3.69 VPxCodec.Decoder Class Reference

Inherits [IDecoderDirect< B >](#).

### Public Member Functions

- **Decoder** ([ILogger](#) logger)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

### Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< [ImageBufferNative](#) > [Output](#) [get, set]  
*Callback to call when a new decoded data buffer is available.*

### 3.69.1 Member Function Documentation

#### 3.69.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call `buf.Retain()` to prevent the caller from disposing the buffer. In this case, the callee should call `buf.Release()` when buffer is no longer needed.

Implements [IDecoder](#).

#### 3.69.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

##### Parameters

|                   |                                          |
|-------------------|------------------------------------------|
| <code>info</code> | Properties of the data stream to decode. |
|-------------------|------------------------------------------|

Implements [IDecoder](#).

### 3.69.2 Property Documentation

#### 3.69.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

#### 3.69.2.2 Output

```
Action<ImageBufferNative> Output [get], [set]
```

Callback to call when a new decoded data buffer is available.

Implements [IDecoderDirect< B >](#).

## 3.70 OpusCodec.Decoder< T > Class Template Reference

Inherits [IDecoder](#).

## Public Member Functions

- **Decoder** (Action< FrameOut< T > > output, [ILogger](#) logger)
- void [Open](#) ([VoiceInfo](#) i)  
*Open (initialize) the decoder.*
- void **Dispose** ()
- void [Input](#) (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*

## Protected Attributes

- OpusDecoder< T > **decoder**

## Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*

### 3.70.1 Member Function Documentation

#### 3.70.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

#### 3.70.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|                                              |                                          |
|----------------------------------------------|------------------------------------------|
| <div><div></div><div><i>info</i></div></div> | Properties of the data stream to decode. |
|----------------------------------------------|------------------------------------------|

Implements [IDecoder](#).

## 3.70.2 Property Documentation

### 3.70.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.71 RawCodec.Decoder< T > Class Template Reference

Inherits [IDecoder](#).

### Public Member Functions

- **Decoder** (Action< FrameOut< T > > output)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) byteBuf)  
*Consumes the given encoded data.*
- void **Dispose** ()

### Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*

## 3.71.1 Member Function Documentation

### 3.71.1.1 Input()

```
void Input (  
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

### 3.71.1.2 Open()

```
void Open (  
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|                   |                                          |
|-------------------|------------------------------------------|
| <code>info</code> | Properties of the data stream to decode. |
|-------------------|------------------------------------------|

Implements [IDecoder](#).

### 3.71.2 Property Documentation

#### 3.71.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.72 DecoderConfigFrame Class Reference

Stores the config frame and prevents other frames decoding until the decoder is ready.

Inherits [IDisposable](#).

#### Public Member Functions

- **DecoderConfigFrame** ([ILogger](#) logger, [IDecoder](#) decoder)
- bool [TryConfigure](#) (ref [FrameBuffer](#) buf, bool decoderReady)  
*Call it in Input().*
- void **Dispose** ()

### 3.72.1 Detailed Description

Stores the config frame and prevents other frames decoding until the decoder is ready.

### 3.72.2 Member Function Documentation

#### 3.72.2.1 TryConfigure()

```
bool TryConfigure (  
    ref FrameBuffer buf,  
    bool decoderReady)
```

Call it in Input().

## Parameters

|                     |                               |             |
|---------------------|-------------------------------|-------------|
|                     | <i>buf</i>                    | Data frame. |
| <i>decoderReady</i> | True if the decoder is ready. |             |

## Returns

True if decoder is allowed to decode the current frame.

## 3.73 DeviceEnumerator Class Reference

Inherits [DeviceEnumeratorBase](#).

Inherited by [AudioInEnumerator](#), and [VideoInEnumerator](#).

### Public Member Functions

- **DeviceEnumerator** ([ILogger](#) logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## 3.73.1 Member Function Documentation

### 3.73.1.1 Refresh()

```
override void Refresh ()
```

Implements [IDeviceEnumerator](#).

## 3.74 DeviceEnumeratorBase Class Reference

Inherits [IDeviceEnumerator](#).

Inherited by , , [AudioInEnumerator](#), [VideoInEnumerator](#), [AudioInEnumerator](#), [DeviceEnumerator](#), [AndroidVideoInEnumerator](#), [AudioInEnumerator](#), and [VideoInEnumerator](#).

### Public Member Functions

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- [IEnumerator](#)< [DeviceInfo](#) > **GetEnumerator** ()
- void **Refresh** ()
- void **Dispose** ()

### Protected Attributes

- [List](#)< [DeviceInfo](#) > **devices** = new [List](#)<[DeviceInfo](#)>()
- [ILogger](#) logger

### Properties

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## 3.74.1 Member Function Documentation

### 3.74.1.1 Refresh()

```
void Refresh () [abstract]
```

Implements [IDeviceEnumerator](#).

## 3.74.2 Property Documentation

### 3.74.2.1 Error

```
virtual string Error [get], [protected set]
```

Implements [IDeviceEnumerator](#).

### 3.74.2.2 IsSupported

```
virtual bool IsSupported [get]
```

Implements [IDeviceEnumerator](#).



### 3.74.2.3 OnReady

Action OnReady [protected get], [set]

Implements [IDeviceEnumerator](#).

## 3.75 DeviceFeatures Class Reference

### Public Member Functions

- **DeviceFeatures** (CameraFacing facing)

### Properties

- CameraFacing **CameraFacing** [get]

## 3.76 DeviceInfo Struct Reference

### Public Member Functions

- **DeviceInfo** (int id, string name, [DeviceFeatures](#) features=null)
- **DeviceInfo** (string id, string name, [DeviceFeatures](#) features=null)
- **DeviceInfo** (string name, [DeviceFeatures](#) features=null)
- override bool **Equals** (object obj)
- override int **GetHashCode** ()
- override string **ToString** ()

### Static Public Member Functions

- static bool **operator==** ([DeviceInfo](#) d1, [DeviceInfo](#) d2)
- static bool **operator!=** ([DeviceInfo](#) d1, [DeviceInfo](#) d2)

### Public Attributes

- [DeviceFeatures](#) **features**

### Static Public Attributes

- static readonly [DeviceInfo](#) **Default** = new [DeviceInfo](#)(true, -128, "", "[Default]")

### Properties

- bool **IsDefault** [get]
- int **IDInt** [get]
- string **IDString** [get]
- string **Name** [get]
- [DeviceFeatures](#) **Features** [get]

## 3.77 RecorderPreset.DSP Struct Reference

### Public Attributes

- bool **AEC**
- bool **VAD**

## 3.78 VPxCodec.Encoder Class Reference

Inherits [IEncoderDirectImage](#).

### Public Member Functions

- **Encoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- [ArraySegment](#)< byte > **DequeueOutput** (out [FrameFlags](#) flags)  
*Returns next encoded data frame (if such output supported).*
- void **Input** ([ImageBufferNative](#) imBuf)
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- [I](#) **GetPlatformAPI**< [I](#) > ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect](#)< [B](#) >

- void **Input** ([B](#) buf)  
*Consumes the given raw data.*

### Properties

- string **Error** [get]  
*If not null, the object is in invalid state.*
- [ImageFormat](#) **ImageFormat** [get]  
*Recommended encoder input image format. [Encoder](#) may support other formats.*
- [Action](#)< [ArraySegment](#)< byte >, [FrameFlags](#) > **Output** [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### 3.78.1 Member Function Documentation

#### 3.78.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.78.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.78.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

## 3.78.2 Property Documentation

### 3.78.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.78.2.2 ImageFormat

```
ImageFormat ImageFormat [get]
```

Recommended encoder input image format. [Encoder](#) may support other formats.

Implements [IEncoderDirectImage](#).

### 3.78.2.3 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.79 OpusCodec.Encoder< T > Class Template Reference

Inherits [IEncoderDirect< B >](#).

Inherited by [OpusCodec.EncoderFloat](#), and [OpusCodec.EncoderShort](#).

### Public Member Functions

- void **Input** (T[] buf)
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- I [GetPlatformAPI< I >](#) ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect< B >](#)

- void [Input](#) (B buf)  
*Consumes the given raw data.*

### Protected Member Functions

- **Encoder** ([VoiceInfo](#) i, [ILogger](#) logger)
- void **encodeTyped** (T[] buf)

### Protected Attributes

- OpusEncoder **encoder**
- bool **disposed**

### Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.79.1 Member Function Documentation

### 3.79.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.79.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.79.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

## 3.79.2 Property Documentation

### 3.79.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.79.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.80 RawCodec.Encoder< T > Class Template Reference

Inherits [IEncoderDirect< B >](#).

## Public Member Functions

- `ArraySegment< byte > DequeueOutput` (out `FrameFlags flags`)  
*Returns next encoded data frame (if such output supported).*
- `void EndOfStream` ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`  
*Returns an platform-specific interface.*
- `void Dispose` ()
- `void Input` (`T[] buf`)

## Public Member Functions inherited from `IEncoderDirect< B >`

- `void Input` (`B buf`)  
*Consumes the given raw data.*

## Properties

- `string Error` [get]  
*If not null, the object is in invalid state.*
- `Action< ArraySegment< byte >, FrameFlags > Output` [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.80.1 Member Function Documentation

### 3.80.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.80.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.80.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

Type Constraints

*I* : *class*

## 3.80.2 Property Documentation

### 3.80.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.80.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.81 OpusCodec.EncoderFloat Class Reference

Inherits [OpusCodec.Encoder< T >](#).

### Protected Member Functions

- override void **encodeTyped** (float[] buf)

### Protected Member Functions inherited from [OpusCodec.Encoder< T >](#)

- **Encoder** ([VoicelInfo](#) i, [ILogger](#) logger)
- void **encodeTyped** (T[] buf)

### Additional Inherited Members

#### Public Member Functions inherited from [OpusCodec.Encoder< T >](#)

- void **Input** (T[] buf)
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

#### Public Member Functions inherited from [IEncoderDirect< B >](#)

- void [Input](#) (B buf)  
*Consumes the given raw data.*

#### Protected Attributes inherited from [OpusCodec.Encoder< T >](#)

- OpusEncoder **encoder**
- bool **disposed**

#### Properties inherited from [OpusCodec.Encoder< T >](#)

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.82 OpusCodec.EncoderShort Class Reference

Inherits [OpusCodec.Encoder< T >](#).

### Protected Member Functions

- override void **encodeTyped** (short[] buf)

#### Protected Member Functions inherited from [OpusCodec.Encoder< T >](#)

- **Encoder** ([VoicelInfo](#) i, [ILogger](#) logger)
- void **encodeTyped** (T[] buf)



**Additional Inherited Members****Public Member Functions inherited from [OpusCodec.Encoder< T >](#)**

- void **Input** (T[] buf)
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

**Public Member Functions inherited from [IEncoderDirect< B >](#)**

- void [Input](#) (B buf)  
*Consumes the given raw data.*

**Protected Attributes inherited from [OpusCodec.Encoder< T >](#)**

- OpusEncoder **encoder**
- bool **disposed**

**Properties inherited from [OpusCodec.Encoder< T >](#)**

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.83 OpusCodec.Factory Class Reference

**Static Public Member Functions**

- static [IEncoder](#) **CreateEncoder**< B > ([VoiceInfo](#) i, [ILogger](#) logger)

## 3.84 Flip Struct Reference

**Public Member Functions**

- override bool **Equals** (object obj)
- override int **GetHashCode** ()

### Static Public Member Functions

- static bool **operator==** ([Flip](#) f1, [Flip](#) f2)
- static bool **operator!=** ([Flip](#) f1, [Flip](#) f2)
- static [Flip](#) **operator\*** ([Flip](#) f1, [Flip](#) f2)

### Static Public Attributes

- static [Flip](#) **None**
- static [Flip](#) **Vertical** = new [Flip](#)() { IsVertical = true }
- static [Flip](#) **Horizontal** = new [Flip](#)() { IsHorizontal = true }
- static [Flip](#) **Both** = Vertical \* Horizontal

### Properties

- bool **IsVertical** [get]
- bool **IsHorizontal** [get]

## 3.85 FMODRecorderSetup Class Reference

Inherits [VoiceComponent](#).

### Protected Member Functions

- override void [Awake](#) ()

### Additional Inherited Members

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### 3.85.1 Member Function Documentation

#### 3.85.1.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

## 3.86 FrameBuffer Struct Reference

### Public Member Functions

- **FrameBuffer** (byte[] array, int offset, int count, FrameFlags flags, byte frameNum, IDisposable disposer)
- **FrameBuffer** (byte[] array, FrameFlags flags, byte frameNum)
- **FrameBuffer** ([FrameBuffer](#) from, int offset, int count, FrameFlags flags, byte frameNum)
- void **Retain** ()
- void **Release** ()
- override string **ToString** ()

### Public Attributes

- readonly byte[] **array**
- readonly int **offset**
- readonly int **count**
- readonly IDisposable **disposer**
- bool **disposed**
- int **refCnt**
- GCHandle **gcHandle**
- IntPtr **ptr**
- bool **pinned**

### Properties

- IntPtr **Ptr** [get]
- byte[] **Array** [get]
- int **Length** [get]
- int **Offset** [get]
- FrameFlags **Flags** [get]
- byte **FrameNum** [get]
- bool **IsFEC** [get]
- bool **IsConfig** [get]
- bool **IsKeyframe** [get]

## 3.87 FrameOut< T > Class Template Reference

### Public Member Functions

- **FrameOut** (T[] buf, bool endOfStream)
- **FrameOut**< T > **Set** (T[] buf, bool endOfStream)

### Properties

- T[] **Buf** [get]
- bool **EndOfStream** [get]

## 3.88 Framer< T > Class Template Reference

Utility class to re-frame packets.

Inherited by [FramerResampler< T >](#).

### Public Member Functions

- **Framer** (int frameSize)  
*Create new Framer instance.*
- virtual IEnumerable< T[] > [Frame](#) (T[] buf)  
*Append arbitrary-sized buffer and return available full frames.*

### Protected Attributes

- T[] **frame**
- int **sizeofT**
- int **framePos** = 0

### 3.88.1 Detailed Description

Utility class to re-frame packets.

### 3.88.2 Member Function Documentation

#### 3.88.2.1 Frame()

```
virtual IEnumerable< T[] > Frame (
    T[] buf) [virtual]
```

Append arbitrary-sized buffer and return available full frames.

#### Parameters

|                  |                          |
|------------------|--------------------------|
| <code>buf</code> | Array of samples to add. |
|------------------|--------------------------|

#### Returns

Enumerator of full frames (might be none).

Reimplemented in [FramerResampler< T >](#).

## 3.89 FramerResampler< T > Class Template Reference

Inherits [Framer< T >](#).

## Public Member Functions

- **FramerResampler** (int frameSize, int channels, int resampleNum, int resampleDen, bool interpolate)
- override IEnumerable< T[] > [Frame](#) (T[] bufT)

*Append arbitrary-sized buffer and return available full frames.*

## Public Member Functions inherited from [Framer< T >](#)

- **Framer** (int frameSize)

*Create new Framer instance.*

## Protected Attributes

- bool **TisFloat**
- bool **interpolate**
- int **channels**
- int **resampleNum**
- int **resampleDen**
- float **resampleRatioInv**
- int **delta**

## Protected Attributes inherited from [Framer< T >](#)

- T[] **frame**
- int **sizeofT**
- int **framePos** = 0

## 3.89.1 Member Function Documentation

### 3.89.1.1 Frame()

```
override IEnumerable< T[] > Frame (
    T[] buf) [virtual]
```

Append arbitrary-sized buffer and return available full frames.

#### Parameters

|            |                          |
|------------|--------------------------|
| <i>buf</i> | Array of samples to add. |
|------------|--------------------------|

#### Returns

Enumerator of full frames (might be none).

Reimplemented from [Framer< T >](#).

## 3.90 FusionVoiceClient Class Reference

Inherits [VoiceFollowClient](#), and [INetworkRunnerCallbacks](#).

### Public Attributes

- bool **UseFusionAppSettings** = true  
Whether or not to use the [Voice](#) AppId and all the other AppSettings from [Fusion's RealtimeAppSettings Scriptable](#)↔ Object singleton in the [Voice](#) client/app.
- bool **UseFusionAuthValues** = true  
Whether or not to use the same AuthenticationValues used in [Fusion](#) client/app in [Voice](#) client/app as well. This means that the same UserID will be used in both clients. If custom authentication is used and setup in [Fusion](#) AppId from dashboard, the same configuration should be done for the [Voice](#) AppId.

### Public Attributes inherited from [VoiceFollowClient](#)

- bool **AutoConnectAndJoin** = true  
Auto connect voice client and join a voice room when Leader client is joined to a Leader room.

### Public Attributes inherited from [VoiceConnection](#)

- AppSettings **Settings**  
Settings to be used by this [Voice](#) Client.

### Protected Member Functions

- override void [Start](#) ()
- override void **Awake** ()
- override void [OnDestroy](#) ()
- override [Speaker InstantiateSpeakerForRemoteVoice](#) (int playerId, byte voiceId, object userData)
- override string **GetVoiceRoomName** ()
- override bool **ConnectVoice** ()

### Protected Member Functions inherited from [VoiceFollowClient](#)

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- override void [OnDestroy](#) ()
- override void [OnOperationResponseReceived](#) (OperationResponse operationResponse)
- void **LeaderStateChanged** ([ClientState](#) toState)
- override void [OnVoiceStateChanged](#) ([ClientState](#) fromState, [ClientState](#) toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

### Protected Member Functions inherited from [VoiceConnection](#)

- override void **Awake** ()
- virtual void **Update** ()
- void **CalcStatistics** ()

## Properties

- override bool **LeaderInRoom** [get]
- override bool **LeaderOfflineMode** [get]

## Properties inherited from [VoiceFollowClient](#)

- bool **LeaderInRoom** [get]
- bool **LeaderOfflineMode** [get]

## Properties inherited from [VoiceConnection](#)

- virtual bool **AlwaysUsePrimaryRecorder** [get]
- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- new [LoadBalancingTransport](#) **Client** [get]
- [VoiceClient](#) **VoiceClient** [get]  
*Returns underlying [Photon Voice](#) client.*
- ClientState **ClientState** [get]  
*Returns [Photon Voice](#) client state.*
- float **FramesReceivedPerSecond** [get]  
*Number of frames received per second.*
- float **FramesLostPerSecond** [get]  
*Number of frames lost per second.*
- float **FramesLostPercent** [get]  
*Percentage of lost frames.*
- GameObject **SpeakerPrefab** [get, set]  
*Prefab that contains [Speaker](#) component to be instantiated when receiving a new remote audio source info.*
- [Recorder](#) **PrimaryRecorder** [get, set]  
*Primary [Recorder](#) to be used by [VoiceConnection](#) implementations directly or via integration objects.*
- bool **UsePrimaryRecorder** [get]  
*Use [VoiceConnection.PrimaryRecorder](#) directly.*
- string **BestRegionSummaryInPreferences** [get, set]  
*Used to store and access the "Best Region Summary" in the Player Preferences.*

## Additional Inherited Members

## Public Member Functions inherited from [VoiceFollowClient](#)

- bool [ConnectAndJoinRoom](#) ()  
*Connect voice client to [Photon](#) servers and join a [Voice](#) room.*
- void **Disconnect** ()  
*Disconnect voice client from all [Photon](#) servers.*

## Public Member Functions inherited from [VoiceConnection](#)

- virtual bool [ConnectUsingSettings](#) (AppSettings overwriteSettings=null)  
*Connect to [Photon](#) server using [Settings](#)*
- bool [AddSpeaker](#) ([Speaker](#) speaker, object userData)  
*Tries to link local [Speaker](#) with remote voice stream using [UserData](#). Useful if [Speaker](#) created after stream is started.*
- [Speaker](#) [InstantiateSpeakerPrefab](#) (GameObject parent, bool destroyOnRemove)  
*Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.*
- bool [AddRecorder](#) ([Recorder](#) rec)
- void [RemoveRecorder](#) ([Recorder](#) rec)

## Static Public Attributes inherited from [VoiceConnection](#)

- const int [ChannelAudio](#) = 1  
*Recommended [Photon](#) Transport channel for audio. Chosen not to interfere with video and default channel.*
- const int [ChannelVideo](#) = 2  
*Recommended [Photon](#) Transport channel for video. Chosen not to interfere with audio and default channel.*

## Events inherited from [VoiceConnection](#)

- Action< [Speaker](#) > [SpeakerLinked](#)  
*Fires when a speaker has been linked to a remote audio stream.*
- Action< [RemoteVoiceLink](#) > [RemoteVoiceAdded](#)  
*Fires when a remote voice stream is added.*

### 3.90.1 Member Function Documentation

#### 3.90.1.1 [InstantiateSpeakerForRemoteVoice\(\)](#)

```
override Speaker InstantiateSpeakerForRemoteVoice (
    int playerId,
    byte voiceId,
    object userData) [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.90.1.2 [OnDestroy\(\)](#)

```
override void OnDestroy () [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.90.1.3 [Start\(\)](#)

```
override void Start () [protected], [virtual]
```

Reimplemented from [VoiceFollowClient](#).



## 3.91 AudioUtil.GeneratorPusher< T > Class Template Reference

IAudioPusher that provides a constant tone signal.

Inherits [IAudioPusher< T >](#).

Inherited by [AudioUtil.ToneAudioPusher< T >](#), and [AudioUtil.WaveformAudioPusher< T >](#).

### Public Member Functions

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*
- void **Dispose** ()

### Protected Member Functions

- int **Gen** (T[] buf, long timeSamples)

### Protected Attributes

- long **timeSamples**

### Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.91.1 Detailed Description

IAudioPusher that provides a constant tone signal.

### 3.91.2 Member Function Documentation

#### 3.91.2.1 SetCallback()

```
void SetCallback (
    Action< T[]> callback,
    ObjectFactory< T[], int > bufferFactory,
    int optimalFrameSize)
```

Set the callback function used for pushing data.

**Parameters**

|  |                      |                                                                         |
|--|----------------------|-------------------------------------------------------------------------|
|  | <i>callback</i>      | Callback function to use                                                |
|  | <i>bufferFactory</i> | Buffer factory used to create the buffer that is pushed to the callback |

Implements [IAudioPusher< T >](#).

### 3.91.3 Property Documentation

#### 3.91.3.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.91.3.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

#### 3.91.3.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.92 AudioUtil.GeneratorReader< T > Class Template Reference

Inherits [IAudioReader< T >](#).

Inherited by [AudioUtil.ToneAudioReader< T >](#), and [AudioUtil.WaveformAudioReader< T >](#).

### Public Member Functions

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool **Read** (T[] buf)

*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

## Protected Member Functions

- int **Gen** (T[] buf, long timeSamples)

## Properties

- int **Channels** [get]  
*Number of channels in the audio signal.*
- int **SamplingRate** [get]  
*Sampling rate of the audio signal (in Hz).*
- string **Error** [get]  
*If not null, audio object is in invalid state.*

## 3.92.1 Member Function Documentation

### 3.92.1.1 Read()

```
bool Read (  
    T[] buffer)
```

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

#### Parameters

|                     |                 |
|---------------------|-----------------|
| <code>buffer</code> | Buffer to fill. |
|---------------------|-----------------|

#### Returns

True if buffer was filled successfully, false otherwise.

Implements [IDataReader< T >](#).

## 3.92.2 Property Documentation

### 3.92.2.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.92.2.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.92.2.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.93 IAudioDesc Interface Reference

Audio Source interface.

Inherits [IDisposable](#).

Inherited by [AudioDesc](#), [IAudioPusher< T >](#), and [IAudioReader< T >](#).

### Properties

- [int SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- [int Channels](#) [get]  
*Number of channels in the audio signal.*
- [string Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.93.1 Detailed Description

Audio Source interface.

### 3.93.2 Property Documentation

#### 3.93.2.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implemented in [AudioDesc](#), [AudioUtil.GeneratorPusher< T >](#), [AudioUtil.GeneratorReader< T >](#), [AudioInReader< T >](#), [AudioInPusher](#), [AudioInReader](#), [AudioInPusher](#), [AudioInReader](#), [AndroidAudioInAEC](#), [AudioClipWrapper](#), [MicWrapper](#), [MicWrapperPusher](#), [WebAudioMicIn](#), [AudioInPusher](#), and [WindowsAudioInPusher](#).

#### 3.93.2.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implemented in [AudioDesc](#), [AudioUtil.GeneratorPusher< T >](#), [AudioUtil.GeneratorReader< T >](#), [AudioInReader< T >](#), [AudioInPusher](#), [AudioInReader](#), [AudioInPusher](#), [AudioInReader](#), [AndroidAudioInAEC](#), [AudioClipWrapper](#), [MicWrapper](#), [MicWrapperPusher](#), [WebAudioMicIn](#), [AudioInPusher](#), and [WindowsAudioInPusher](#).

### 3.93.2.3 `SamplingRate`

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implemented in [AudioDesc](#), [AudioUtil.GeneratorPusher< T >](#), [AudioUtil.GeneratorReader< T >](#), [AudioInReader< T >](#), [AudioInPusher](#), [AudioInReader](#), [AudioInPusher](#), [AudioInReader](#), [AndroidAudioInAEC](#), [AudioClipWrapper](#), [MicWrapper](#), [MicWrapperPusher](#), [WebAudioMicIn](#), [AudioInPusher](#), and [WindowsAudioInPusher](#).

## 3.94 `IAudioInChangeNotifier` Interface Reference

Inherits `IDisposable`.

Inherited by [AudioInChangeNotifierNotSupported](#), [AudioInChangeNotifier](#), and [AudioInChangeNotifier](#).

### Properties

- bool **IsSupported** [get]
- string **Error** [get]

### 3.94.1 Property Documentation

#### 3.94.1.1 **Error**

```
string Error [get]
```

Implemented in [AudioInChangeNotifier](#), and [AudioInChangeNotifier](#).

## 3.95 `IAudioOut< T >` Interface Template Reference

Inherited by [AudioOutDelayControl< T >](#), and [AudioOutDummy< T >](#).

### Public Member Functions

- void **Start** (int frequency, int channels, int frameSamplesPerChannel)
- void **Flush** ()
- void **Stop** ()
- void **Push** (T[] frame)
- void **Service** ()

### Properties

- bool **IsPlaying** [get]
- int **Lag** [get]

## 3.96 IAudioPusher< T > Interface Template Reference

Audio Pusher interface.

Inherits [IAudioDesc](#).

Inherited by [AudioUtil.GeneratorPusher< T >](#), [AudioInPusher](#), [AudioInPusher](#), [AudioInPusher](#), [AndroidAudioInAEC](#), [MicWrapperPusher](#), [WebAudioMicIn](#), and [WindowsAudioInPusher](#).

### Public Member Functions

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

### Additional Inherited Members

### Properties inherited from [IAudioDesc](#)

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.96.1 Detailed Description

Audio Pusher interface.

Opposed to an [IAudioReader](#) (which will deliver audio data when it is "pulled"), an [IAudioPusher](#) will push its audio data whenever it is ready,

### 3.96.2 Member Function Documentation

#### 3.96.2.1 SetCallback()

```
void SetCallback (
    Action< T[]> callback,
    ObjectFactory< T[], int > bufferFactory,
    int optimalFrameSize)
```

Set the callback function used for pushing data.

#### Parameters

|                               |                                                                         |
|-------------------------------|-------------------------------------------------------------------------|
| <a href="#">callback</a>      | Callback function to use.                                               |
| <a href="#">bufferFactory</a> | Buffer factory used to create the buffer that is pushed to the callback |

Implemented in [AudioUtil.GeneratorPusher< T >](#).

## 3.97 IAudioReader< T > Interface Template Reference

Audio Reader interface.

Inherits [IDataReader< T >](#), and [IAudioDesc](#).

Inherited by [AudioUtil.GeneratorReader< T >](#), [AudioInReader< T >](#), [AudioInReader](#), [AudioInReader](#), [AudioClipWrapper](#), and [MicWrapper](#).

### Additional Inherited Members

### Public Member Functions inherited from [IDataReader< T >](#)

- bool [Read](#) (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Properties inherited from [IAudioDesc](#)

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.97.1 Detailed Description

Audio Reader interface.

Opposed to an [IAudioPusher](#) (which will push its audio data whenever it is ready), an [IAudioReader](#) will deliver audio data when it is "pulled" (it's [Read](#) function is called).

## 3.98 IDataReader< T > Interface Template Reference

Interface for pulling data, in case this is more appropriate than pushing it.

Inherits [IDisposable](#).

Inherited by [IAudioReader< T >](#).

### Public Member Functions

- bool [Read](#) (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### 3.98.1 Detailed Description

Interface for pulling data, in case this is more appropriate than pushing it.

### 3.98.2 Member Function Documentation

#### 3.98.2.1 Read()

```
bool Read (
    T[] buffer)
```

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

**Parameters**

|                     |                 |
|---------------------|-----------------|
| <code>buffer</code> | Buffer to fill. |
|---------------------|-----------------|

**Returns**

True if buffer was filled successfully, false otherwise.

Implemented in [AudioUtil.GeneratorReader< T >](#), and [AudioInReader< T >](#).

## 3.99 IDecoder Interface Reference

Generic decoder interface.

Inherits [IDisposable](#).

Inherited by [ByteStreamDecoder](#), [IDecoderDirect< B >](#), [VideoDecoder](#), [OpusCodec.Decoder< T >](#), [RawCodec.Decoder< T >](#), [VideoDecoder](#), [AndroidVideoDecoder](#), and [WebCodecsVideoDecoderUnityTexture](#).

**Public Member Functions**

- void [Open](#) ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void [Input](#) (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*

**Properties**

- string [Error](#) [get]  
*If not null, the object is in invalid state.*

### 3.99.1 Detailed Description

Generic decoder interface.

### 3.99.2 Member Function Documentation

#### 3.99.2.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call `buf.Retain()` to prevent the caller from disposing the buffer. In this case, the callee should call `buf.Release()` when buffer is no longer needed.

Implemented in [ByteStreamDecoder](#), [VideoDecoder](#), [VideoDecoder](#), [OpusCodec.Decoder< T >](#), [RawCodec.Decoder< T >](#), [AndroidVideoDecoder](#), [WebCodecsVideoDecoderUnityTexture](#), [VideoDecoder](#), [VPxCodec.Decoder](#), and [MFTCodec.VideoDecoder](#).

#### 3.99.2.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.



## Parameters

|                   |                                          |
|-------------------|------------------------------------------|
| <code>info</code> | Properties of the data stream to decode. |
|-------------------|------------------------------------------|

Implemented in [ByteStreamDecoder](#), [VideoDecoder](#), [VideoDecoder](#), [OpusCodec.Decoder< T >](#), [RawCodec.Decoder< T >](#), [AndroidVideoDecoder](#), [WebCodecsVideoDecoderUnityTexture](#), [VideoDecoder](#), [VideoDecoderMediaPlayerElement](#), [VPxCodec.Decoder](#), and [MFTCodec.VideoDecoder](#).

### 3.99.3 Property Documentation

#### 3.99.3.1 Error

`string Error [get]`

If not null, the object is in invalid state.

Implemented in [ByteStreamDecoder](#), [VideoDecoder](#), [VideoDecoder](#), [OpusCodec.Decoder< T >](#), [RawCodec.Decoder< T >](#), [AndroidVideoDecoder](#), [WebCodecsVideoDecoderUnityTexture](#), [VideoDecoder](#), [VPxCodec.Decoder](#), and [MFTCodec.VideoDecoder](#).

## 3.100 IDecoderDirect< B > Interface Template Reference

Interface for an decoder which outputs data via explicit call.

Inherits [IDecoder](#).

Inherited by [VideoDecoder](#), [VPxCodec.Decoder](#), and [MFTCodec.VideoDecoder](#).

### Properties

- `Action< B > Output [set]`  
*Callback to call when a new decoded data buffer is available.*

### Properties inherited from [IDecoder](#)

- `string Error [get]`  
*If not null, the object is in invalid state.*

### Additional Inherited Members

### Public Member Functions inherited from [IDecoder](#)

- `void Open (VoicInfo info)`  
*Open (initialize) the decoder.*
- `void Input (ref FrameBuffer buf)`  
*Consumes the given encoded data.*

### 3.100.1 Detailed Description

Interface for an decoder which outputs data via explicit call.

### 3.100.2 Property Documentation

#### 3.100.2.1 Output

```
Action<B> Output [set]
```

Callback to call when a new decoded data buffer is available.

Implemented in [VideoDecoder](#), [VPxCodec.Decoder](#), and [MFTCodec.VideoDecoder](#).

## 3.101 IDeviceEnumerator Interface Reference

Inherits IDisposable, and IEnumerable.

Inherited by [DeviceEnumeratorBase](#).

### Public Member Functions

- void [Refresh](#) ()

### Properties

- bool **IsSupported** [get]
- Action **OnReady** [set]
- string **Error** [get]

### 3.101.1 Member Function Documentation

#### 3.101.1.1 Refresh()

```
void Refresh ()
```

Implemented in [VideoInEnumerator](#), [AudioInEnumerator](#), [AndroidVideoInEnumerator](#), and [AudioInEnumerator](#).

## 3.102 IEncoder Interface Reference

Generic encoder interface.

Inherits IDisposable.

Inherited by [IEncoderDirect< B >](#), [VideoEncoder](#), [VideoEncoder](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), and [WebCodecsVideoEncoder](#).

## Public Member Functions

- `ArraySegment< byte > DequeueOutput (out FrameFlags flags)`  
*Returns next encoded data frame (if such output supported).*
- `void EndOfStream ()`  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`  
*Returns an platform-specific interface.*

## Properties

- `string Error [get]`  
*If not null, the object is in invalid state.*
- `Action< ArraySegment< byte >, FrameFlags > Output [set]`  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### 3.102.1 Detailed Description

Generic encoder interface.

Depending on implementation, encoder should either call `Output` on each data frame or return next data frame in `DequeueOutput()` call.

### 3.102.2 Member Function Documentation

#### 3.102.2.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implemented in [ByteStreamEncoder](#), [VideoEncoder](#), [VideoEncoder](#), [OpusCodec.Encoder< T >](#), [RawCodec.Encoder< T >](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), [WebCodecsVideoEncoder](#), [VideoEncoder](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

#### 3.102.2.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with `EndOfStream` flag (in output queue).

Implemented in [ByteStreamEncoder](#), [VideoEncoder](#), [VideoEncoder](#), [OpusCodec.Encoder< T >](#), [RawCodec.Encoder< T >](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), [WebCodecsVideoEncoder](#), [VideoEncoder](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

### 3.102.2.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implemented in [ByteStreamEncoder](#), [VideoEncoder](#), [VideoEncoder](#), [OpusCodec.Encoder< T >](#), [RawCodec.Encoder< T >](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), [WebCodecsVideoEncoder](#), [VideoEncoder](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

#### Type Constraints

**I : class**

## 3.102.3 Property Documentation

### 3.102.3.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implemented in [ByteStreamEncoder](#), [VideoEncoder](#), [VideoEncoder](#), [OpusCodec.Encoder< T >](#), [RawCodec.Encoder< T >](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), [WebCodecsVideoEncoder](#), [VideoEncoder](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

### 3.102.3.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implemented in [ByteStreamEncoder](#), [VideoEncoder](#), [VideoEncoder](#), [OpusCodec.Encoder< T >](#), [RawCodec.Encoder< T >](#), [AndroidNDKVideoEncoder](#), [AndroidVideoEncoder](#), [WebCodecsVideoEncoder](#), [VideoEncoder](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

## 3.103 IEncoderDirect< B > Interface Template Reference

Interface for an encoder which consumes input data via explicit call.

Inherits [IEncoder](#).

Inherited by [ByteStreamEncoder](#), [IEncoderDirectImage](#), [OpusCodec.Encoder< T >](#), and [RawCodec.Encoder< T >](#).

#### Public Member Functions

- void [Input](#) (B buf)  
*Consumes the given raw data.*

## Public Member Functions inherited from IEncoder

- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*

## Additional Inherited Members

## Properties inherited from IEncoder

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

### 3.103.1 Detailed Description

Interface for an encoder which consumes input data via explicit call.

### 3.103.2 Member Function Documentation

#### 3.103.2.1 Input()

```
void Input (
    B buf)
```

Consumes the given raw data.

#### Parameters

|                     |                                                 |
|---------------------|-------------------------------------------------|
| <a href="#">buf</a> | Array containing raw data (e.g. audio samples). |
|---------------------|-------------------------------------------------|

## 3.104 IEncoderDirectImage Interface Reference

Interface for an encoder which consumes images via explicit call.

Inherits [IEncoderDirect](#)< B >.

Inherited by [VideoEncoder](#), [WebCodecsVideoEncoderDirectImage](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

## Properties

- ImageFormat [ImageFormat](#) [get]  
*Recommended encoder input image format. Encoder may support other formats.*

## Properties inherited from [IEncoder](#)

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## Additional Inherited Members

## Public Member Functions inherited from [IEncoderDirect< B >](#)

- void [Input](#) (B buf)  
*Consumes the given raw data.*

## Public Member Functions inherited from [IEncoder](#)

- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI< I >](#) ()  
*Returns an platform-specific interface.*

### 3.104.1 Detailed Description

Interface for an encoder which consumes images via explicit call.

### 3.104.2 Property Documentation

#### 3.104.2.1 ImageFormat

`ImageFormat ImageFormat [get]`

Recommended encoder input image format. Encoder may support other formats.

Implemented in [VideoEncoder](#), [WebCodecsVideoEncoderDirectImage](#), [VPxCodec.Encoder](#), and [MFTCodec.VideoEncoder](#).

## 3.105 AudioUtil.ILevelMeter Interface Reference

Audio Level Metering interface.

Inherited by [AudioUtil.LevelMeter< T >](#), and [AudioUtil.LevelMeterDummy](#).

## Public Member Functions

- void [ResetAccumAvgPeakAmp](#) ()  
*Reset [AccumAvgPeakAmp](#).*

## Properties

- float [CurrentAvgAmp](#) [get]  
*Average amplitude value over last half second.*
- float [CurrentPeakAmp](#) [get]  
*Maximum amplitude value over last half second sec.*
- float [AccumAvgPeakAmp](#) [get]  
*Average of CurrentPeakAmps since last reset.*

## 3.105.1 Detailed Description

Audio Level Metering interface.

## 3.105.2 Member Function Documentation

### 3.105.2.1 ResetAccumAvgPeakAmp()

```
void ResetAccumAvgPeakAmp ()
```

Reset [AccumAvgPeakAmp](#).

Implemented in [AudioUtil.LevelMeter< T >](#), and [AudioUtil.LevelMeterDummy](#).

## 3.105.3 Property Documentation

### 3.105.3.1 AccumAvgPeakAmp

```
float AccumAvgPeakAmp [get]
```

Average of CurrentPeakAmps since last reset.

Implemented in [AudioUtil.LevelMeter< T >](#), and [AudioUtil.LevelMeterDummy](#).

### 3.105.3.2 CurrentAvgAmp

```
float CurrentAvgAmp [get]
```

Average amplitude value over last half second.

Implemented in [AudioUtil.LevelMeter< T >](#), and [AudioUtil.LevelMeterDummy](#).

### 3.105.3.3 CurrentPeakAmp

```
float CurrentPeakAmp [get]
```

Maximum amplitude value over last half second sec.

Implemented in [AudioUtil.LevelMeter< T >](#), and [AudioUtil.LevelMeterDummy](#).

## 3.106 ILocalVoiceAudio Interface Reference

Interface for an outgoing audio stream.

Inherited by [LocalVoiceAudio< T >](#), and [LocalVoiceAudioDummy](#).

### Public Member Functions

- void [VoiceDetectorCalibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Trigger voice detector calibration process.*

### Properties

- AudioUtil.IVoiceDetector [VoiceDetector](#) [get]  
*The VoiceDetector in use.*
- AudioUtil.ILevelMeter [LevelMeter](#) [get]  
*The LevelMeter utility in use.*
- bool [VoiceDetectorCalibrating](#) [get]  
*If true, voice detector calibration is in progress.*

### 3.106.1 Detailed Description

Interface for an outgoing audio stream.

A [LocalVoice](#) always brings a LevelMeter and a VoiceDetector, which you can access using this interface.

### 3.106.2 Member Function Documentation

#### 3.106.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (  
    int durationMs,  
    Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. [Voice](#) detector sets threshold based on measured background noise level.



**Parameters**

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <code>durationMs</code>   | Duration of calibration (in milliseconds).                             |
| <code>onCalibrated</code> | Called when calibration is complete. Parameter is new threshold value. |

Implemented in [LocalVoiceAudio< T >](#), and [LocalVoiceAudioDummy](#).

**3.106.3 Property Documentation****3.106.3.1 LevelMeter**

```
AudioUtil.ILevelMeter LevelMeter [get]
```

The LevelMeter utility in use.

Implemented in [LocalVoiceAudio< T >](#), and [LocalVoiceAudioDummy](#).

**3.106.3.2 VoiceDetector**

```
AudioUtil.IVoiceDetector VoiceDetector [get]
```

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implemented in [LocalVoiceAudio< T >](#), and [LocalVoiceAudioDummy](#).

**3.106.3.3 VoiceDetectorCalibrating**

```
bool VoiceDetectorCalibrating [get]
```

If true, voice detector calibration is in progress.

Implemented in [LocalVoiceAudio< T >](#), and [LocalVoiceAudioDummy](#).

**3.107 ILogger Interface Reference**

Inherited by [Logger](#).

**Public Member Functions**

- void **Log** (LogLevel level, string msg)
- void **Log** (LogLevel level, string fmt, params object[] args)

## Properties

- LogLevel **Level** [get]

## 3.108 ImageBufferInfo Struct Reference

### Classes

- struct [StrideSet](#)

### Public Member Functions

- **ImageBufferInfo** (int width, int height, [StrideSet](#) stride, ImageFormat format)

### Properties

- int **Width** [get]
- int **Height** [get]
- [StrideSet](#) **Stride** [get]
- ImageFormat **Format** [get]
- Rotation **Rotation** [get, set]
- [Flip](#) **Flip** [get, set]

## 3.109 ImageBufferNative Class Reference

Inherited by [ImageBufferNativeAlloc](#), [ImageBufferNativeGCHandleBytes](#), and [ImageBufferNativeGCHandleSinglePlane](#).

### Classes

- struct [PlaneSet](#)

### Public Member Functions

- **ImageBufferNative** ([ImageBufferInfo](#) info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()
- virtual void **Dispose** ()

### Public Attributes

- [ImageBufferInfo](#) **Info**
- [PlaneSet](#) **Planes**

### Protected Member Functions

- virtual void **Reset** ()
- virtual void **Free** ()

## 3.110 ImageBufferNativeAlloc Class Reference

Inherits [ImageBufferNative](#), and `IDisposable`.

### Public Member Functions

- **ImageBufferNativeAlloc** (ImageBufferNativePool< [ImageBufferNativeAlloc](#) > pool, [ImageBufferInfo](#) info)
- override void [Dispose](#) ()

### Public Member Functions inherited from [ImageBufferNative](#)

- **ImageBufferNative** ([ImageBufferInfo](#) info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()

### Protected Member Functions

- override void [Free](#) ()

### Protected Member Functions inherited from [ImageBufferNative](#)

- virtual void **Reset** ()

### Additional Inherited Members

### Public Attributes inherited from [ImageBufferNative](#)

- [ImageBufferInfo](#) Info
- [PlaneSet](#) Planes

## 3.110.1 Member Function Documentation

### 3.110.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [ImageBufferNative](#).

### 3.110.1.2 Free()

```
override void Free () [protected], [virtual]
```

Reimplemented from [ImageBufferNative](#).

## 3.111 ImageBufferNativeGCHandleBytes Class Reference

Inherits [ImageBufferNative](#), and `IDisposable`.

### Public Member Functions

- **ImageBufferNativeGCHandleBytes** (`ImageBufferNativePool`< [ImageBufferNativeGCHandleBytes](#) > pool, [ImageBufferInfo](#) info)
- override void [Dispose](#) ()

### Public Member Functions inherited from [ImageBufferNative](#)

- **ImageBufferNative** ([ImageBufferInfo](#) info)
- **ImageBufferNative** (`IntPtr` buf, `int` width, `int` height, `int` stride, `ImageFormat` imageFormat)
- void **Retain** ()
- void **Retain** (`int` times)
- void **Release** ()

### Protected Member Functions

- override void [Free](#) ()

### Protected Member Functions inherited from [ImageBufferNative](#)

- virtual void **Reset** ()

### Properties

- `byte[][]` **PlaneBytes** [get]

### Additional Inherited Members

### Public Attributes inherited from [ImageBufferNative](#)

- [ImageBufferInfo](#) **Info**
- [PlaneSet](#) **Planes**

### 3.111.1 Member Function Documentation

#### 3.111.1.1 Dispose()

`override void Dispose () [virtual]`

Reimplemented from [ImageBufferNative](#).

#### 3.111.1.2 Free()

`override void Free () [protected], [virtual]`

Reimplemented from [ImageBufferNative](#).

## 3.112 ImageBufferNativeGCHandleSinglePlane Class Reference

Inherits [ImageBufferNative](#), and `IDisposable`.

### Public Member Functions

- **ImageBufferNativeGCHandleSinglePlane** ([ImageBufferNativePool](#)< [ImageBufferNativeGCHandleSinglePlane](#) > pool, [ImageBufferInfo](#) info)
- void **PinPlane** (byte[] plane)
- override void [Dispose](#) ()

### Public Member Functions inherited from [ImageBufferNative](#)

- **ImageBufferNative** ([ImageBufferInfo](#) info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()

### Protected Member Functions

- override void [Free](#) ()

### Protected Member Functions inherited from [ImageBufferNative](#)

- virtual void **Reset** ()

### Additional Inherited Members

### Public Attributes inherited from [ImageBufferNative](#)

- [ImageBufferInfo](#) Info
- [PlaneSet](#) Planes

### 3.112.1 Member Function Documentation

#### 3.112.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [ImageBufferNative](#).

#### 3.112.1.2 Free()

```
override void Free () [protected], [virtual]
```

Reimplemented from [ImageBufferNative](#).

## 3.113 ImageBufferNativePool< T > Class Template Reference

Inherits [ObjectPool< TType, TInfo >](#).

### Public Member Functions

- delegate T **Factory** (ImageBufferNativePool< T > pool, [ImageBufferInfo](#) info)
- **ImageBufferNativePool** (int capacity, Factory factory, string name, [ImageBufferInfo](#) info)

### Public Member Functions inherited from [ObjectPool< TType, TInfo >](#)

- [ObjectPool](#) (int capacity, string name, TInfo info)  
*Create a new ObjectPool instance with the given info structure.*
- TType [New](#) ()  
*Acquire an existing object, or create a new one if none are available.*
- TType [New](#) (TInfo info)  
*Acquire an existing object (if info matches), or create a new one from the passed info.*
- virtual bool [Free](#) (TType obj, TInfo objInfo)  
*Returns object to pool.*
- virtual bool [Free](#) (TType obj)  
*Returns object to pool, or destroys it if the pool is full.*
- void **Dispose** ()  
*Free resources associated with this ObjectPool.*

### Protected Member Functions

- override T **createObject** ([ImageBufferInfo](#) info)
- override void **destroyObject** (T obj)
- override bool **infosMatch** ([ImageBufferInfo](#) i0, [ImageBufferInfo](#) i1)

**Protected Member Functions inherited from [ObjectPool< TType, TInfo >](#)**

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

**Additional Inherited Members****Protected Attributes inherited from [ObjectPool< TType, TInfo >](#)**

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

**Properties inherited from [ObjectPool< TType, TInfo >](#)**

- TInfo **Info** [get]  
*The property (info) that objects in this Pool must match.*

## 3.114 IMediaCapture Interface Reference

**Properties**

- MediaCapture **MediaCapture** [get]

**Events**

- MediaCaptureInitCompleted **MediaCaptureInitCompleted**

## 3.115 IMixedRealityCapture Interface Reference

**Properties**

- bool **HologramCompositionEnabled** [get, set]

## 3.116 IPreviewManager Interface Reference

Inherited by [PreviewManager](#), and [PreviewManagerScreenQuadTexture](#).

## Public Member Functions

- void **AddView** (object id, [IVideoPreview](#) view)
- void **RemoveView** (object id)
- bool **Has** (object id)
- void **SetBounds** (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), [Rotation](#) rot=[Rotation.Rotate0](#))

*Positions the preview of the view / image. Not optimized to be called every frame.*

## 3.116.1 Member Function Documentation

### 3.116.1.1 SetBounds()

```
void SetBounds (
    object id,
    int x,
    int y,
    int w,
    int h,
    Flip flip = default(Flip),
    Rotation rot = Rotation.Rotate0)
```

Positions the preview of the view / image. Not optimized to be called every frame.

Implemented in [PreviewManager](#), and [PreviewManagerScreenQuadTexture](#).

## 3.117 IProcessor< T > Interface Template Reference

Processor interface.

Inherits [IDisposable](#).

Inherited by [AudioUtil.LevelMeter< T >](#), [AudioUtil.Resampler< T >](#), [AudioUtil.VoiceDetector< T >](#), [AudioUtil.VoiceDetectorCalibration< T >](#), [AudioUtil.VoiceLevelDetectCalibrate< T >](#), [MicAmplifierFloat](#), [MicAmplifierShort](#), and [WebRTCAudioProcessor](#).

## Public Member Functions

- [T\[\]](#) **Process** ([T\[\]](#) buf)  
*Process a frame of data.*

### 3.117.1 Detailed Description

Processor interface.

## 3.117.2 Member Function Documentation

### 3.117.2.1 Process()

```
T\[\] Process (
    T\[\] buf)
```

Process a frame of data.



## Parameters

|                  |                              |
|------------------|------------------------------|
| <code>buf</code> | Buffer containing input data |
|------------------|------------------------------|

## Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implemented in [AudioUtil.LevelMeter< T >](#), [AudioUtil.Resampler< T >](#), [AudioUtil.VoiceDetector< T >](#), [AudioUtil.VoiceDetectorCalibration< T >](#), and [AudioUtil.VoiceLevelDetectCalibrate< T >](#).

## 3.118 IResettable Interface Reference

Inherited by [AudioInPusher](#), [AudioInReader](#), and [AndroidAudioInAEC](#).

## Public Member Functions

- void **Reset** ()

## 3.119 IServiceable Interface Reference

Interface for classes that want their [Service\(\)](#) function to be called regularly in the context of a [LocalVoice](#).

Inherited by [BufferReaderPushAdapterBase< T >](#).

## Public Member Functions

- void [Service](#) ([LocalVoice](#) localVoice)  
*Service function that should be called regularly.*

### 3.119.1 Detailed Description

Interface for classes that want their [Service\(\)](#) function to be called regularly in the context of a [LocalVoice](#).

### 3.119.2 Member Function Documentation

#### 3.119.2.1 Service()

```
void Service (
    LocalVoice localVoice)
```

Service function that should be called regularly.

Implemented in [BufferReaderPushAdapterAsyncPool< T >](#), [BufferReaderPushAdapterAsyncPoolFloatToShort](#), [BufferReaderPushAdapterAsyncPoolShortToFloat](#), and [BufferReaderPushAdapterBase< T >](#).

## 3.120 IStreamReceiver Interface Reference

Interface for component relying on a [LoadBalancingTransport](#) to receive a stream of data (audio, video, ...)

Inherited by [AudioStreamReceiver](#).

### Public Member Functions

- void [OnRemoteVoiceAdd](#) (int channelId, int playerId, byte voiceId, [VoiceInfo](#) i, ref [RemoteVoiceOptions](#) options)

*Called when a remote stream (voice) has been detected.*

### 3.120.1 Detailed Description

Interface for component relying on a [LoadBalancingTransport](#) to receive a stream of data (audio, video, ...)

### 3.120.2 Member Function Documentation

#### 3.120.2.1 OnRemoteVoiceAdd()

```
void OnRemoteVoiceAdd (  
    int channelId,  
    int playerId,  
    byte voiceId,  
    VoiceInfo i,  
    ref RemoteVoiceOptions options)
```

Called when a remote stream (voice) has been detected.

Implemented in [AudioStreamReceiver](#).

## 3.121 IStreamSender Interface Reference

Interface for component relying on a [LoadBalancingTransport](#) to send a stream of data (audio, video, ...)

Inherited by [AudioStreamSender](#).

### Public Member Functions

- void [OnRoomJoined](#) ()  
*Called when the [Photon Voice](#) room is joined, and creating streams (voice) is possible.*
- void [OnLeftRoom](#) ()

### 3.121.1 Detailed Description

Interface for component relying on a [LoadBalancingTransport](#) to send a stream of data (audio, video, ...)

### 3.121.2 Member Function Documentation

#### 3.121.2.1 OnRoomJoined()

```
void OnRoomJoined ()
```

Called when the [Photon Voice](#) room is joined, and creating streams (voice) is possible.

Implemented in [AudioStreamSender](#).

## 3.122 IStreamTransportManager Interface Reference

Interface for components capable of providing a [LoadBalancingTransport](#) for an [IStreamSender](#) and/or [IStreamReceiver](#) Those components should forward the stream callbacks to their registered listeners:

Inherited by [StreamTransportClient](#), and [StreamTransportObserver](#).

### Public Member Functions

- LogLevel **StreamLogLevel** ()
- void **RegisterStreamSender** ([IStreamSender](#) streamSender)
- void **RegisterStreamReceiver** ([IStreamReceiver](#) streamReceiver)

### Properties

- StreamTransport **Transport** [get]
- long **RoundTripTime** [get]
- long **BytesOut** [get]
- long **PacketsOut** [get]
- long **BytesIn** [get]
- long **PacketsIn** [get]

### 3.122.1 Detailed Description

Interface for components capable of providing a [LoadBalancingTransport](#) for an [IStreamSender](#) and/or [IStreamReceiver](#) Those components should forward the stream callbacks to their registered listeners:

- OnRoomJoined / OnLeftRoom to [IStreamSender](#)
- OnRemoteVoiceAdd to [IStreamReceiver](#)

## 3.123 IStreamTransportManagerListener Interface Reference

Inherited by [StreamTransportManagerListener](#).

**Properties**

- GameObject **gameObject** [get]
- GameObject **StreamTransportGameObject** [get, set]
- [IStreamTransportManager](#) **StreamTransportManager** [get, set]

**3.124 IVideoPlayer Interface Reference**

Inherits [IVideoPreview](#).

Inherited by [VideoPlayerUnityTexture](#), [VideoPlayerUnityTexture](#), [AndroidVideoPlayerUnityTexture](#), [VideoPlayerUnity](#), [WebCodecsVideoPlayerUnityTexture](#), and [VideoPlayer](#).

**Properties**

- [IDecoder](#) **Decoder** [get]

**Properties inherited from [IVideoPreview](#)**

- object **PlatformView** [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]

**3.125 IVideoPreview Interface Reference**

Inherits [IDisposable](#).

Inherited by [IVideoPlayer](#), and [IVideoRecorder](#).

**Properties**

- object **PlatformView** [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]

**3.126 IVideoRecorder Interface Reference**

Inherits [IVideoPreview](#).

Inherited by [VideoRecorderLayer](#), [VideoRecorderUnityTexture](#), [VideoRecorderUnityTextureInput](#), [IVideoRecorderPusher](#), [VideoRecorderUnityTextureInput](#), [VideoRecorderMediaPlayerElement](#), [VideoRecorderUnityTexture](#), [AndroidNDKVideoRecorderSurfaceView](#), [AndroidNDKVideoRecorderUnityTexture](#), [AndroidVideoRecorderSurfaceView](#), [AndroidVideoRecorderTextureInput](#), [AndroidVideoRecorderUnityTexture](#), and [WebCodecsVideoRecorderUnityTexture](#).

### Properties

- [IEncoder](#) **Encoder** [get]
- string **Error** [get]

### Properties inherited from [IVideoPreview](#)

- object **PlatformView** [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]

## 3.127 IVideoRecorderPusher Interface Reference

Inherits [IVideoRecorder](#).

Inherited by [ScreenRecorder](#), and [VideoRecorderUnity](#).

### Properties

- [IVideoSink](#) **VideoSink** [set]

### Properties inherited from [IVideoRecorder](#)

- [IEncoder](#) **Encoder** [get]
- string **Error** [get]

### Properties inherited from [IVideoPreview](#)

- object **PlatformView** [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]

## 3.128 IVideoSink Interface Reference

Inherited by [LocalVoiceVideo](#), and [VideoSinkArray](#).

### Public Member Functions

- void **PushImageAsync** ([ImageBufferNative](#) buf)
- void **PushImage** ([ImageBufferNative](#) buf)

## Properties

- int **PushImageQueueCount** [get]

## 3.129 AudioUtil.IVoiceDetector Interface Reference

Voice Activity Detector interface.

Inherited by [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

## Properties

- bool **On** [get, set]  
*If true, voice detection enabled.*
- float **Threshold** [get, set]  
*Voice detected as soon as signal level exceeds threshold.*
- bool **Detected** [get]  
*If true, voice detected.*
- DateTime **DetectedTime** [get]  
*Last time when switched to detected state.*
- int **ActivityDelayMs** [get, set]  
*Keep detected state during this time after signal level dropped below threshold.*

## Events

- Action **OnDetected**  
*Called when switched to detected state.*

### 3.129.1 Detailed Description

Voice Activity Detector interface.

### 3.129.2 Property Documentation

#### 3.129.2.1 ActivityDelayMs

```
int ActivityDelayMs [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implemented in [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

### 3.129.2.2 Detected

```
bool Detected [get]
```

If true, voice detected.

Implemented in [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

### 3.129.2.3 DetectedTime

```
DateTime DetectedTime [get]
```

Last time when switched to detected state.

Implemented in [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

### 3.129.2.4 On

```
bool On [get], [set]
```

If true, voice detection enabled.

Implemented in [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

### 3.129.2.5 Threshold

```
float Threshold [get], [set]
```

[Voice](#) detected as soon as signal level exceeds threshold.

Implemented in [AudioUtil.VoiceDetector< T >](#), and [AudioUtil.VoiceDetectorDummy](#).

## 3.130 IVoiceTransport Interface Reference

Inherited by [LoadBalancingTransport](#).

### Public Member Functions

- bool **IsChannelJoined** (int channelId)
- void **SendVoiceInfo** ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void **SendVoiceRemove** ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void **SendFrame** (ArraySegment< byte > data, FrameFlags flags, ushort evNumber, byte frNumber, byte voiceld, int channelId, [SendFrameParams](#) par)
- string **ChannelIdStr** (int channelId)
- string **PlayerIdStr** (int playerId)
- int **GetPayloadFragmentSize** ([SendFrameParams](#) par)

### 3.131 AudioUtil.LevelMeter< T > Class Template Reference

Audio Level Meter.

Inherits [IProcessor< T >](#), and [AudioUtil.ILevelMeter](#).

Inherited by [AudioUtil.LevelMeterFloat](#), and [AudioUtil.LevelMeterShort](#).

#### Public Member Functions

- void [ResetAccumAvgPeakAmp](#) ()  
*Reset [AccumAvgPeakAmp](#).*
- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

#### Protected Attributes

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

#### Properties

- float [CurrentAvgAmp](#) [get]  
*Average amplitude value over last half second.*
- float [CurrentPeakAmp](#) [get, protected set]  
*Maximum amplitude value over last half second sec.*
- float [AccumAvgPeakAmp](#) [get]  
*Average of CurrentPeakAmps since last reset.*

#### 3.131.1 Detailed Description

Audio Level Meter.

#### 3.131.2 Member Function Documentation

##### 3.131.2.1 Process()

```
T[] Process (
    T[] buf) [abstract]
```

Process a frame of data.



#### Parameters

|                  |                              |
|------------------|------------------------------|
| <code>buf</code> | Buffer containing input data |
|------------------|------------------------------|

#### Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements [IProcessor< T >](#).

#### 3.131.2.2 ResetAccumAvgPeakAmp()

```
void ResetAccumAvgPeakAmp ()
```

Reset [AccumAvgPeakAmp](#).

Implements [AudioUtil.ILevelMeter](#).

### 3.131.3 Property Documentation

#### 3.131.3.1 AccumAvgPeakAmp

```
float AccumAvgPeakAmp [get]
```

Average of CurrentPeakAmps since last reset.

Implements [AudioUtil.ILevelMeter](#).

#### 3.131.3.2 CurrentAvgAmp

```
float CurrentAvgAmp [get]
```

Average amplitude value over last half second.

Implements [AudioUtil.ILevelMeter](#).

#### 3.131.3.3 CurrentPeakAmp

```
float CurrentPeakAmp [get], [protected set]
```

Maximum amplitude value over last half second sec.

Implements [AudioUtil.ILevelMeter](#).

## 3.132 AudioUtil.LevelMeterDummy Class Reference

Dummy Audio Level Meter that doesn't actually do anything.

Inherits [AudioUtil.ILevelMeter](#).

## Public Member Functions

- void [ResetAccumAvgPeakAmp](#) ()  
*Reset [AccumAvgPeakAmp](#).*

## Properties

- float [CurrentAvgAmp](#) [get]  
*Average amplitude value over last half second.*
- float [CurrentPeakAmp](#) [get]  
*Maximum amplitude value over last half second sec.*
- float [AccumAvgPeakAmp](#) [get]  
*Average of [CurrentPeakAmps](#) since last reset.*

### 3.132.1 Detailed Description

Dummy Audio Level Meter that doesn't actually do anything.

### 3.132.2 Member Function Documentation

#### 3.132.2.1 [ResetAccumAvgPeakAmp\(\)](#)

```
void ResetAccumAvgPeakAmp ()
```

Reset [AccumAvgPeakAmp](#).

Implements [AudioUtil.ILevelMeter](#).

### 3.132.3 Property Documentation

#### 3.132.3.1 [AccumAvgPeakAmp](#)

```
float AccumAvgPeakAmp [get]
```

Average of [CurrentPeakAmps](#) since last reset.

Implements [AudioUtil.ILevelMeter](#).

#### 3.132.3.2 [CurrentAvgAmp](#)

```
float CurrentAvgAmp [get]
```

Average amplitude value over last half second.

Implements [AudioUtil.ILevelMeter](#).

### 3.132.3.3 CurrentPeakAmp

float CurrentPeakAmp [get]

Maximum amplitude value over last half second sec.

Implements [AudioUtil.ILevelMeter](#).

## 3.133 AudioUtil.LevelMeterFloat Class Reference

LevelMeter specialization for float audio.

Inherits [AudioUtil.LevelMeter< T >](#).

### Public Member Functions

- [LevelMeterFloat](#) (int samplingRate, int numChannels)  
*Create new [LevelMeterFloat](#) instance.*
- override float[] **Process** (float[] buf)

### Public Member Functions inherited from [AudioUtil.LevelMeter< T >](#)

- void [ResetAccumAvgPeakAmp](#) ()  
*Reset [AccumAvgPeakAmp](#).*
- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [AudioUtil.LevelMeter< T >](#)

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

### Properties inherited from [AudioUtil.LevelMeter< T >](#)

- float [CurrentAvgAmp](#) [get]  
*Average amplitude value over last half second.*
- float [CurrentPeakAmp](#) [get, protected set]  
*Maximum amplitude value over last half second sec.*
- float [AccumAvgPeakAmp](#) [get]  
*Average of CurrentPeakAmps since last reset.*

### 3.133.1 Detailed Description

LevelMeter specialization for float audio.

### 3.133.2 Constructor & Destructor Documentation

#### 3.133.2.1 LevelMeterFloat()

```
LevelMeterFloat (
    int samplingRate,
    int numChannels)
```

Create new [LevelMeterFloat](#) instance.

##### Parameters

|                     |                                            |
|---------------------|--------------------------------------------|
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz). |
| <i>numChannels</i>  | Number of channels in the audio signal.    |

## 3.134 AudioUtil.LevelMeterShort Class Reference

LevelMeter specialization for short audio.

Inherits [AudioUtil.LevelMeter< T >](#).

### Public Member Functions

- [LevelMeterShort](#) (int samplingRate, int numChannels)  
*Create new [LevelMeterShort](#) instance.*
- override short[] **Process** (short[] buf)

### Public Member Functions inherited from [AudioUtil.LevelMeter< T >](#)

- void [ResetAccumAvgPeakAmp](#) ()  
*Reset [AccumAvgPeakAmp](#).*
- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [AudioUtil.LevelMeter< T >](#)

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

## Properties inherited from [AudioUtil.LevelMeter< T >](#)

- float [CurrentAvgAmp](#) [get]  
*Average amplitude value over last half second.*
- float [CurrentPeakAmp](#) [get, protected set]  
*Maximum amplitude value over last half second sec.*
- float [AccumAvgPeakAmp](#) [get]  
*Average of CurrentPeakAmps since last reset.*

### 3.134.1 Detailed Description

LevelMeter specialization for short audio.

### 3.134.2 Constructor & Destructor Documentation

#### 3.134.2.1 LevelMeterShort()

```
LevelMeterShort (
    int samplingRate,
    int numChannels)
```

Create new [LevelMeterShort](#) instance.

#### Parameters

|                     |                                            |
|---------------------|--------------------------------------------|
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz). |
| <i>numChannels</i>  | Number of channels in the audio signal.    |

## 3.135 LoadBalancingTransport Class Reference

Extends [LoadBalancingClient](#) with media streaming functionality.

Inherits [LoadBalancingClient](#), [IVoiceTransport](#), and [IDisposable](#).

Inherited by [LoadBalancingTransport2](#).

### Public Member Functions

- virtual int [GetPayloadFragmentSize](#) ([SendFrameParams](#) par)
- bool [IsChannelJoined](#) (int channelId)
- [LoadBalancingTransport](#) ([ILogger](#) logger=null, [ConnectionProtocol](#) connectionProtocol=[ConnectionProtocol.Udp](#), bool cppCompatibilityMode=false)  
*Initializes a new [LoadBalancingTransport](#).*
- new void **Service** ()  
*This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2 to 20 times a second).*
- virtual bool **ChangeAudioGroups** (byte[] groupsToRemove, byte[] groupsToAdd)
- void [SendVoiceInfo](#) ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void [SendVoiceRemove](#) ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void [SendFrame](#) ([ArraySegment< byte >](#) data, [FrameFlags](#) flags, ushort evNumber, byte frNumber, byte voiceld, int channelId, [SendFrameParams](#) par)
- string [ChannelIdStr](#) (int channelId)
- string [PlayerIdStr](#) (int playerId)
- void **Dispose** ()  
*Releases all resources used by the [LoadBalancingTransport](#) instance.*

### Protected Member Functions

- virtual object **buildFrameMessage** (byte voiceId, ushort evNumber, byte frNumber, ArraySegment< byte > data, FrameFlags flags)
- virtual void **onEventActionVoiceClient** (EventData ev)

### Protected Attributes

- [VoiceClient](#) **voiceClient**
- readonly bool **cppCompatibilityMode**
- readonly [ILogger](#) **logger**

### Properties

- [VoiceClient](#) **VoiceClient** [get]  
The [VoiceClient](#) implementation associated with this [LoadBalancingTransport](#).
- virtual byte **FrameCode** [get]

## 3.135.1 Detailed Description

Extends LoadBalancingClient with media streaming functionality.

Use your normal LoadBalancing workflow to join a [Voice](#) room. All standard LoadBalancing features are available. Use [VoiceClient](#) to work with media streams.

## 3.135.2 Constructor & Destructor Documentation

### 3.135.2.1 LoadBalancingTransport()

```
LoadBalancingTransport (
    ILogger logger = null,
    ConnectionProtocol connectionProtocol = ConnectionProtocol::Udp,
    bool cppCompatibilityMode = false)
```

Initializes a new [LoadBalancingTransport](#).

#### Parameters

|  |                             |                                                                                                                        |
|--|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
|  | <i>logger</i>               | <a href="#">ILogger</a> instance. If null, this instance LoadBalancingClient.DebugReturn implementation is used. Conne |
|  | <i>connectionProtocol</i>   | Connection protocol (UDP or TCP). Conne                                                                                |
|  | <i>cppCompatibilityMode</i> | Use a protocol compatible with V                                                                                       |

## 3.135.3 Member Function Documentation

### 3.135.3.1 ChannelIdStr()

```
string ChannelIdStr (
    int channelId)
```

Implements [IVoiceTransport](#).

### 3.135.3.2 GetPayloadFragmentSize()

```
virtual int GetPayloadFragmentSize (  
    SendFrameParams par) [virtual]
```

Implements [IVoiceTransport](#).

### 3.135.3.3 IsChannelJoined()

```
bool IsChannelJoined (  
    int channelId)
```

Implements [IVoiceTransport](#).

### 3.135.3.4 PlayerIdStr()

```
string PlayerIdStr (  
    int playerId)
```

Implements [IVoiceTransport](#).

### 3.135.3.5 SendFrame()

```
void SendFrame (  
    ArraySegment< byte > data,  
    FrameFlags flags,  
    ushort evNumber,  
    byte frNumber,  
    byte voiceId,  
    int channelId,  
    SendFrameParams par)
```

Implements [IVoiceTransport](#).

### 3.135.3.6 SendVoiceInfo()

```
void SendVoiceInfo (  
    LocalVoice voice,  
    int channelId,  
    bool targetMe,  
    int[] targetPlayers)
```

Implements [IVoiceTransport](#).

### 3.135.3.7 SendVoiceRemove()

```
void SendVoiceRemove (  
    LocalVoice voice,  
    int channelId,  
    bool targetMe,  
    int[] targetPlayers)
```

Implements [IVoiceTransport](#).

### 3.136 LoadBalancingTransport2 Class Reference

Variant of [LoadBalancingTransport](#). Aims to be non-alloc at the cost of breaking compatibility with older clients.

Inherits [LoadBalancingTransport](#).

#### Public Member Functions

- **LoadBalancingTransport2** ([ILogger](#) logger=null, [ConnectionProtocol](#) connectionProtocol=[ConnectionProtocol.Udp](#), bool cppCompatibilityMode=false)
- override int [GetPayloadFragmentSize](#) ([SendFrameParams](#) par)

#### Public Member Functions inherited from [LoadBalancingTransport](#)

- bool [IsChannelJoined](#) (int channelId)
- [LoadBalancingTransport](#) ([ILogger](#) logger=null, [ConnectionProtocol](#) connectionProtocol=[ConnectionProtocol.Udp](#), bool cppCompatibilityMode=false)  
*Initializes a new [LoadBalancingTransport](#).*
- new void **Service** ()  
*This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2 to 20 times a second).*
- virtual bool **ChangeAudioGroups** (byte[] groupsToRemove, byte[] groupsToAdd)
- void [SendVoiceInfo](#) ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void [SendVoiceRemove](#) ([LocalVoice](#) voice, int channelId, bool targetMe, int[] targetPlayers)
- void [SendFrame](#) (ArraySegment< byte > data, [FrameFlags](#) flags, ushort evNumber, byte frNumber, byte voiceld, int channelId, [SendFrameParams](#) par)
- string [ChannelIdStr](#) (int channelId)
- string [PlayerIdStr](#) (int playerId)
- void **Dispose** ()  
*Releases all resources used by the [LoadBalancingTransport](#) instance.*

#### Protected Member Functions

- override object [buildFrameMessage](#) (byte voiceld, ushort evNumber, byte frNumber, ArraySegment< byte > data, [FrameFlags](#) flags)
- override void [onEventActionVoiceClient](#) ([EventData](#) ev)

#### Properties

- override byte **FrameCode** [get]

#### Properties inherited from [LoadBalancingTransport](#)

- [VoiceClient](#) **VoiceClient** [get]  
*The [VoiceClient](#) implementation associated with this [LoadBalancingTransport](#).*
- virtual byte **FrameCode** [get]



## Additional Inherited Members

## Protected Attributes inherited from [LoadBalancingTransport](#)

- [VoiceClient](#) **voiceClient**
- readonly bool **cppCompatibilityMode**
- readonly [ILogger](#) **logger**

### 3.136.1 Detailed Description

Variant of [LoadBalancingTransport](#). Aims to be non-alloc at the cost of breaking compatibility with older clients.

### 3.136.2 Member Function Documentation

#### 3.136.2.1 buildFrameMessage()

```
override object buildFrameMessage (  
    byte voiceId,  
    ushort evNumber,  
    byte frNumber,  
    ArraySegment< byte > data,  
    FrameFlags flags) [protected], [virtual]
```

Reimplemented from [LoadBalancingTransport](#).

#### 3.136.2.2 GetPayloadFragmentSize()

```
override int GetPayloadFragmentSize (  
    SendFrameParams par) [virtual]
```

Reimplemented from [LoadBalancingTransport](#).

#### 3.136.2.3 onEventActionVoiceClient()

```
override void onEventActionVoiceClient (  
    EventData ev) [protected], [virtual]
```

Reimplemented from [LoadBalancingTransport](#).

## 3.137 LocalVoice Class Reference

Represents outgoing data stream.

Inherits [IDisposable](#).

Inherited by [LocalVoiceAudioDummy](#), [LocalVoiceFramed< T >](#), and [LocalVoiceVideo](#).

## Public Member Functions

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#).*
- virtual void **Dispose** ()

## Static Public Attributes

- const int **DATA\_POOL\_CAPACITY** = 50

## Protected Member Functions

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

## Protected Attributes

- int[] **targetPlayers\_**
- [VoiceInfo](#) **info**
- [IEncoder](#) **encoder**
- [VoiceClient](#) **voiceClient**
- bool **threadingEnabled**
- [ArraySegment< byte >](#) **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

## Properties

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*

- int **FEC** [get, set]  
Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).
- [IServiceable](#) **LocalUserServiceable** [get, set]  
Optional user object attached to [LocalVoice](#). its `Service()` will be called at each [VoiceClient.Service\(\)](#) call.
- byte **Group** [get, set]
- byte [InterestGroup](#) [get, set]  
If `InterestGroup != 0`, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).
- bool [DebugEchoMode](#) [get, set]  
If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).
- int[] [TargetPlayers](#) [get, set]  
If `TargetPlayers` is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

### 3.137.1 Detailed Description

Represents outgoing data stream.

### 3.137.2 Member Function Documentation

#### 3.137.2.1 Dispose()

```
virtual void Dispose () [virtual]
```

Reimplemented in [LocalVoiceFramed< T >](#).

### 3.137.3 Property Documentation

#### 3.137.3.1 DebugEchoMode

```
bool DebugEchoMode [get], [set]
```

If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).

This functionality availability depends on transport.

### 3.137.3.2 InterestGroup

```
byte InterestGroup [get], [set]
```

If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).

A remote voice is created even if the remote player is not subscribed to the InterestGroup. Use [VoiceCreateOptions.TargetPlayers](#) and [TargetPlayers](#) to completely hide the existence of [LocalVoice](#) from the remote player./>

### 3.137.3.3 TargetPlayers

```
int [] TargetPlayers [get], [set]
```

If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).

TargetPlayers update triggers the sending of voice info to added players and voice remove to removed. Depending on the transport, TargetPlayers may disregard [InterestGroup](#): the remote player whose number is in TargetPlayers, receives the stream even if not subscribed to the interest group. If the local player number is in TargetPlayers, it works like [DebugEchoMode](#). Using both DebugEchoMode and TargetPlayers at the same time to route the stream back to the sender leads to an inconsistent state after one of the options is switched off: the voice is removed but the frames are still delivered.

## 3.138 LocalVoiceAudio< T > Class Template Reference

Outgoing audio stream.

Inherits [LocalVoiceFramed< T >](#), and [ILocalVoiceAudio](#).

Inherited by [LocalVoiceAudioFloat](#), and [LocalVoiceAudioShort](#).

### Public Member Functions

- void [VoiceDetectorCalibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Trigger voice detector calibration process.*

### Public Member Functions inherited from [LocalVoiceFramed< T >](#)

- void [AddPostProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors after any built-in processors and everything added with AddPreProcessor.*
- void [AddPreProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void [RemoveProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void **ClearProcessors** ()  
*Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.*
- void **PushDataAsync** (T[] buf)  
*Asynchronously push data into this stream.*
- void **PushData** (T[] buf)  
*Synchronously push data into this stream.*
- override void [Dispose](#) ()  
*Releases resources used by the [LocalVoiceFramed<T>](#) instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.*

## Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#)).*

## Protected Member Functions

- void **initBuiltinProcessors** ()

## Protected Member Functions inherited from LocalVoiceFramed< T >

- T[] **processFrame** (T[] buf, int p0, int p1)

## Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

## Protected Attributes

- AudioUtil.VoiceDetector< T > **voiceDetector**
- AudioUtil.VoiceDetectorCalibration< T > **voiceDetectorCalibration**
- AudioUtil.LevelMeter< T > **levelMeter**
- int **channels**

## Protected Attributes inherited from LocalVoice

- int[] **targetPlayers\_**
- [VoiceInfo](#) **info**
- [IEncoder](#) **encoder**
- [VoiceClient](#) **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

## Properties

- virtual AudioUtil.IVoiceDetector [VoiceDetector](#) [get]  
*The VoiceDetector in use.*
- virtual AudioUtil.ILevelMeter [LevelMeter](#) [get]  
*The LevelMeter utility in use.*
- bool [VoiceDetectorCalibrating](#) [get]  
*True if the VoiceDetector is currently calibrating.*

## Properties inherited from [LocalVoiceFramed< T >](#)

- int **OptimalSourceFrameSize** [get]
- ObjectFactory< T[], int > **BufferFactory** [get]  
*<see cref="PushData(T[])" and />.*
- bool **PushDataAsyncReady** [get]  
*Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.*

## Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*
- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its Service() will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

## Additional Inherited Members

## Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

### 3.138.1 Detailed Description

Outgoing audio stream.

### 3.138.2 Member Function Documentation

#### 3.138.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
    int durationMs,
    Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. [Voice](#) detector sets threshold basing on measured background noise level.

#### Parameters

|                     |                                                                        |
|---------------------|------------------------------------------------------------------------|
| <i>durationMs</i>   | Duration of calibration in milliseconds.                               |
| <i>onCalibrated</i> | Called when calibration is complete. Parameter is new threshold value. |

Implements [ILocalVoiceAudio](#).

### 3.138.3 Property Documentation

#### 3.138.3.1 LevelMeter

```
virtual AudioUtil.ILevelMeter LevelMeter [get]
```

The LevelMeter utility in use.

Implements [ILocalVoiceAudio](#).

#### 3.138.3.2 VoiceDetector

```
virtual AudioUtil.IVoiceDetector VoiceDetector [get]
```

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implements [ILocalVoiceAudio](#).

### 3.138.3.3 VoiceDetectorCalibrating

bool VoiceDetectorCalibrating [get]

True if the VoiceDetector is currently calibrating.

Implements [ILocalVoiceAudio](#).

## 3.139 LocalVoiceAudioDummy Class Reference

Dummy LocalVoiceAudio.

Inherits [LocalVoice](#), and [ILocalVoiceAudio](#).

### Public Member Functions

- void [VoiceDetectorCalibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Trigger voice detector calibration process.*

### Public Member Functions inherited from [LocalVoice](#)

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#).*
- virtual void [Dispose](#) ()

### Static Public Attributes

- static [LocalVoiceAudioDummy](#) **Dummy** = new [LocalVoiceAudioDummy](#)()  
*A Dummy LocalVoiceAudio instance.*

### Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

### Properties

- AudioUtil.IVoiceDetector [VoiceDetector](#) [get]  
*The VoiceDetector in use.*
- AudioUtil.ILevelMeter [LevelMeter](#) [get]  
*The LevelMeter utility in use.*
- bool [VoiceDetectorCalibrating](#) [get]  
*If true, voice detector calibration is in progress.*



## Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*
- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its Service() will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

## Additional Inherited Members

## Protected Member Functions inherited from [LocalVoice](#)

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

## Protected Attributes inherited from [LocalVoice](#)

- `int[] targetPlayers_`
- [VoiceInfo](#) `info`
- [IEncoder](#) `encoder`
- [VoiceClient](#) `voiceClient`
- `bool threadingEnabled`
- `ArraySegment< byte > configFrame`
- `volatile bool disposed`
- `object disposeLock = new object()`

### 3.139.1 Detailed Description

Dummy LocalVoiceAudio.

For testing, this LocalVoiceAudio implementation features a [AudioUtil.VoiceDetectorDummy](#) and a [AudioUtil.LevelMeterDummy](#)

### 3.139.2 Member Function Documentation

#### 3.139.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
    int durationMs,
    Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. [Voice](#) detector sets threshold based on measured background noise level.

#### Parameters

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <code>durationMs</code>   | Duration of calibration (in milliseconds).                             |
| <code>onCalibrated</code> | Called when calibration is complete. Parameter is new threshold value. |

Implements [ILocalVoiceAudio](#).

### 3.139.3 Property Documentation

#### 3.139.3.1 LevelMeter

```
AudioUtil.ILevelMeter LevelMeter [get]
```

The LevelMeter utility in use.

Implements [ILocalVoiceAudio](#).

### 3.139.3.2 VoiceDetector

AudioUtil.IVoiceDetector VoiceDetector [get]

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implements [ILocalVoiceAudio](#).

### 3.139.3.3 VoiceDetectorCalibrating

bool VoiceDetectorCalibrating [get]

If true, voice detector calibration is in progress.

Implements [ILocalVoiceAudio](#).

## 3.140 LocalVoiceAudioFloat Class Reference

Specialization of [LocalVoiceAudio<T>](#) for float audio.

Inherits [LocalVoiceAudio< T >](#).

### Additional Inherited Members

### Public Member Functions inherited from [LocalVoiceAudio< T >](#)

- void [VoiceDetectorCalibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Trigger voice detector calibration process.*

### Public Member Functions inherited from [LocalVoiceFramed< T >](#)

- void [AddPostProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors after any built-in processors and everything added with AddPreProcessor.*
- void [AddPreProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void [RemoveProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void **ClearProcessors** ()  
*Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.*
- void **PushDataAsync** (T[] buf)  
*Asynchronously push data into this stream.*
- void **PushData** (T[] buf)  
*Synchronously push data into this stream.*
- override void [Dispose](#) ()  
*Releases resources used by the [LocalVoiceFramed<T>](#) instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.*

## Public Member Functions inherited from [LocalVoice](#)

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#)).*

## Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

## Protected Member Functions inherited from [LocalVoiceAudio< T >](#)

- void **initBuiltinProcessors** ()

## Protected Member Functions inherited from [LocalVoiceFramed< T >](#)

- T[] **processFrame** (T[] buf, int p0, int p1)

## Protected Member Functions inherited from [LocalVoice](#)

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

## Protected Attributes inherited from [LocalVoiceAudio< T >](#)

- AudioUtil.VoiceDetector< T > **voiceDetector**
- AudioUtil.VoiceDetectorCalibration< T > **voiceDetectorCalibration**
- AudioUtil.LevelMeter< T > **levelMeter**
- int **channels**

## Protected Attributes inherited from [LocalVoice](#)

- int[] **targetPlayers\_**
- [VoiceInfo](#) **info**
- [IEncoder](#) **encoder**
- [VoiceClient](#) **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

## Properties inherited from [LocalVoiceAudio< T >](#)

- virtual AudioUtil.IVoiceDetector [VoiceDetector](#) [get]  
*The VoiceDetector in use.*
- virtual AudioUtil.ILevelMeter [LevelMeter](#) [get]  
*The LevelMeter utility in use.*
- bool [VoiceDetectorCalibrating](#) [get]  
*True if the VoiceDetector is currently calibrating.*

## Properties inherited from [LocalVoiceFramed< T >](#)

- int **OptimalSourceFrameSize** [get]
- ObjectFactory< T[], int > **BufferFactory** [get]  
*<see cref="PushData(T[])" and />.*
- bool **PushDataAsyncReady** [get]  
*Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.*

## Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*
- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its Service() will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

### 3.140.1 Detailed Description

Specialization of [LocalVoiceAudio<T>](#) for float audio.

## 3.141 LocalVoiceAudioShort Class Reference

Specialization of [LocalVoiceAudio<T>](#) for short audio.

Inherits [LocalVoiceAudio< T >](#).

### Additional Inherited Members

### Public Member Functions inherited from [LocalVoiceAudio< T >](#)

- void [VoiceDetectorCalibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Trigger voice detector calibration process.*

### Public Member Functions inherited from [LocalVoiceFramed< T >](#)

- void [AddPostProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors after any built-in processors and everything added with AddPreProcessor.*
- void [AddPreProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void [RemoveProcessor](#) (params IProcessor< T >[] processors)  
*Adds processors before built-in processors and everything added with AddPostProcessor.*
- void **ClearProcessors** ()  
*Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.*
- void **PushDataAsync** (T[] buf)  
*Asynchronously push data into this stream.*
- void **PushData** (T[] buf)  
*Synchronously push data into this stream.*
- override void [Dispose](#) ()  
*Releases resources used by the [LocalVoiceFramed<T>](#) instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.*

### Public Member Functions inherited from [LocalVoice](#)

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#)).*

### Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

### Protected Member Functions inherited from [LocalVoiceAudio< T >](#)

- void **initBuiltinProcessors** ()

### Protected Member Functions inherited from [LocalVoiceFramed< T >](#)

- [T\[\]](#) **processFrame** ([T\[\]](#) buf, int p0, int p1)

### Protected Member Functions inherited from [LocalVoice](#)

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

### Protected Attributes inherited from [LocalVoiceAudio< T >](#)

- [AudioUtil.VoiceDetector< T >](#) **voiceDetector**
- [AudioUtil.VoiceDetectorCalibration< T >](#) **voiceDetectorCalibration**
- [AudioUtil.LevelMeter< T >](#) **levelMeter**
- int **channels**

### Protected Attributes inherited from [LocalVoice](#)

- int[] **targetPlayers\_**
- [VoiceInfo](#) **info**
- [IEncoder](#) **encoder**
- [VoiceClient](#) **voiceClient**
- bool **threadingEnabled**
- [ArraySegment< byte >](#) **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

### Properties inherited from [LocalVoiceAudio< T >](#)

- virtual [AudioUtil.IVoiceDetector](#) [VoiceDetector](#) [get]  
*The VoiceDetector in use.*
- virtual [AudioUtil.ILevelMeter](#) [LevelMeter](#) [get]  
*The LevelMeter utility in use.*
- bool [VoiceDetectorCalibrating](#) [get]  
*True if the VoiceDetector is currently calibrating.*

### Properties inherited from [LocalVoiceFramed< T >](#)

- int **OptimalSourceFrameSize** [get]
- [ObjectFactory< T\[\], int >](#) **BufferFactory** [get]  
*<see cref="PushData(T[])" and />.*
- bool **PushDataAsyncReady** [get]  
*Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.*

## Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*
- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its Service() will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

### 3.141.1 Detailed Description

Specialization of [LocalVoiceAudio<T>](#) for short audio.



## 3.142 LocalVoiceFramed< T > Class Template Reference

Typed re-framing [LocalVoice](#).

Inherits [LocalVoice](#).

Inherited by [LocalVoiceAudio< T >](#).

### Public Member Functions

- void [AddPostProcessor](#) (params [IProcessor< T >\[\]](#) processors)  
*Adds processors after any built-in processors and everything added with [AddPreProcessor](#).*
- void [AddPreProcessor](#) (params [IProcessor< T >\[\]](#) processors)  
*Adds processors before built-in processors and everything added with [AddPostProcessor](#).*
- void [RemoveProcessor](#) (params [IProcessor< T >\[\]](#) processors)  
*Adds processors before built-in processors and everything added with [AddPostProcessor](#).*
- void **ClearProcessors** ()  
*Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.*
- void **PushDataAsync** ([T\[\]](#) buf)  
*Asynchronously push data into this stream.*
- void **PushData** ([T\[\]](#) buf)  
*Synchronously push data into this stream.*
- override void [Dispose](#) ()  
*Releases resources used by the [LocalVoiceFramed< T >](#) instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.*

### Public Member Functions inherited from [LocalVoice](#)

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#)).*

### Protected Member Functions

- [T\[\]](#) **processFrame** ([T\[\]](#) buf, int p0, int p1)

### Protected Member Functions inherited from [LocalVoice](#)

- bool **targetExits** (bool targetMe, [int\[\]](#) targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, [int\[\]](#) targetPlayers)
- void **sendVoiceRemove** (bool targetMe, [int\[\]](#) targetPlayers)

### Properties

- int **OptimalSourceFrameSize** [get]
- [ObjectFactory< T\[\], int >](#) **BufferFactory** [get]  
*< see cref="PushData(T[])" and />.*
- bool **PushDataAsyncReady** [get]  
*Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.*

## Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*
- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its Service() will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

## Additional Inherited Members

## Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

## Protected Attributes inherited from [LocalVoice](#)

- `int[] targetPlayers_`
- [VoiceInfo](#) `info`
- [IEncoder](#) `encoder`
- [VoiceClient](#) `voiceClient`
- `bool threadingEnabled`
- `ArraySegment< byte > configFrame`
- `volatile bool disposed`
- `object disposeLock = new object()`

### 3.142.1 Detailed Description

Typed re-framing [LocalVoice](#).

Consumes data in array buffers of arbitrary length. Repacks them in frames of [VoiceInfo.FrameSize](#) length for further processing and encoding.

### 3.142.2 Member Function Documentation

#### 3.142.2.1 AddPostProcessor()

```
void AddPostProcessor (
    params IProcessor< T >[] processors)
```

Adds processors after any built-in processors and everything added with AddPreProcessor.

Parameters

|                         |
|-------------------------|
| <code>processors</code> |
|-------------------------|

#### 3.142.2.2 AddPreProcessor()

```
void AddPreProcessor (
    params IProcessor< T >[] processors)
```

Adds processors before built-in processors and everything added with AddPostProcessor.

Parameters

|                         |
|-------------------------|
| <code>processors</code> |
|-------------------------|

#### 3.142.2.3 Dispose()

```
override void Dispose () [virtual]
```

Releases resources used by the [LocalVoiceFramed<T>](#) instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Reimplemented from [LocalVoice](#).

#### 3.142.2.4 RemoveProcessor()

```
void RemoveProcessor (
    params IProcessor< T >[] processors)
```

Adds processors before built-in processors and everything added with AddPostProcessor.

## Parameters

|                         |
|-------------------------|
| <code>processors</code> |
|-------------------------|

### 3.143 LocalVoiceVideo Class Reference

Inherits [LocalVoice](#), and [IVideoSink](#).

#### Public Member Functions

- void [PushImageAsync](#) ([ImageBufferNative](#) buf)
- void [PushImage](#) ([ImageBufferNative](#) buf)
- override void [Dispose](#) ()

#### Public Member Functions inherited from [LocalVoice](#)

- void [SendSpacingProfileStart](#) ()
- void [RemoveSelf](#) ()  
*Remove this voice from it's [VoiceClient](#) (using [VoiceClient.RemoveLocalVoice](#).*

#### Properties

- int [PushImageQueueCount](#) [get]

#### Properties inherited from [LocalVoice](#)

- [VoiceInfo](#) **Info** [get]  
*Returns Info structure assigned on local voice cration.*
- bool **TransmitEnabled** [get, set]  
*If true, stream data broadcasted.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if stream broadcasts.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentFragmented** [get]  
*Sent fragmented frames counter.*
- int **FramesSentFragments** [get]  
*Sent frames fragments counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- bool **Reliable** [get, set]  
*Send data reliable. See also [VoiceCreateOptions.Reliable](#).*
- bool **Encrypt** [get, set]  
*Send data encrypted. See also [VoiceCreateOptions.Encrypt](#).*
- bool **Fragment** [get, set]  
*Split frames into fragments according to the size provided by the Transport. See also [VoiceCreateOptions.Fragment](#).*

- int **FEC** [get, set]  
*Forward Error Correction control. See also [VoiceCreateOptions.FEC](#).*
- [IServiceable](#) **LocalUserServiceable** [get, set]  
*Optional user object attached to [LocalVoice](#). its [Service\(\)](#) will be called at each [VoiceClient.Service\(\)](#) call.*
- byte **Group** [get, set]
- byte **InterestGroup** [get, set]  
*If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also [VoiceCreateOptions.InterestGroup](#).*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams. See also [VoiceCreateOptions.DebugEchoMode](#).*
- int[] **TargetPlayers** [get, set]  
*If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also [VoiceCreateOptions.TargetPlayers](#).*
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]  
*Logs input frames time spacing profiling results. Do not call frequently.*
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

#### Additional Inherited Members

#### Static Public Attributes inherited from [LocalVoice](#)

- const int **DATA\_POOL\_CAPACITY** = 50

#### Protected Member Functions inherited from [LocalVoice](#)

- bool **targetExits** (bool targetMe, int[] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[] targetPlayers)

#### Protected Attributes inherited from [LocalVoice](#)

- int[] **targetPlayers\_**
- [VoiceInfo](#) **info**
- [IEncoder](#) **encoder**
- [VoiceClient](#) **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

### 3.143.1 Member Function Documentation

#### 3.143.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [LocalVoice](#).

#### 3.143.1.2 PushImage()

```
void PushImage (  
    ImageBufferNative buf)
```

Implements [IVideoSink](#).

#### 3.143.1.3 PushImageAsync()

```
void PushImageAsync (  
    ImageBufferNative buf)
```

Implements [IVideoSink](#).

### 3.143.2 Property Documentation

#### 3.143.2.1 PushImageQueueCount

```
int PushImageQueueCount [get]
```

Implements [IVideoSink](#).

## 3.144 Logger Class Reference

Inherits [ILogger](#).

### Public Member Functions

- **Logger** (LogLevel level=LogLevel.Debug)
- void [Log](#) (LogLevel level, string msg)
- void [Log](#) (LogLevel level, string fmt, params object[] args)

### Properties

- LogLevel [Level](#) [get, set]

### 3.144.1 Member Function Documentation

#### 3.144.1.1 Log() [1/2]

```
void Log (
    LogLevel level,
    string fmt,
    params object[] args)
```

Implements [ILogger](#).

#### 3.144.1.2 Log() [2/2]

```
void Log (
    LogLevel level,
    string msg)
```

Implements [ILogger](#).

### 3.144.2 Property Documentation

#### 3.144.2.1 Level

```
LogLevel Level [get], [set]
```

Implements [ILogger](#).

## 3.145 MFTCodec Class Reference

### Classes

- class [VideoDecoder](#)
- class [VideoEncoder](#)

### Public Member Functions

- delegate void **LogCallbackDelegate** (int instanceID, int level, IntPtr str)

## 3.146 MicAmplifier Class Reference

Inherits [VoiceComponent](#).

### Properties

- float **AmplificationFactor** [get, set]

### Properties inherited from [VoiceComponent](#)

- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### Additional Inherited Members

### Protected Member Functions inherited from [VoiceComponent](#)

- virtual void **Awake** ()

## 3.147 MicAmplifierFloat Class Reference

Inherits [IProcessor< T >](#).

### Public Member Functions

- **MicAmplifierFloat** (float amplificationFactor)
- float[] **Process** (float[] buf)
- void **Dispose** ()

### Public Member Functions inherited from [IProcessor< T >](#)

- T[] [Process](#) (T[] buf)  
*Process a frame of data.*

### Properties

- float **AmplificationFactor** [get, set]
- bool **Disabled** [get, set]

## 3.148 MicAmplifierShort Class Reference

Inherits [IProcessor< T >](#).

### Public Member Functions

- **MicAmplifierShort** (float amplificationFactor)
- short[] **Process** (short[] buf)
- void **Dispose** ()



## Public Member Functions inherited from [IProcessor< T >](#)

- [T\[\] Process](#) (T[] buf)  
*Process a frame of data.*

## Properties

- float **AmplificationFactor** [get, set]
- bool **Disabled** [get, set]

## 3.149 MicrophonePermission Class Reference

Helper to request Microphone permission on Android or iOS.

Inherits [VoiceComponent](#).

## Public Member Functions

- void **InitVoice** ()

## Protected Member Functions

- override void [Awake](#) ()

## Properties

- bool **HasPermission** [get]

## Properties inherited from [VoiceComponent](#)

- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## Events

- static Action< bool > **MicrophonePermissionCallback**

### 3.149.1 Detailed Description

Helper to request Microphone permission on Android or iOS.

### 3.149.2 Member Function Documentation

#### 3.149.2.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

## 3.150 MicWrapper Class Reference

Inherits [IAudioReader< T >](#).

### Public Member Functions

- **MicWrapper** (string device, int suggestedFrequency, [ILogger](#) logger)
- void **Dispose** ()
- bool **Read** (float[] buffer)

### Public Member Functions inherited from [IDataReader< T >](#)

- bool [Read](#) (T[] buffer)  
*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Properties

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.150.1 Property Documentation

#### 3.150.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.150.1.2 Error

string Error [get]

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.150.1.3 SamplingRate

int SamplingRate [get]

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.151 MicWrapperPusher Class Reference

Inherits [IAudioPusher< T >](#).

### Public Member Functions

- **MicWrapperPusher** (GameObject parent, string device, int suggestedFrequency, [ILogger](#) logger)
- void **SetCallback** (Action< float[]> callback, ObjectFactory< float[], int > bufferFactory, int optimalFrameSize)
- void **Dispose** ()

### Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

### Properties

- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.151.1 Property Documentation

#### 3.151.1.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

#### 3.151.1.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

#### 3.151.1.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.152 MicWrapperPusherOnAudioFilterRead Class Reference

Inherits MonoBehaviour.

### Events

- `Action< float[], int > OnAudioFrame`

## 3.153 MonoPInvokeCallbackAttribute Class Reference

Inherits Attribute.

### Public Member Functions

- `MonoPInvokeCallbackAttribute (Type t)`

## 3.154 ObjectFactory< TType, TInfo > Interface Template Reference

Object factory with optional info useful in combination with [ObjectPool<TType, TInfo>](#).

Inherits IDisposable.

Inherited by [ArrayPoolSet< T >](#), [ArraySegmentPool< T >](#), and [ObjectPool< TType, TInfo >](#).

### Public Member Functions

- TType [New](#) ()
- TType [New](#) (TInfo info)
- bool [Free](#) (TType obj)
- bool [Free](#) (TType obj, TInfo info)

### 3.154.1 Detailed Description

Object factory with optional info useful in combination with [ObjectPool<TType, TInfo>](#).

#### Template Parameters

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| <i>TType</i> | Object type.                                                                       |
| <i>TInfo</i> | Type of property used to check 2 objects identity (like integral length of array). |

### 3.154.2 Member Function Documentation

#### 3.154.2.1 Free() [1/2]

```
bool Free (
    TType obj)
```

Implemented in [ObjectPool< TType, TInfo >](#).

#### 3.154.2.2 Free() [2/2]

```
bool Free (
    TType obj,
    TInfo info)
```

Implemented in [ObjectPool< TType, TInfo >](#).

#### 3.154.2.3 New() [1/2]

```
TType New ()
```

Implemented in [ObjectPool< TType, TInfo >](#).

### 3.154.2.4 New() [2/2]

```
TType New (
    TInfo info)
```

Implemented in [ObjectPool< TType, TInfo >](#).

## 3.155 ObjectPool< TType, TInfo > Class Template Reference

Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).

Inherits [ObjectFactory< TType, TInfo >](#).

Inherited by [ArrayPool< T >](#), and [ImageBufferNativePool< T >](#).

### Public Member Functions

- [ObjectPool](#) (int capacity, string name, TInfo info)  
*Create a new ObjectPool instance with the given info structure.*
- TType [New](#) ()  
*Acquire an existing object, or create a new one if none are available.*
- TType [New](#) (TInfo info)  
*Acquire an existing object (if info matches), or create a new one from the passed info.*
- virtual bool [Free](#) (TType obj, TInfo objInfo)  
*Returns object to pool.*
- virtual bool [Free](#) (TType obj)  
*Returns object to pool, or destroys it if the pool is full.*
- void **Dispose** ()  
*Free resources associated with this ObjectPool.*

### Protected Member Functions

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

### Protected Attributes

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

### Properties

- TInfo **Info** [get]  
*The property (info) that objects in this Pool must match.*

### 3.155.1 Detailed Description

Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).

## Template Parameters

|  |              |                                                                                     |
|--|--------------|-------------------------------------------------------------------------------------|
|  | <i>TType</i> | Object type.                                                                        |
|  | <i>TInfo</i> | Type of parameter used to check 2 objects identity (like integral length of array). |

**3.155.2 Member Function Documentation****3.155.2.1 Free()** [1/2]

```
virtual bool Free (
    TType obj) [virtual]
```

Returns object to pool, or destroys it if the pool is full.

## Parameters

|            |                                   |
|------------|-----------------------------------|
| <i>obj</i> | The object to return to the pool. |
|------------|-----------------------------------|

Implements [ObjectFactory< TType, TInfo >](#).

**3.155.2.2 Free()** [2/2]

```
virtual bool Free (
    TType obj,
    TInfo objInfo) [virtual]
```

Returns object to pool.

## Parameters

|                |                                   |
|----------------|-----------------------------------|
| <i>obj</i>     | The object to return to the pool. |
| <i>objInfo</i> | The info structure about obj.     |

*obj* is returned to the pool only if *objInfo* matches this pool's info. Else, it is destroyed.

Implements [ObjectFactory< TType, TInfo >](#).

**3.155.2.3 New()** [1/2]

```
TType New ()
```

Acquire an existing object, or create a new one if none are available.

If it fails to get one from the pool, this will create from the info given in this pool's constructor.

Implements [ObjectFactory< TType, TInfo >](#).

**3.155.2.4 New()** [2/2]

```
TType New (
    TInfo info)
```

Acquire an existing object (if info matches), or create a new one from the passed info.

**Parameters**

|             |                                                       |
|-------------|-------------------------------------------------------|
| <i>info</i> | Info structure to match, or create a new object with. |
|-------------|-------------------------------------------------------|

Implements [ObjectFactory< TType, TInfo >](#).

**3.155.2.5 ObjectPool()**

```
ObjectPool (
    int capacity,
    string name,
    TInfo info)
```

Create a new ObjectPool instance with the given info structure.

**Parameters**

|                 |                                     |
|-----------------|-------------------------------------|
| <i>capacity</i> | Capacity (size) of the object pool. |
| <i>name</i>     | Name of the object pool.            |
| <i>info</i>     | Info about this Pool's objects.     |

**3.156 AndroidCamera.OnReadyCallback Class Reference**

Inherits [AndroidJavaProxy](#).

**Public Member Functions**

- **OnReadyCallback** (Action< [AndroidJavaObject](#) > callback)
- void **onReady** ([AndroidJavaObject](#) camera)

**3.157 AndroidCameraVideoEncoder.OnReadyCallback Class Reference**

Inherits [AndroidJavaProxy](#).

**Public Member Functions**

- **OnReadyCallback** (Action< [AndroidJavaObject](#) > callback)
- void **onReady** ([AndroidJavaObject](#) camera)

**3.158 AndroidSurfaceView.OnReadyCallback Class Reference**

Inherits [AndroidJavaProxy](#).



**Public Member Functions**

- **OnReadyCallback** (Action< AndroidJavaObject > callback)
- void **onReady** (AndroidJavaObject surface)

**3.159 OpusCodec Class Reference****Classes**

- class [Decoder< T >](#)
- class [Encoder< T >](#)
- class [EncoderFloat](#)
- class [EncoderShort](#)
- class [Factory](#)
- class [Util](#)

**Public Types**

- enum **FrameDuration**

**Properties**

- static string **Version** [get]

**3.160 PermissionsRequester.PermissionRequest Class Reference**

Launch a permission request.

**Public Member Functions**

- **PermissionRequest** (string permissionName, float delayBeforeRequest=DefaultDelayBeforeRequest)
- async void **CheckPermission** ()
- void **UpdatePermission** ()

**Public Attributes**

- string **permissionName** = ""
- bool **hasBeenChecked** = false  
*Tells if the permission has received an answer. Set by callbacks on Android, and by the HasPermission setter on other platforms.*
- bool **isRequesting** = false
- float **delayBeforeRequest** = DefaultDelayBeforeRequest
- float **delayBeforeCallback** = 0
- UnityEvent< bool > **permissionCallback** = new UnityEvent<bool>()
- UnityEvent **permissionGrantedCallback** = new UnityEvent()
- UnityEvent **permissionCheckStartedCallback** = new UnityEvent()
- bool **isUnityApplicationAuthorization** = false

### Static Public Attributes

- const float **DefaultDelayBeforeRequest** = 0.05f

### Protected Member Functions

- void **AndroidCheckPermission** ()
- IEnumerator **UnityCheckAuthorization** ()  
*UserAuthorization based authorization, for iOS and WebGL.*

### Properties

- bool **HasPermission** [get]  
*Modify hasPermission, and when setting values, ensure to set hasBeenChecked/isRequesting, to consider the request as handled.*

## 3.160.1 Detailed Description

Launch a permission request.

## 3.160.2 Member Function Documentation

### 3.160.2.1 UnityCheckAuthorization()

`IEnumerator UnityCheckAuthorization () [protected]`

UserAuthorization based authorization, for iOS and WebGL.

Returns

## 3.161 PermissionsRequester Class Reference

Launch successively permission requests (Android only for now)

Inherits MonoBehaviour.

### Classes

- class **PermissionRequest**  
*Launch a permission request.*

## Public Member Functions

- string **PermissionNameForUserAuthorization** (UserAuthorization userAuthorization)
- async Task **LaunchPermissionRequests** ()
- bool **TryFindConfiguredPermissionRequest** (string permission, out [PermissionRequest](#) requester)
- void **AddMicrophoneRequest** (UnityAction< bool > permissionCallback=null, UnityAction permissionCheckStartedCallback=null)
- void **AddWebcamRequest** (UnityAction< bool > permissionCallback=null, UnityAction permissionCheckStartedCallback=null)
- void **AddAuthorizationRequest** (UserAuthorization authorization, UnityAction< bool > permissionCallback=null, UnityAction permissionCheckStartedCallback=null)  
*Queue a permission request for authorization UserAuthorization (for iOS or webGL) Will be converted to a string (WebcamAuthorizationPermissionName or MicrophoneAuthorizationPermissionName), and passed to AddPermissionRequest.*
- async void **AddPermissionRequest** (string permission, UnityAction< bool > permissionCallback=null, UnityAction permissionCheckStartedCallback=null)  
*Queue a permission request for permission string (can be either an Android request, or, for iOS and webGL WebcamAuthorizationPermissionName or MicrophoneAuthorizationPermissionName values)*

## Static Public Member Functions

- static bool **TryConvertToUserAuthorization** (string permissionName, out UserAuthorization userAuthorization)  
*Convert a string permission to a UserAuthorization (using MicrophoneAuthorizationPermissionName and WebcamAuthorizationPermissionName values)*
- static bool **HasMicrophonePermission** ()
- static bool **HasWebcamPermission** ()
- static async Task **Delay** (int milliseconds)

## Public Attributes

- List< [PermissionRequest](#) > **permissionRequests** = new List<[PermissionRequest](#)>()
- UnityEvent **allPermissionCallback** = new UnityEvent()
- List< GameObject > **objectsToActivateAfterPermissionChecks** = new List<GameObject>()
- bool **requestOnStart** = true
- bool **isLaunchingRequests** = false

## Static Public Attributes

- const int **EXECUTION\_ORDER** = -20\_000
- static [PermissionsRequester](#) **\_sharedInstance**
- const string **MicrophoneAuthorizationPermissionName** = MicrophoneAndroidPermissionName
- const string **WebcamAuthorizationPermissionName** = WebcamAndroidPermissionName
- const string **MicrophoneAndroidPermissionName** = "android.permission.RECORD\_AUDIO"
- const string **WebcamAndroidPermissionName** = "android.permission.CAMERA"

## Properties

- static [PermissionsRequester](#) **SharedInstance** [get]

### 3.161.1 Detailed Description

Launch successively permission requests (Android only for now)

## 3.162 PermissionWaiter Class Reference

Wait for a permission to be granted.

Inherits MonoBehaviour.

Inherited by [VoiceConnectionPermissionWaiter](#).

### Public Member Functions

- virtual void [OnPermissionGranted](#) ()

*Should be called manually (for instance in a [PermissionsRequester](#) callback) if `checkPermisisonGrantAutomatically` is not checked Otherwise, automatically called when relevant in `Update`.*

### Public Attributes

- bool **checkPermisisonGrantAutomatically** = true
- UnityEvent< bool > **onPermissionChanged** = new UnityEvent<bool>()
- UnityEvent **onPermissionGranted** = new UnityEvent()

### Static Public Attributes

- const int **EXECUTION\_ORDER** = PermissionsRequester.EXECUTION\_ORDER + 1

### Protected Member Functions

- virtual void **Awake** ()
- virtual void [OnPermissionRequired](#) ()  
*Called when the permission is required.*
- virtual void **OnPermissionChanged** (bool isGranted)
- virtual void **Update** ()

### Protected Attributes

- bool **waitingForPermission** = false

### Properties

- bool **IsWaitingForPermission** [get]
- virtual string **PermissionName** [get]
- bool **IsPermissionGranted** [get]

### 3.162.1 Detailed Description

Wait for a permission to be granted.

Should be used alongside a [PermissionsRequester](#), to align permissions on the desired timing (to avoid conflicting requests) Can create one if missing

Subclass should have a [DefaultExecutionOrder(PermissionWaiter.EXECUTION\_ORDER)] (PermissionWaiter.EXECUTION\_ORDER or any higher value - to wait for the Permissionsrequester Awake, so that the singleton is detected) tag to be sure to trigger early, if needed

### 3.162.2 Member Function Documentation

#### 3.162.2.1 OnPermissionGranted()

```
virtual void OnPermissionGranted () [virtual]
```

Should be called manually (for instance in a [PermissionsRequester](#) callback) if checkPermisisonGrantAutomatically is not checked Otherwise, automatically called when relevant in Update.

Reimplemented in [VoiceConnectionPermissionWaiter](#).

#### 3.162.2.2 OnPermissionRequired()

```
virtual void OnPermissionRequired () [protected], [virtual]
```

Called when the permission is required.

Reimplemented in [VoiceConnectionPermissionWaiter](#).

## 3.163 PhotonAppSettings Class Reference

Collection of connection-relevant settings, used internally by PhotonNetwork.ConnectUsingSettings.

Inherits ScriptableObject.

### Public Member Functions

- void **UseCloud** (string cloudAppid, string code="")  
*Sets appid and region code in the AppSettings. Used in Editor.*
- override string **ToString** ()  
*String summary of the AppSettings.*

### Static Public Member Functions

- static void **LoadOrCreateSettings** ()

### Public Attributes

- AppSettings **AppSettings**

### Properties

- static [PhotonAppSettings](#) **Instance** [get]

## 3.163.1 Detailed Description

Collection of connection-relevant settings, used internally by PhotonNetwork.ConnectUsingSettings.

Includes the AppSettings class from the Realtime APIs plus some other, PUN-relevant, settings.

## 3.164 PhotonVoiceCreatedParams Class Reference

### Properties

- Voice.LocalVoice **Voice** [get, set]
- Voice.IAudioDesc **AudioDesc** [get, set]

## 3.165 PhotonVoiceLagSimulationGui Class Reference

Inherits MonoBehaviour.

### Public Member Functions

- void **Start** ()

## 3.166 PhotonVoiceStatsGui Class Reference

Basic GUI to show traffic and health statistics of the connection to [Photon](#), toggled by shift+tab.

Inherits MonoBehaviour.

### 3.166.1 Detailed Description

Basic GUI to show traffic and health statistics of the connection to [Photon](#), toggled by shift+tab.

The shown health values can help identify problems with connection losses or performance. Example: If the time delta between two consecutive SendOutgoingCommands calls is a second or more, chances rise for a disconnect being caused by this (because acknowledgments to the server need to be sent in due time).

## 3.167 PhotonVoiceView Class Reference

Component that should be attached to a networked [PUN](#) prefab that has PhotonView. It will bind remote [Recorder](#) with local [Speaker](#) of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.

Inherits [VoiceComponent](#).

### Protected Member Functions

- override void [Awake](#) ()

### Properties

- [Recorder](#) **RecorderInUse** [get]  
*The [Recorder](#) component currently used by this [PhotonVoiceView](#).*
- [Speaker](#) **SpeakerInUse** [get]  
*The [Speaker](#) component currently used by this [PhotonVoiceView](#).*
- bool **IsSpeaking** [get]  
*If true, this [PhotonVoiceView](#) has a [Speaker](#) that is currently playing received audio frames from remote audio source.*
- bool **IsRecording** [get]  
*If true, this [PhotonVoiceView](#) has a [Recorder](#) that is currently transmitting audio stream from local audio source.*

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### 3.167.1 Detailed Description

Component that should be attached to a networked [PUN](#) prefab that has PhotonView. It will bind remote [Recorder](#) with local [Speaker](#) of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.

### 3.167.2 Member Function Documentation

#### 3.167.2.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

### 3.168 ImageBufferNative.PlaneSet Struct Reference

#### Public Member Functions

- **PlaneSet** (int length, IntPtr p0=default(IntPtr), IntPtr p1=default(IntPtr), IntPtr p2=default(IntPtr), IntPtr p3=default(IntPtr))

#### Properties

- IntPtr **this[int key]** [get, set]
- int **Length** [get]

### 3.169 Platform Class Reference

#### Static Public Member Functions

- static [IDeviceEnumerator](#) **CreateAudioInEnumerator** ([ILogger](#) logger)
- static [IAudioInChangeNotifier](#) **CreateAudioInChangeNotifier** (Action callback, [ILogger](#) logger)
- static [IEncoder](#) **CreateDefaultAudioEncoder**< T > ([ILogger](#) logger, [VoiceInfo](#) info)
- static [IAudioDesc](#) **CreateDefaultAudioSource** ([ILogger](#) logger, [DeviceInfo](#) dev, int samplingRate, int channels, object otherParams=null)
- static [IAudioOut](#)< float > **CreateUnityAudioOut** (UnityEngine.AudioSource audioSource, AudioOutDelayControl.PlayDelayConfig playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- static [IDeviceEnumerator](#) **CreateVideoInEnumerator** ([ILogger](#) logger)
- static [IEncoderDirectImage](#) **CreateDefaultVideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- static [IDecoderDirect](#)< [ImageBufferNative](#) > **CreateDefaultVideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- static [IVideoRecorder](#) **CreateDefaultVideoRecorder** ([ILogger](#) logger, [VoiceInfo](#) info, [DeviceInfo](#) camDevice, Action< [IVideoRecorder](#) > onReady)
- static [IVideoPlayer](#) **CreateDefaultVideoPlayer** ([ILogger](#) logger, [VoiceInfo](#) info, Action< [IVideoPlayer](#) > onReady)
- static [IPreviewManager](#) **CreateDefaultPreviewManager** ([ILogger](#) logger)
- static [IVideoRecorder](#) **CreateVideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, [DeviceInfo](#) camDevice, Action< [IVideoRecorder](#) > onReady)
- static [IVideoPlayer](#) **CreateVideoPlayerUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, Action< [IVideoPlayer](#) > onReady)
- static [IPreviewManager](#) **CreatePreviewManagerUnityTexture** ([ILogger](#) logger)

### 3.170 AudioOutDelayControl.PlayDelayConfig Struct Reference

#### Public Attributes

- int **Low**
- int **High**
- int **Max**
- int **SpeedUpPerc**

#### Static Public Attributes

- static [PlayDelayConfig](#) **Default**



## Properties

- int **Delay** [get, set]

## 3.170.1 Member Data Documentation

### 3.170.1.1 Default

`PlayDelayConfig` Default [static]

#### Initial value:

```
= new PlayDelayConfig()
{
    Low = 200,
    High = 200,
    Max = 1000,
    SpeedUpPerc = 5,
}
```

## 3.171 PreviewManager Class Reference

Inherits [IPreviewManager](#).

Inherited by [PreviewManagerLayer](#), [PreviewManagerMediaPlayerElement](#), [AndroidPreviewManagerSurfaceView](#), and [PreviewManagerUnityGUI](#).

## Classes

- class [ViewState](#)

## Public Member Functions

- virtual void [AddView](#) (object id, [IVideoPreview](#) view)
- virtual void [RemoveView](#) (object id)
- virtual bool [Has](#) (object id)
- virtual void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), Rotation rot=Rotation.Rotate0)

*Positions the preview of the view / image. Not optimized to be called every frame.*

## Protected Member Functions

- void **Apply** ([ViewState](#) v)

## Protected Attributes

- Dictionary< object, [ViewState](#) > **views** = new Dictionary<object, [ViewState](#)>()

### 3.171.1 Member Function Documentation

#### 3.171.1.1 AddView()

```
virtual void AddView (  
    object id,  
    IVideoPreview view) [virtual]
```

Implements [IPreviewManager](#).

#### 3.171.1.2 Has()

```
virtual bool Has (  
    object id) [virtual]
```

Implements [IPreviewManager](#).

#### 3.171.1.3 RemoveView()

```
virtual void RemoveView (  
    object id) [virtual]
```

Implements [IPreviewManager](#).

#### 3.171.1.4 SetBounds()

```
virtual void SetBounds (  
    object id,  
    int x,  
    int y,  
    int w,  
    int h,  
    Flip flip = default(Flip),  
    Rotation rot = Rotation::Rotate0) [virtual]
```

Positions the preview of the view / image. Not optimized to be called every frame.

call at the end of override

Implements [IPreviewManager](#).

## 3.172 PreviewManagerLayer Class Reference

Inherits [PreviewManager](#).

### Public Member Functions

- **PreviewManagerLayer** ([ILogger](#) logger)

**Public Member Functions inherited from [PreviewManager](#)**

- virtual void [AddView](#) (object id, [IVideoPreview](#) view)
- virtual void [RemoveView](#) (object id)
- virtual bool [Has](#) (object id)
- virtual void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), Rotation rot=Rotation.Rotate0)

*Positions the preview of the view / image. Not optimized to be called every frame.*

**Protected Member Functions**

- override void **Apply** ([ViewState](#) v)

**Protected Member Functions inherited from [PreviewManager](#)**

- void **Apply** ([ViewState](#) v)

**Additional Inherited Members****Protected Attributes inherited from [PreviewManager](#)**

- Dictionary< object, [ViewState](#) > **views** = new Dictionary<object, [ViewState](#)>()

## 3.173 PreviewManagerMediaPlayerElement Class Reference

Inherits [PreviewManager](#).

**Public Member Functions**

- **PreviewManagerMediaPlayerElement** ([ILogger](#) logger)
- override void [AddView](#) (object id, [IVideoPreview](#) view)
- override void [RemoveView](#) (object id)

**Public Member Functions inherited from [PreviewManager](#)**

- virtual bool [Has](#) (object id)
- virtual void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), Rotation rot=Rotation.Rotate0)

*Positions the preview of the view / image. Not optimized to be called every frame.*

**Protected Member Functions**

- override void **Apply** ([ViewState](#) v)

**Protected Member Functions inherited from [PreviewManager](#)**

- void **Apply** ([ViewState](#) v)

## Additional Inherited Members

### Protected Attributes inherited from [PreviewManager](#)

- Dictionary< object, [ViewState](#) > **views** = new Dictionary<object, [ViewState](#)>()

## 3.173.1 Member Function Documentation

### 3.173.1.1 AddView()

```
override void AddView (
    object id,
    IVideoPreview view) [virtual]
```

Reimplemented from [PreviewManager](#).

### 3.173.1.2 RemoveView()

```
override void RemoveView (
    object id) [virtual]
```

Reimplemented from [PreviewManager](#).

## 3.174 PreviewManagerScreenQuadTexture Class Reference

Inherits [IPreviewManager](#).

### Classes

- class [QuadView](#)

### Public Member Functions

- **PreviewManagerScreenQuadTexture** ([ILogger](#) logger)
- void [AddView](#) (object id, [IVideoPreview](#) view)
- void [RemoveView](#) (object id)
- bool [Has](#) (object id)
- void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), [Rotation](#) rot=[Rotation.Rotate0](#))

*Positions the preview of the view / image. Not optimized to be called every frame.*

### Protected Attributes

- Dictionary< object, [QuadView](#) > **views** = new Dictionary<object, [QuadView](#)>()

### 3.174.1 Member Function Documentation

#### 3.174.1.1 AddView()

```
void AddView (
    object id,
    IVideoPreview view)
```

Implements [IPreviewManager](#).

#### 3.174.1.2 Has()

```
bool Has (
    object id)
```

Implements [IPreviewManager](#).

#### 3.174.1.3 RemoveView()

```
void RemoveView (
    object id)
```

Implements [IPreviewManager](#).

#### 3.174.1.4 SetBounds()

```
void SetBounds (
    object id,
    int x,
    int y,
    int w,
    int h,
    Flip flip = default(Flip),
    Rotation rot = Rotation::Rotate0)
```

Positions the preview of the view / image. Not optimized to be called every frame.

Implements [IPreviewManager](#).

## 3.175 PreviewManagerUnityGUI Class Reference

Inherits [PreviewManager](#).

### Public Member Functions

- void **OnGUI** ()

### Public Member Functions inherited from [PreviewManager](#)

- virtual void [AddView](#) (object id, [IVideoPreview](#) view)
- virtual void [RemoveView](#) (object id)
- virtual bool [Has](#) (object id)
- virtual void [SetBounds](#) (object id, int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), [Rotation](#) rot=[Rotation.Rotate0](#))

*Positions the preview of the view / image. Not optimized to be called every frame.*

### Protected Member Functions

- override void [Apply](#) ([ViewState](#) v)

### Protected Member Functions inherited from [PreviewManager](#)

- void [Apply](#) ([ViewState](#) v)

### Additional Inherited Members

### Protected Attributes inherited from [PreviewManager](#)

- Dictionary< object, [ViewState](#) > **views** = new Dictionary<object, [ViewState](#)>()

## 3.176 PunVoiceClient Class Reference

This class can be used to automatically sync client states between [PUN](#) and [Voice](#). It also finds the [Speaker](#) component for a character's voice. For this to work attach a [PhotonVoiceView](#) next to the [PhotonView](#) of your player's prefab.

Inherits [VoiceFollowClient](#).

### Static Public Attributes

- const string **VoiceRoomNameSuffix** = "\_voice\_"  
*Suffix for voice room names appended to Leader room names.*

### Static Public Attributes inherited from [VoiceConnection](#)

- const int **ChannelAudio** = 1  
*Recommended [Photon](#) Transport channel for audio. Chosen not to interfere with video and default channel.*
- const int **ChannelVideo** = 2  
*Recommended [Photon](#) Transport channel for video. Chosen not to interfere with audio and default channel.*

### Protected Member Functions

- override void [Start](#) ()
- override void [OnDestroy](#) ()
- override [Speaker InstantiateSpeakerForRemoteVoice](#) (int playerId, byte voiceId, object userData)
- override string [GetVoiceRoomName](#) ()
- override bool [ConnectVoice](#) ()

## Protected Member Functions inherited from [VoiceFollowClient](#)

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- override void **OnDestroy** ()
- override void **OnOperationResponseReceived** (OperationResponse operationResponse)
- void **LeaderStateChanged** (ClientState toState)
- override void **OnVoiceStateChanged** (ClientState fromState, ClientState toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

## Protected Member Functions inherited from [VoiceConnection](#)

- override void **Awake** ()
- virtual void **Update** ()
- void **CalcStatistics** ()

## Properties

- override bool **LeaderInRoom** [get]
- override bool **LeaderOfflineMode** [get]
- static [PunVoiceClient](#) **Instance** [get]  
*Singleton instance for [PunVoiceClient](#).*
- bool **UsePunAppSettings** [get, set]  
*Whether or not to use the [Voice](#) AppId and all the other AppSettings from [PUN](#)'s PhotonServerSettings Scriptable↔ Object singleton in the [Voice](#) client/app.*
- bool **UsePunAuthValues** [get, set]  
*Whether or not to use the same PhotonNetwork.AuthValues in PunVoiceClient.Instance.Client.AuthValues. This means that the same UserID will be used in both clients. If custom authentication is used and setup in [PUN](#) app, the same configuration should be done for the [Voice](#) app.*

## Properties inherited from [VoiceFollowClient](#)

- bool **LeaderInRoom** [get]
- bool **LeaderOfflineMode** [get]

## Properties inherited from [VoiceConnection](#)

- virtual bool **AlwaysUsePrimaryRecorder** [get]
- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- new [LoadBalancingTransport](#) **Client** [get]
- [VoiceClient](#) **VoiceClient** [get]  
*Returns underlying [Photon Voice](#) client.*
- ClientState **ClientState** [get]  
*Returns [Photon Voice](#) client state.*
- float **FramesReceivedPerSecond** [get]  
*Number of frames received per second.*
- float **FramesLostPerSecond** [get]  
*Number of frames lost per second.*
- float **FramesLostPercent** [get]

*Percentage of lost frames.*

- GameObject **SpeakerPrefab** [get, set]  
*Prefab that contains [Speaker](#) component to be instantiated when receiving a new remote audio source info.*
- [Recorder](#) **PrimaryRecorder** [get, set]  
*Primary [Recorder](#) to be used by [VoiceConnection](#) implementations directly or via integration objects.*
- bool **UsePrimaryRecorder** [get]  
*Use [VoiceConnection.PrimaryRecorder](#) directly.*
- string **BestRegionSummaryInPreferences** [get, set]  
*Used to store and access the "Best Region Summary" in the Player Preferences.*

## Additional Inherited Members

## Public Member Functions inherited from [VoiceFollowClient](#)

- bool [ConnectAndJoinRoom](#) ()  
*Connect voice client to [Photon](#) servers and join a [Voice](#) room.*
- void **Disconnect** ()  
*Disconnect voice client from all [Photon](#) servers.*

## Public Member Functions inherited from [VoiceConnection](#)

- virtual bool [ConnectUsingSettings](#) (AppSettings overwriteSettings=null)  
*Connect to [Photon](#) server using [Settings](#)*
- bool [AddSpeaker](#) ([Speaker](#) speaker, object userData)  
*Tries to link local [Speaker](#) with remote voice stream using [UserData](#). Useful if [Speaker](#) created after stream is started.*
- [Speaker](#) [InstantiateSpeakerPrefab](#) (GameObject parent, bool destroyOnRemove)  
*Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.*
- bool **AddRecorder** ([Recorder](#) rec)
- void **RemoveRecorder** ([Recorder](#) rec)

## Public Attributes inherited from [VoiceFollowClient](#)

- bool **AutoConnectAndJoin** = true  
*Auto connect voice client and join a voice room when Leader client is joined to a Leader room.*

## Public Attributes inherited from [VoiceConnection](#)

- AppSettings **Settings**  
*Settings to be used by this [Voice](#) Client.*

## Events inherited from [VoiceConnection](#)

- Action< [Speaker](#) > **SpeakerLinked**  
*Fires when a speaker has been linked to a remote audio stream.*
- Action< [RemoteVoiceLink](#) > **RemoteVoiceAdded**  
*Fires when a remote voice stream is added.*



### 3.176.1 Detailed Description

This class can be used to automatically sync client states between [PUN](#) and [Voice](#). It also finds the [Speaker](#) component for a character's voice. For this to work attach a [PhotonVoiceView](#) next to the PhotonView of your player's prefab.

### 3.176.2 Member Function Documentation

#### 3.176.2.1 InstantiateSpeakerForRemoteVoice()

```
override Speaker InstantiateSpeakerForRemoteVoice (
    int playerId,
    byte voiceId,
    object userData) [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.176.2.2 OnDestroy()

```
override void OnDestroy () [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.176.2.3 Start()

```
override void Start () [protected], [virtual]
```

Reimplemented from [VoiceFollowClient](#).

## 3.177 PreviewManagerScreenQuadTexture.QuadView Class Reference

### Public Member Functions

- **QuadView** (Shader shaderScreen, [IVideoPreview](#) view)
- void **SetBounds** (int x, int y, int w, int h, [Flip](#) flip=default([Flip](#)), Rotation rot=Rotation.Rotate0)
- void **Dispose** ()

## 3.178 RawCodec Class Reference

### Classes

- class [Decoder< T >](#)
- class [Encoder< T >](#)
- class [ShortToFloat](#)

## 3.179 Recorder Class Reference

Component representing outgoing audio stream in scene.

Inherits [VoiceComponent](#).

### Public Types

- enum **InputSourceType**
- enum **MicType**

### Public Member Functions

- bool **RestartRecording** ()  
*Restarts recording if [Recorder.RecordingEnabled](#) is true.*
- void **VoiceDetectorCalibrate** (int durationMs, Action< float > detectionEndedCallback=null)  
*Trigger voice detector calibration process. While calibrating, keep silence. [Voice](#) detector sets threshold basing on measured background noise level.*
- bool **SetIosAudioSessionParameters** (IOS.AudioSessionParameters asp)  
*Sets the AudioSessionParameters for iOS audio initialization when [Photon](#) MicrophoneType is used.*
- bool **SetIosAudioSessionParameters** (IOS.AudioSessionCategory category, IOS.AudioSessionMode mode, IOS.AudioSessionCategoryOption[] options)  
*Sets the AudioSessionParameters for iOS audio initialization when [Photon](#) MicrophoneType is used.*
- bool **SetAndroidNativeMicrophoneSettings** (bool aec=false, bool agc=false, bool ns=false)  
*Sets the native Android audio input settings when the [Photon](#) microphone type is used.*
- bool **ResetLocalAudio** ()  
*Resets audio session and parameters locally to fix broken recording due to system configuration modifications or audio interruptions or audio routing changes.*

### Static Public Attributes

- const int **MIN\_OPUS\_BITRATE** = 6000
- const int **MAX\_OPUS\_BITRATE** = 510000

### Protected Member Functions

- virtual void **SendPhotonVoiceCreatedMessage** ()
- void **Update** ()

### Protected Member Functions inherited from [VoiceComponent](#)

- virtual void **Awake** ()

## Properties

- bool **TransmitEnabled** [get, set]  
*If true, audio transmission is enabled.*
- bool **Encrypt** [get, set]  
*If true, voice stream is sent encrypted.*
- bool **DebugEchoMode** [get, set]  
*If true, outgoing stream routed back to client via server same way as for remote client's streams.*
- bool **ReliableMode** [get, set]  
*If true, stream data sent in reliable mode.*
- bool **VoiceDetection** [get, set]  
*If true, voice detection enabled.*
- float **VoiceDetectionThreshold** [get, set]  
*Voice detection threshold (0..1, where 1 is full amplitude).*
- int **VoiceDetectionDelayMs** [get, set]  
*Keep detected state during this time after signal level dropped below threshold. Default is 500ms.*
- object **UserData** [get, set]  
*Custom user object to be sent in the voice stream info event.*
- Func< [IAudioDesc](#) > **InputFactory** [get, set]  
*Set the method returning new [Voice.IAudioDesc](#) instance to be assigned to a new voice created with Source set to Factory.*
- AudioUtil.IVoiceDetector **VoiceDetector** [get]  
*Returns voice activity detector for recorder's audio stream.*
- byte **InterestGroup** [get, set]  
*Target interest group that will receive transmitted audio.*
- int[] **TargetPlayers** [get, set]  
*Target players which will receive transmitted audio.*
- bool **IsCurrentlyTransmitting** [get]  
*Returns true if audio stream broadcasts.*
- AudioUtil.ILevelMeter **LevelMeter** [get]  
*Level meter utility.*
- bool **VoiceDetectorCalibrating** [get]  
*If true, voice detector calibration is in progress.*
- [ILocalVoiceAudio](#) **voiceAudio** [get]
- InputSourceType **SourceType** [get, set]  
*Audio data source.*
- MicType **MicrophoneType** [get, set]  
*Which microphone API to use when the Source is set to Microphone.*
- AudioClip **AudioClip** [get, set]  
*Source audio clip.*
- bool **LoopAudioClip** [get, set]  
*Loop playback for audio clip sources.*
- SamplingRate **SamplingRate** [get, set]  
*Outgoing audio stream sampling rate.*
- OpusCodec.FrameDuration **FrameDuration** [get, set]  
*Outgoing audio stream encoder delay.*
- int **Bitrate** [get, set]  
*Outgoing audio stream bitrate.*
- bool **RecordingEnabled** [get, set]  
*Gets or sets whether this [Recorder](#) is recording audio to be transmitted.*
- bool **StopRecordingWhenPaused** [get, set]

- *If true, stop recording when paused resume/restart when un-paused.*  
 • bool **UseOnAudioFilterRead** [get, set]
- *If true, recording will make use of [Unity](#)'s OnAudioFiltterRead callback from a muted local AudioSource.*  
 • bool **UseMicrophoneTypeFallback** [get, set]
- *If true, if recording fails to start with [Unity](#) microphone type, [Photon](#) microphone type is used -if available- as a fallback and vice versa.*  
 • bool **RecordWhenJoined** [get, set]
- *If true, recording starts when joining the room and stops when leaving the room.*  
 • [DeviceInfo](#) **MicrophoneDevice** [get, set]
- bool **AndroidMicrophoneAGC** [get]
- bool **AndroidMicrophoneAEC** [get]
- bool **AndroidMicrophoneNS** [get]

## Properties inherited from [VoiceComponent](#)

- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### 3.179.1 Detailed Description

Component representing outgoing audio stream in scene.

### 3.179.2 Member Function Documentation

#### 3.179.2.1 ResetLocalAudio()

```
bool ResetLocalAudio ()
```

Resets audio session and parameters locally to fix broken recording due to system configuration modifications or audio interruptions or audio routing changes.

#### Returns

If reset is done.

#### 3.179.2.2 SetAndroidNativeMicrophoneSettings()

```
bool SetAndroidNativeMicrophoneSettings (
    bool aec = false,
    bool agc = false,
    bool ns = false)
```

Sets the native Android audio input settings when the [Photon](#) microphone type is used.

#### Parameters

|            |                            |
|------------|----------------------------|
| <i>aec</i> | Acoustic Echo Cancellation |
| <i>agc</i> | Automatic Gain Control     |
| <i>ns</i>  | Noise Suppression          |

#### Returns

If a change has been made.

**3.179.2.3 SetIosAudioSessionParameters()** [1/2]

```
bool SetIosAudioSessionParameters (
    IOS.AudioSessionCategory category,
    IOS.AudioSessionMode mode,
    IOS.AudioSessionCategoryOption[] options)
```

Sets the AudioSessionParameters for iOS audio initialization when [Photon MicrophoneType](#) is used.

**Parameters**

|                 |                                           |
|-----------------|-------------------------------------------|
| <i>category</i> | Audio session category to be used.        |
| <i>mode</i>     | Audio session mode to be used.            |
| <i>options</i>  | Audio session category options to be used |

**Returns**

If a change has been made.

**3.179.2.4 SetIosAudioSessionParameters()** [2/2]

```
bool SetIosAudioSessionParameters (
    IOS.AudioSessionParameters asp)
```

Sets the AudioSessionParameters for iOS audio initialization when [Photon MicrophoneType](#) is used.

**Parameters**

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| <i>asp</i> | You can use custom value or one from presets, <a href="#">IOS.AudioSessionParametersPresets</a> |
|------------|-------------------------------------------------------------------------------------------------|

**Returns**

If a change has been made.

**3.179.2.5 VoiceDetectorCalibrate()**

```
void VoiceDetectorCalibrate (
    int durationMs,
    Action< float > detectionEndedCallback = null)
```

Trigger voice detector calibration process. While calibrating, keep silence. [Voice](#) detector sets threshold basing on measured background noise level.

**Parameters**

|                               |                                          |
|-------------------------------|------------------------------------------|
| <i>durationMs</i>             | Duration of calibration in milliseconds. |
| <i>detectionEndedCallback</i> | Callback when VAD calibration ends.      |

### 3.179.3 Property Documentation

#### 3.179.3.1 DebugEchoMode

```
bool DebugEchoMode [get], [set]
```

If true, outgoing stream routed back to client via server same way as for remote client's streams.

Initialized with serialized field and can be updated by Editor at runtime.

#### 3.179.3.2 Encrypt

```
bool Encrypt [get], [set]
```

If true, voice stream is sent encrypted.

Initialized with serialized field and can be updated by Editor at runtime.

#### 3.179.3.3 InterestGroup

```
byte InterestGroup [get], [set]
```

Target interest group that will receive transmitted audio.

If InterestGroup != 0, recorder's audio data is sent only to clients listening to this group. Initialized with serialized field and can be updated by Editor at runtime.

#### 3.179.3.4 ReliableMode

```
bool ReliableMode [get], [set]
```

If true, stream data sent in reliable mode.

Initialized with serialized field and can be updated by Editor at runtime.

#### 3.179.3.5 TargetPlayers

```
int [] TargetPlayers [get], [set]
```

Target players which will receive transmitted audio.

Initialized with serialized field and can be updated by Editor at runtime

#### 3.179.3.6 UseOnAudioFilterRead

```
bool UseOnAudioFilterRead [get], [set]
```

If true, recording will make use of [Unity's](#) OnAudioFiltlerRead callback from a muted local AudioSource.

If enabled, 3D sounds and voice positioning can be lost.

## 3.180 RecorderPreset Class Reference

Inherits [VoiceComponent](#).

### Classes

- struct [DSP](#)

### Public Attributes

- RuntimePlatform **Platform**
- MicType **MicrophoneType**
- bool **DSPEnabled**
- [DSP](#) **DSPSettings**

### Protected Member Functions

- override void [Awake](#) ()

### Additional Inherited Members

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## 3.180.1 Member Function Documentation

### 3.180.1.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

## 3.181 RemoteVoiceInfo Class Reference

Information about a remote voice (incoming stream).

## Properties

- [VoiceInfo](#) **Info** [get]  
*Remote voice info.*
- int **ChannelId** [get]  
*ID of channel used for transmission.*
- int **PlayerId** [get]  
*Player ID of voice owner.*
- byte **Voiceld** [get]  
*[Voice](#) ID (unique in the room).*

### 3.181.1 Detailed Description

Information about a remote voice (incoming stream).

## 3.182 RemoteVoiceLink Class Reference

### Public Member Functions

- **RemoteVoiceLink** ([VoiceInfo](#) info, int playerId, byte voiceld, int channelId, ref [RemoteVoiceOptions](#) options)
- override string **ToString** ()

### Public Attributes

- readonly [VoiceInfo](#) **VoiceInfo**
- readonly int **PlayerId**
- readonly byte **Voiceld**
- readonly int **ChannelId**

### Events

- Action< FrameOut< float > > **FloatFrameDecoded**
- Action **RemoteVoiceRemoved**

## 3.183 RemoteVoiceOptions Struct Reference

Event Actions and other options for a remote voice (incoming stream).

### Public Member Functions

- **RemoteVoiceOptions** ([ILogger](#) logger, string logPrefix, [VoiceInfo](#) voiceInfo)
- void **SetOutput** (Action< FrameOut< float > > output)  
*Create default audio decoder and register a method to be called when a data frame is decoded.*
- void **SetOutput** (Action< FrameOut< short > > output)  
*Create default audio decoder and register a method to be called when a data frame is decoded.*



## Properties

- Action **OnRemoteVoiceRemoveAction** [get, set]  
*Register a method to be called when the remote voice is removed.*
- [IDecoder](#) **Decoder** [get, set]  
*Remote voice data decoder. Use to set decoder options or override it with user decoder.*

### 3.183.1 Detailed Description

Event Actions and other options for a remote voice (incoming stream).

## 3.184 AudioUtil.Resampler< T > Class Template Reference

Sample-rate conversion Audio Processor.

Inherits [IProcessor< T >](#).

## Public Member Functions

- [Resampler](#) (int dstSize, int channels)  
*Create a new Resampler instance.*
- [T\[\] Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

## Protected Attributes

- T[] **frameResampled**

### 3.184.1 Detailed Description

Sample-rate conversion Audio Processor.

This processor converts the sample-rate of the source stream. Internally, it uses `AudioUtil.Resample<T>(T[], T[], int, int)`.

### 3.184.2 Member Function Documentation

#### 3.184.2.1 Process()

```
T[] Process (
    T[] buf)
```

Process a frame of data.

**Parameters**

|                  |                              |
|------------------|------------------------------|
| <code>buf</code> | Buffer containing input data |
|------------------|------------------------------|

**Returns**

Buffer containing output data or null if frame has been discarded (VAD)

Implements [IProcessor< T >](#).

**3.184.2.2 Resampler()**

```
Resampler (
    int dstSize,
    int channels)
```

Create a new Resampler instance.

**Parameters**

|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| <code>dstSize</code>  | Frame size of a destination frame. Determines output rate. |
| <code>channels</code> | Number of audio channels expected in both in- and output.  |

**3.185 SaveIncomingStreamToFile Class Reference**

Inherits [VoiceComponent](#).

**Protected Member Functions**

- override void [Awake](#) ()

**Additional Inherited Members****Properties inherited from [VoiceComponent](#)**

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

**3.185.1 Member Function Documentation****3.185.1.1 Awake()**

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

## 3.186 SaveOutgoingStreamToFile Class Reference

Inherits [VoiceComponent](#).

### Additional Inherited Members

### Protected Member Functions inherited from [VoiceComponent](#)

- virtual void **Awake** ()

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## 3.187 ScreenRecorder Class Reference

Inherits MonoBehaviour, and [IVideoRecorderPusher](#).

### Public Member Functions

- void **Init** (ILogger logger, [VoiceInfo](#) info)
- void **Dispose** ()

### Public Attributes

- int **encoderFPS** = 30

### Properties

- object [PlatformView](#) [get]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]
- [IEncoder](#) **Encoder** [get]
- string **Error** [get]
- [IVideoSink](#) **VideoSink** [set]
- Action< [IVideoRecorder](#) > **OnReady** [set]

### 3.187.1 Property Documentation

#### 3.187.1.1 Encoder

`IEncoder` Encoder [get]

Implements [IVideoRecorder](#).

#### 3.187.1.2 Error

`string` Error [get]

Implements [IVideoRecorder](#).

#### 3.187.1.3 Flip

`Flip` Flip [get]

Implements [IVideoPreview](#).

#### 3.187.1.4 Height

`int` Height [get]

Implements [IVideoPreview](#).

#### 3.187.1.5 PlatformView

`object` PlatformView [get]

Implements [IVideoPreview](#).

#### 3.187.1.6 Rotation

`Rotation` Rotation [get]

Implements [IVideoPreview](#).

#### 3.187.1.7 VideoSink

`IVideoSink` VideoSink [set]

Implements [IVideoRecorderPusher](#).

### 3.187.1.8 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.188 SendFrameParams Struct Reference

### Public Member Functions

- **SendFrameParams** (bool targetMe, int[] targetPlayers, byte interestGroup, bool reliable, bool encrypt)

### Properties

- bool **TargetMe** [get]
- int[] **TargetPlayers** [get]
- byte **InterestGroup** [get]
- bool **Reliable** [get]
- bool **Encrypt** [get]

## 3.189 VideoTexture.Shader3D Class Reference

### Static Public Member Functions

- static Material **MakeMaterial** (Texture tex, [Flip](#) flip)

### Properties

- static string **Name** [get]

## 3.190 VideoTexture.ShaderScreen Class Reference

### Properties

- static string **Name** [get]

## 3.191 RawCodec.ShortToFloat Class Reference

### Public Member Functions

- **ShortToFloat** (Action< FrameOut< float > > output)
- void **Output** (FrameOut< short > shortBuf)

## 3.192 Speaker Class Reference

Inherits [VoiceComponent](#).

Inherited by [SpeakerFMOD](#), and [SpeakerAudioFilterRead](#).

### Public Member Functions

- void [RestartPlayback](#) ()  
*Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.*

### Protected Member Functions

- override void [Awake](#) ()
- virtual IAudioOut< float > [CreateAudioOut](#) ()
- virtual void [OnDestroy](#) ()
- void [Update](#) ()

### Protected Attributes

- IAudioOut< float > [audioOutput](#)
- AudioOutDelayControl.PlayDelayConfig [playDelayConfig](#) = AudioOutDelayControl.PlayDelayConfig.Default
- bool [restartOnDeviceChange](#) = true

### Properties

- bool [IsPlaying](#) [get]  
*Is the speaker playing right now.*
- int [Lag](#) [get]  
*The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.*
- Action< [Speaker](#) > [OnRemoteVoiceRemoveAction](#) [get, set]  
*Register a method to be called when remote voice removed.*
- [RemoteVoiceLink](#) [RemoteVoice](#) [get]
- bool [IsLinked](#) [get]  
*Whether or not this [Speaker](#) has been linked to a remote voice stream.*
- AudioOutDelayControl.PlayDelayConfig [PlayDelayConfig](#) [get, set]  
*Gets or sets jitter buffer config.*
- int [PlayDelay](#) [get, set]  
*Gets or sets jitter buffer size in ms.*
- bool [RestartOnDeviceChange](#) [get, set]

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger [Logger](#) [get]
- [VoiceLogger](#) [VoiceLogger](#) [get]
- string [Name](#) [set]

### 3.192.1 Member Function Documentation

#### 3.192.1.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [VoiceComponent](#).

#### 3.192.1.2 RestartPlayback()

```
void RestartPlayback ()
```

Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.

##### Returns

True if playback is successfully restarted.

### 3.192.2 Property Documentation

#### 3.192.2.1 PlayDelay

```
int PlayDelay [get], [set]
```

Gets or sets jitter buffer size in ms.

The method updates PlayDelayConfig with reasonable values based on the single value provided. Use [PlayDelayConfig](#) for more precise control.

#### 3.192.2.2 PlayDelayConfig

```
AudioOutDelayControl.PlayDelayConfig PlayDelayConfig [get], [set]
```

Gets or sets jitter buffer config.

Make sure that the new value is fully initialized or built from AudioOutDelayControl.PlayDelayConfig.Default.

## 3.193 SpeakerAudioFilterRead Class Reference

Inherits [Speaker](#).

### Protected Member Functions

- override IAudioOut< float > [CreateAudioOut](#) ()

## Protected Member Functions inherited from [Speaker](#)

- override void [Awake](#) ()
- virtual void [OnDestroy](#) ()
- void [Update](#) ()

## Additional Inherited Members

## Public Member Functions inherited from [Speaker](#)

- void [RestartPlayback](#) ()  
*Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.*

## Protected Attributes inherited from [Speaker](#)

- IAudioOut< float > **audioOutput**
- AudioOutDelayControl.PlayDelayConfig **playDelayConfig** = AudioOutDelayControl.PlayDelayConfig.Default
- bool **restartOnDeviceChange** = true

## Properties inherited from [Speaker](#)

- bool **IsPlaying** [get]  
*Is the speaker playing right now.*
- int **Lag** [get]  
*The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.*
- Action< [Speaker](#) > **OnRemoteVoiceRemoveAction** [get, set]  
*Register a method to be called when remote voice removed.*
- [RemoteVoiceLink](#) **RemoteVoice** [get]
- bool **IsLinked** [get]  
*Whether or not this [Speaker](#) has been linked to a remote voice stream.*
- AudioOutDelayControl.PlayDelayConfig **PlayDelayConfig** [get, set]  
*Gets or sets jitter buffer config.*
- int **PlayDelay** [get, set]  
*Gets or sets jitter buffer size in ms.*
- bool **RestartOnDeviceChange** [get, set]

## Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## 3.193.1 Member Function Documentation

### 3.193.1.1 CreateAudioOut()

```
override IAudioOut< float > CreateAudioOut () [protected], [virtual]
```

Reimplemented from [Speaker](#).



## 3.194 SpeakerFMOD Class Reference

Inherits [Speaker](#).

### Protected Member Functions

- override IAudioOut< float > [CreateAudioOut](#) ()

### Protected Member Functions inherited from [Speaker](#)

- override void [Awake](#) ()
- virtual void **OnDestroy** ()
- void **Update** ()

### Additional Inherited Members

### Public Member Functions inherited from [Speaker](#)

- void [RestartPlayback](#) ()  
*Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.*

### Protected Attributes inherited from [Speaker](#)

- IAudioOut< float > **audioOutput**
- AudioOutDelayControl.PlayDelayConfig **playDelayConfig** = AudioOutDelayControl.PlayDelayConfig.Default
- bool **restartOnDeviceChange** = true

### Properties inherited from [Speaker](#)

- bool **IsPlaying** [get]  
*Is the speaker playing right now.*
- int **Lag** [get]  
*The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.*
- Action< [Speaker](#) > **OnRemoteVoiceRemoveAction** [get, set]  
*Register a method to be called when remote voice removed.*
- [RemoteVoiceLink](#) **RemoteVoice** [get]
- bool **IsLinked** [get]  
*Whether or not this [Speaker](#) has been linked to a remote voice stream.*
- AudioOutDelayControl.PlayDelayConfig **PlayDelayConfig** [get, set]  
*Gets or sets jitter buffer config.*
- int **PlayDelay** [get, set]  
*Gets or sets jitter buffer size in ms.*
- bool **RestartOnDeviceChange** [get, set]

## Properties inherited from [VoiceComponent](#)

- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### 3.194.1 Member Function Documentation

#### 3.194.1.1 CreateAudioOut()

```
override IAudioOut< float > CreateAudioOut () [protected], [virtual]
```

Reimplemented from [Speaker](#).

## 3.195 StreamMatchmakingSettings Class Reference

### Public Member Functions

- virtual PhotonJoinRandomRoomArgs **RandomRoomParams** ()
- virtual PhotonEnterRoomArgs **EnterRoomParams** ()

### Public Attributes

- int **maxPlayers** = 20
- List< [StringSessionProperty](#) > **matchmakingSessionProperties** = new List<[StringSessionProperty](#)>()

## 3.196 StreamTransportClient Class Reference

Basic implementation of a [IRegisterableStreamTransportManager](#), initiating a [LoadBalancingTransport](#) and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using [FusionVoiceClient](#), [UnityVoiceClient](#), ... aka the main [VoiceConnection](#) subclasses (that manage recorders, offer a deeper app life cycle management, ...)

Inherits [MonoBehaviour](#), [IStreamTransportManager](#), [IConnectionCallbacks](#), and [IMatchmakingCallbacks](#).

**Public Member Functions**

- virtual void **Disconnect** ()
- virtual void **StartConnectionWithDefaultSettings** ()
- virtual void **StartConnection** ([StreamTransportClientSettings](#) startVideoClientSettings)
- virtual void **StartConnection** (AppSettings appSettings)
- virtual AppSettings **FindAppSettings** ()  
*Find the AppSettings used to connect. Behaviour:*
- virtual AppSettings **FindAppSettings** ([StreamTransportClientSettings](#) startVideoClientSettings)  
*Find the AppSettings used to connect. Behaviour:*
- LogLevel **StreamLogLevel** ()
- virtual void **RegisterStreamSender** (IStreamSender streamSender)
- virtual void **RegisterStreamReceiver** (IStreamReceiver streamReceiver)
- virtual void **JoinRoom** ()  
*Join the room once connected to the server. The default implementation relies on random matchmaking, using EnterRoomParams parameters.*
- virtual void **OnConnectedToMaster** ()
- virtual void **OnDisconnected** (DisconnectCause cause)
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)

**Public Attributes**

- bool **autoStartWithAppSettings** = true
- bool **UseSharedAppSettings** = false
- [StreamTransportClientSettings](#) **transportClientSettings** = new [StreamTransportClientSettings](#)()
- [StreamMatchmakingSettings](#) **matchmakingSettings** = new [StreamMatchmakingSettings](#)()
- LogLevel **logLevel** = LogLevel.Info
- UnityEvent **onStartConnection** = new UnityEvent()
- bool **isRoomJoined** = false
- string **joinedRoomName** = ""
- List< [StringSessionProperty](#) > **roomCustomPropertiesPreview** = new List<[StringSessionProperty](#)>()

**Protected Member Functions**

- virtual void **Awake** ()
- virtual void **Start** ()
- virtual void **Update** ()
- virtual void **OnApplicationQuit** ()
- virtual void **OnDestroy** ()
- virtual void **Service** ()
- virtual void **NotifyJoinedRoom** ()
- virtual void **NotifyLeftRoom** ()
- virtual void **NotifyRemoteVoiceAdd** (int channelId, int playerId, byte voiceld, [VoiceInfo](#) i, ref [RemoteVoiceOptions](#) options)
- virtual void **CheckJoinStatus** ()
- virtual void **OnRemoteVoiceAdd** (int channelId, int playerId, byte voiceld, [VoiceInfo](#) i, ref [RemoteVoiceOptions](#) options)

### Protected Attributes

- [Photon.Voice.Unity.Logger](#) **\_logger**
- StreamTransport **\_client** = new StreamTransport()
- List< [IStreamSender](#) > **\_registeredStreamSenders** = new List<[IStreamSender](#)>()
- List< [IStreamReceiver](#) > **\_registeredStreamReceivers** = new List<[IStreamReceiver](#)>()

### Properties

- ConnectionProtocol **Protocol** [get]
- ConnectionProtocol **UsedProtocol** [get]
- virtual AuthenticationValues **AuthValues** [get]
- StreamTransport [Transport](#) [get]

### Properties inherited from [IStreamTransportManager](#)

- long **RoundTripTime** [get]
- long **BytesOut** [get]
- long **PacketsOut** [get]
- long **BytesIn** [get]
- long **PacketsIn** [get]

## 3.196.1 Detailed Description

Basic implementation of a IRegisterableStreamTransportManager, initiating a [LoadBalancingTransport](#) and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using FusionVoice↔ Client, UnityVoiceClient, ... aka the main VoiceConnection subclasses (that manage recorders, offer a deeper app life cycle management, ...)

## 3.196.2 Member Function Documentation

### 3.196.2.1 FindAppSettings() [1/2]

```
virtual AppSettings FindAppSettings () [virtual]
```

Find the AppSettings used to connect. Behaviour:

- if UseSharedAppSettings == true => use the global VoiceAppSettings ([PhotonAppSettings](#)). If its AppId is empty, log an error and return null.
- if UseSharedAppSettings == false => use transportClientSettings. If its appId is empty, log an error and return null.

### Returns

**3.196.2.2 FindAppSettings() [2/2]**

```
virtual AppSettings FindAppSettings (
    StreamTransportClientSettings startVideoClientSettings) [virtual]
```

Find the AppSettings used to connect. Behaviour:

- if UseSharedAppSettings == true => use the global VoiceAppSettings ([PhotonAppSettings](#)). If its AppId is empty, log an error and return null.
- if UseSharedAppSettings == false => use the local startVideoClientSettings. If its appid is empty, log an error and return null.

**3.196.2.3 RegisterStreamReceiver()**

```
virtual void RegisterStreamReceiver (
    IStreamReceiver streamReceiver) [virtual]
```

Implements [IStreamTransportManager](#).

**3.196.2.4 RegisterStreamSender()**

```
virtual void RegisterStreamSender (
    IStreamSender streamSender) [virtual]
```

Implements [IStreamTransportManager](#).

**3.196.2.5 StreamLogLevel()**

```
LogLevel StreamLogLevel ()
```

Implements [IStreamTransportManager](#).

**3.196.3 Property Documentation****3.196.3.1 Transport**

```
StreamTransport Transport [get]
```

Implements [IStreamTransportManager](#).

**3.197 StreamTransportClientSettings Class Reference****Public Attributes**

- string **appId**
- string **appVersion** = "1"
- string **region** = ""
- ConnectionProtocol **protocol** = ConnectionProtocol.Udp

## 3.198 StreamTransportManagerListener Class Reference

Base class for stream senders/receivers which rely on a [IRegisterableStreamTransportManager](#) to provide the underlying [LoadBalancingTransport](#) upon which the stream run Automatically tries to find [\\_streamTransportManager](#) [IRegisterableStreamTransportManager](#), through [StreamTransportObserver.FindStreamTransportManager](#).

Inherits [MonoBehaviour](#), and [IStreamTransportManagerListener](#).

Inherited by [AudioStreamReceiver](#), and [AudioStreamSender](#).

### Public Attributes

- [GameObject](#) **streamTransportGameObject** = null

### Protected Member Functions

- virtual void **Awake** ()
- virtual void **OnDestroy** ()
- virtual IEnumerator [FindStreamTransportManagerCoroutine](#) ()  
*Try to find a [IRegisterableStreamTransportManager](#), using [StreamTransportObserver.FindStreamTransportManager](#), that will fill the [\\_streamTransportManager](#) field if found Retries a few times (around 10s) if not found, before giving up.*
- void **OnStreamTransportManagerFound** ()  
*Called when a [\\_streamTransportManager](#) is set, and so a trnasport can be used.*

### Protected Attributes

- [IStreamTransportManager](#) **\_streamTransportManager** = null
- [Photon.Voice.Unity.Logger](#) **\_logger** = new [Photon.Voice.Unity.Logger](#)(LogLevel.Info)

### Properties

- [GameObject](#) [StreamTransportGameObject](#) [get, set]
- [IStreamTransportManager](#) [StreamTransportManager](#) [get, set]
- [StreamTransport](#) **Transport** [get]

### Properties inherited from [IStreamTransportManagerListener](#)

- [GameObject](#) **gameObject** [get]

#### 3.198.1 Detailed Description

Base class for stream senders/receivers which rely on a [IRegisterableStreamTransportManager](#) to provide the underlying [LoadBalancingTransport](#) upon which the stream run Automatically tries to find [\\_streamTransportManager](#) [IRegisterableStreamTransportManager](#), through [StreamTransportObserver.FindStreamTransportManager](#).

[OnStreamTransportManagerFound](#) is called when the [IRegisterableStreamTransportManager](#) is found

### 3.198.2 Member Function Documentation

#### 3.198.2.1 FindStreamTransportManagerCoroutine()

```
virtual IEnumerator FindStreamTransportManagerCoroutine () [protected], [virtual]
```

Try to find a [IRegisterableStreamTransportManager](#), using [StreamTransportObserver.FindStreamTransportManager](#), that will fill the `_streamTransportManager` field if found. Retries a few times (around 10s) if not found, before giving up.

Returns

### 3.198.3 Property Documentation

#### 3.198.3.1 StreamTransportGameObject

```
GameObject StreamTransportGameObject [get], [set]
```

Implements [IStreamTransportManagerListener](#).

#### 3.198.3.2 StreamTransportManager

```
IStreamTransportManager StreamTransportManager [get], [set]
```

Implements [IStreamTransportManagerListener](#).

## 3.199 StreamTransportObserver Class Reference

Wrapper handling [LoadBalancingTransport](#) callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the [IRegisterableStreamTransportManager](#) interface.

Inherits [IStreamTransportManager](#), [IDisposable](#), [IConnectionCallbacks](#), and [IMatchmakingCallbacks](#).

### Public Member Functions

- LogLevel [StreamLogLevel](#) ()
- **StreamTransportObserver** (StreamTransport transport, LogLevel logLevel=LogLevel.Info)
- void **RegisterOnTransport** ()
- void [RegisterStreamSender](#) ([IStreamSender](#) streamSender)
- void [RegisterStreamReceiver](#) ([IStreamReceiver](#) streamReceiver)
- void **Dispose** ()
- void **OnDisconnected** (DisconnectCause cause)
- void **OnConnectedToMaster** ()
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)

### Static Public Member Functions

- static IEnumerator [FindStreamTransportManager](#) ([IStreamTransportManagerListener](#) listener, bool change↔StreamTransportGameObjectWithResult=true, bool logFailure=true)

*Stream Transport Manager finder function, that will, for a given game object:*

### Public Attributes

- bool **isRoomJoined** = false

### Protected Member Functions

- virtual void **OnRemoteVoiceAdd** (int channelId, int playerId, byte voiceld, [VoiceInfo](#) voiceInfo, ref [RemoteVoiceOptions](#) options)
- void **NotifyJoinedRoom** ()
- void **NotifyLeftRoom** ()
- virtual void **NotifyRemoteVoiceAdd** (int channelId, int playerId, byte voiceld, [VoiceInfo](#) i, ref [RemoteVoiceOptions](#) options)
- virtual void **CheckJoinStatus** ()

### Protected Attributes

- List< [IStreamSender](#) > **\_registeredStreamSenders** = new List<[IStreamSender](#)>()
- List< [IStreamReceiver](#) > **\_registeredStreamReceivers** = new List<[IStreamReceiver](#)>()
- LogLevel **\_logLevel** = LogLevel.Info

### Properties

- StreamTransport [Transport](#) = null [get, set]

### Properties inherited from [IStreamTransportManager](#)

- long **RoundTripTime** [get]
- long **BytesOut** [get]
- long **PacketsOut** [get]
- long **BytesIn** [get]
- long **PacketsIn** [get]

### 3.199.1 Detailed Description

Wrapper handling [LoadBalancingTransport](#) callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the [IRegisterableStreamTransportManager](#) interface.

Provide the [StreamTransportObserver.FindStreamTransportManager](#) helper, that will, for a given game object:

1. find a [IRegisterableStreamTransportManager](#)
2. or wrap a [IStreamTransportManager](#) in its children to enhance it with callback handling
3. or, wrap a [VoiceConnection](#) (like [FusionVoiceClient](#)), if any in the children



## 3.199.2 Member Function Documentation

### 3.199.2.1 FindStreamTransportManager()

```
IEnumerator FindStreamTransportManager (
    IStreamTransportManagerListener listener,
    bool changeStreamTransportGameObjectWithResult = true,
    bool logFailure = true) [static]
```

Stream Transport Manager finder function, that will, for a given game object:

1. find a [IRegisterableStreamTransportManager](#)
2. or wrap a [IStreamTransportManager](#) in its children to enhance it with callback handling
3. or, wrap a [VoiceConnection](#) (like [FusionVoiceClient](#)), if any in the children or in the scene
4. will look for a [StreamTransportClient](#) in the scene

For the found manager, if the Transport is not yet set, wait for it to be set before returning

### 3.199.2.2 RegisterStreamReceiver()

```
void RegisterStreamReceiver (
    IStreamReceiver streamReceiver)
```

Implements [IStreamTransportManager](#).

### 3.199.2.3 RegisterStreamSender()

```
void RegisterStreamSender (
    IStreamSender streamSender)
```

Implements [IStreamTransportManager](#).

### 3.199.2.4 StreamLogLevel()

```
LogLevel StreamLogLevel ()
```

Implements [IStreamTransportManager](#).

## 3.199.3 Property Documentation

### 3.199.3.1 Transport

```
StreamTransport Transport = null [get], [set]
```

Implements [IStreamTransportManager](#).

## 3.200 ImageBufferInfo.StrideSet Struct Reference

### Public Member Functions

- **StrideSet** (int length, int s0=0, int s1=0, int s2=0, int s3=0)

### Properties

- int **this**[int key] [get, set]
- int **Length** [get]

## 3.201 StringSessionProperty Struct Reference

### Public Attributes

- string **propertyName**
- string **value**

## 3.202 AudioUtil.TempoUp< T > Class Template Reference

### Public Member Functions

- void **Begin** (int channels, int changePerc, int skipGroup)
- int **Process** (T[] s, T[] d)
- int **End** (T[] s)
- int **endFloat** (float[] s)
- int **endShort** (short[] s)

## 3.203 TestTone Class Reference

Inherits MonoBehaviour.

## 3.204 TimerWorker Class Reference

### Public Member Functions

- **TimerWorker** (Action callback, int interval)
- void **Stop** ()

### Static Public Member Functions

- static void **CallbackStatic** (int handle)

## 3.205 AudioUtil.ToneAudioPusher< T > Class Template Reference

IAudioPusher that provides a constant tone signal.

Inherits [AudioUtil.GeneratorPusher< T >](#).

### Public Member Functions

- [ToneAudioPusher](#) (int frequency=440, int bufSizeMs=100, int samplingRate=48000, int channels=1)  
*Create a new ToneAudioReader instance.*

### Public Member Functions inherited from [AudioUtil.GeneratorPusher< T >](#)

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*
- void **Dispose** ()

### Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

### Protected Member Functions inherited from [AudioUtil.GeneratorPusher< T >](#)

- int **Gen** (T[] buf, long timeSamples)

### Additional Inherited Members

### Protected Attributes inherited from [AudioUtil.GeneratorPusher< T >](#)

- long **timeSamples**

### Properties inherited from [AudioUtil.GeneratorPusher< T >](#)

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.205.1 Detailed Description

IAudioPusher that provides a constant tone signal.

### 3.205.2 Member Function Documentation

#### 3.205.2.1 ToneAudioPusher()

```
ToneAudioPusher (
    int frequency = 440,
    int bufSizeMs = 100,
    int samplingRate = 48000,
    int channels = 1)
```

Create a new ToneAudioReader instance.

## Parameters

|                     |                                            |
|---------------------|--------------------------------------------|
| <i>frequency</i>    | Frequency of the generated tone (in Hz).   |
| <i>bufSizeMs</i>    | Size of buffers to push (in milliseconds). |
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz). |
| <i>channels</i>     | Number of channels in the audio signal.    |

### 3.206 AudioUtil.ToneAudioReader< T > Class Template Reference

IAudioReader that provides a constant tone signal.

Inherits [AudioUtil.GeneratorReader< T >](#).

#### Public Member Functions

- [ToneAudioReader](#) (Func< double > clockSec=null, double frequency=440, int samplingRate=48000, int channels=1)

*Create a new ToneAudioReader instance.*

#### Public Member Functions inherited from [AudioUtil.GeneratorReader< T >](#)

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool [Read](#) (T[] buf)

*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

#### Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

#### Protected Member Functions inherited from [AudioUtil.GeneratorReader< T >](#)

- int **Gen** (T[] buf, long timeSamples)

#### Additional Inherited Members

#### Properties inherited from [AudioUtil.GeneratorReader< T >](#)

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

### 3.206.1 Detailed Description

IAudioReader that provides a constant tone signal.

Because of current resampling algorithm, the tone is distorted if SamplingRate does not equal encoder sampling rate.

### 3.206.2 Member Function Documentation

#### 3.206.2.1 ToneAudioReader()

```
ToneAudioReader (
    Func< double > clockSec = null,
    double frequency = 440,
    int samplingRate = 48000,
    int channels = 1)
```

Create a new ToneAudioReader instance.

##### Parameters

|                     |                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <i>clockSec</i>     | Function to get current time in seconds. In <a href="#">Unity</a> , pass in '()' => <code>AudioSettings.dspTime</code> for better results. |
| <i>frequency</i>    | Frequency of the generated tone (in Hz).                                                                                                   |
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz).                                                                                                 |
| <i>channels</i>     | Number of channels in the audio signal.                                                                                                    |

## 3.207 UnityAudioOut Class Reference

Inherits [AudioOutDelayControl< T >](#).

### Public Member Functions

- **UnityAudioOut** (AudioSource audioSource, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (float[] data, int offsetSamples)
- override void **Stop** ()

### Public Member Functions inherited from [AudioOutDelayControl< T >](#)

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void **Start** (int frequency, int channels, int frameSamples)
- void **Service** ()
- void **Push** (T[] frame)
- void **Flush** ()

### Protected Attributes

- readonly AudioSource **source**
- AudioClip **clip**

### Protected Attributes inherited from [AudioOutDelayControl< T >](#)

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

### Properties

- override long **OutPos** [get]

### Properties inherited from [AudioOutDelayControl< T >](#)

- long **OutPos** [get]
- int [Lag](#) [get]
- bool **IsFlushed** [get]
- bool [IsPlaying](#) [get]

### Additional Inherited Members

### Static Public Attributes inherited from [AudioOutDelayControl< T >](#)

- const int **FRAME\_POOL\_CAPACITY** = 50

## 3.207.1 Member Function Documentation

### 3.207.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from [AudioOutDelayControl< T >](#).

## 3.208 UnityLogger Class Reference

### Static Public Member Functions

- static void **Log** (LogLevel level, Object obj, string tag, string objName, string msg)
- static void **Log** (LogLevel level, Object obj, string tag, string objName, string fmt, params object[] args)

## 3.209 UnityMicrophone Class Reference

A wrapper around `UnityEngine.Microphone` to be able to safely use `Microphone` and compile for WebGL.

### Static Public Member Functions

- static void **End** (string `deviceName`)
- static void **GetDeviceCaps** (string `deviceName`, out int `minFreq`, out int `maxFreq`)
- static int **GetPosition** (string `deviceName`)
- static bool **IsRecording** (string `deviceName`)
- static AudioClip **Start** (string `deviceName`, bool `loop`, int `lengthSec`, int `frequency`)
- static string **CheckDevice** (Voice.ILogger `logger`, string `logPref`, string `device`, int `suggestedFrequency`, out int `frequency`)

### Properties

- static string[] **devices** [get]

### 3.209.1 Detailed Description

A wrapper around `UnityEngine.Microphone` to be able to safely use `Microphone` and compile for WebGL.

## 3.210 UnityVoiceClient Class Reference

Component that represents a [Voice](#) client and manages a simple [Unity](#) integration: a single [Recorder](#) and multiple remote speakers.

Inherits [VoiceConnection](#).

### Public Member Functions

- override bool [ConnectUsingSettings](#) (AppSettings `overwriteSettings`=null)  
*Connect to [Photon](#) server using [Settings](#)*

### Public Member Functions inherited from [VoiceConnection](#)

- bool [AddSpeaker](#) ([Speaker](#) `speaker`, object `userData`)  
*Tries to link local [Speaker](#) with remote voice stream using [UserData](#). Useful if [Speaker](#) created after stream is started.*
- [Speaker](#) [InstantiateSpeakerPrefab](#) (GameObject `parent`, bool `destroyOnRemove`)  
*Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.*
- bool **AddRecorder** ([Recorder](#) `rec`)
- void **RemoveRecorder** ([Recorder](#) `rec`)

## Public Attributes

- bool **UseVoiceAppSettings** = false  
Whether or not to use the [Voice](#) Appld and all the other AppSettings from [Fusion's RealtimeAppSettings Scriptable](#)↔ Object singleton in the [Voice](#) client/app.

## Public Attributes inherited from [VoiceConnection](#)

- AppSettings **Settings**  
Settings to be used by this [Voice](#) Client.

## Protected Member Functions

- virtual void **Start** ()
- override [Speaker](#) **InstantiateSpeakerForRemoteVoice** (int playerId, byte voiceId, object userData)

## Protected Member Functions inherited from [VoiceConnection](#)

- override void **Awake** ()
- virtual void **Update** ()
- virtual void **OnDestroy** ()
- virtual void **OnVoiceStateChanged** ([ClientState](#) fromState, [ClientState](#) toState)
- void **CalcStatistics** ()
- virtual void **OnOperationResponseReceived** (OperationResponse operationResponse)

## Properties

- override bool **AlwaysUsePrimaryRecorder** [get]

## Properties inherited from [VoiceConnection](#)

- virtual bool **AlwaysUsePrimaryRecorder** [get]
- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- new [LoadBalancingTransport](#) **Client** [get]
- [VoiceClient](#) **VoiceClient** [get]  
Returns underlying [Photon Voice](#) client.
- [ClientState](#) **ClientState** [get]  
Returns [Photon Voice](#) client state.
- float **FramesReceivedPerSecond** [get]  
Number of frames received per second.
- float **FramesLostPerSecond** [get]  
Number of frames lost per second.
- float **FramesLostPercent** [get]  
Percentage of lost frames.
- [GameObject](#) **SpeakerPrefab** [get, set]  
Prefab that contains [Speaker](#) component to be instantiated when receiving a new remote audio source info.
- [Recorder](#) **PrimaryRecorder** [get, set]  
Primary [Recorder](#) to be used by [VoiceConnection](#) implementations directly or via integration objects.
- bool **UsePrimaryRecorder** [get]  
Use [VoiceConnection.PrimaryRecorder](#) directly.
- string **BestRegionSummaryInPreferences** [get, set]  
Used to store and access the "Best Region Summary" in the Player Preferences.



## Additional Inherited Members

## Static Public Attributes inherited from [VoiceConnection](#)

- const int **ChannelAudio** = 1  
*Recommended [Photon](#) Transport channel for audio. Chosen not to interfere with video and default channel.*
- const int **ChannelVideo** = 2  
*Recommended [Photon](#) Transport channel for video. Chosen not to interfere with audio and default channel.*

## Events inherited from [VoiceConnection](#)

- Action< [Speaker](#) > **SpeakerLinked**  
*Fires when a speaker has been linked to a remote audio stream.*
- Action< [RemoteVoiceLink](#) > **RemoteVoiceAdded**  
*Fires when a remote voice stream is added.*

### 3.210.1 Detailed Description

Component that represents a [Voice](#) client and manages a simple [Unity](#) integration: a single [Recorder](#) and multiple remote speakers.

### 3.210.2 Member Function Documentation

#### 3.210.2.1 ConnectUsingSettings()

```
override bool ConnectUsingSettings (
    AppSettings overwriteSettings = null) [virtual]
```

Connect to [Photon](#) server using [Settings](#)

##### Parameters

|                                                         |                                                       |
|---------------------------------------------------------|-------------------------------------------------------|
| <input type="checkbox"/> <code>overwriteSettings</code> | Overwrites <a href="#">Settings</a> before connecting |
|---------------------------------------------------------|-------------------------------------------------------|

##### Returns

If true voice connection command was sent from client

Reimplemented from [VoiceConnection](#).

#### 3.210.2.2 InstantiateSpeakerForRemoteVoice()

```
override Speaker InstantiateSpeakerForRemoteVoice (
    int playerId,
    byte voiceId,
    object userData) [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

## 3.211 UnsupportedCodecException Class Reference

Exception thrown if an unsupported codec is encountered.

Inherits Exception.

### Public Member Functions

- [UnsupportedCodecException](#) (string info, [Codec](#) codec)  
Create a new [UnsupportedCodecException](#).

### 3.211.1 Detailed Description

Exception thrown if an unsupported codec is encountered.

### 3.211.2 Constructor & Destructor Documentation

#### 3.211.2.1 UnsupportedCodecException()

```
UnsupportedCodecException (
    string info,
    Codec codec)
```

Create a new [UnsupportedCodecException](#).

#### Parameters

|                    |                                       |
|--------------------|---------------------------------------|
| <code>info</code>  | The info prepending standard message. |
| <code>codec</code> | The codec actually encountered.       |

## 3.212 UnsupportedPlatformException Class Reference

Exception thrown if an unsupported platform is encountered.

Inherits Exception.

### Public Member Functions

- [UnsupportedPlatformException](#) (string subject, string platform=null)  
Create a new [UnsupportedPlatformException](#).

### 3.212.1 Detailed Description

Exception thrown if an unsupported platform is encountered.

### 3.212.2 Constructor & Destructor Documentation

#### 3.212.2.1 UnsupportedPlatformException()

```
UnsupportedPlatformException (
    string subject,
    string platform = null)
```

Create a new [UnsupportedPlatformException](#).

## Parameters

|                      |                                       |
|----------------------|---------------------------------------|
| <code>subject</code> | The info prepending standard message. |
|----------------------|---------------------------------------|

///

## Parameters

|                       |                         |
|-----------------------|-------------------------|
| <code>platform</code> | Optional platform name. |
|-----------------------|-------------------------|

## 3.213 UnsupportedSampleTypeException Class Reference

Exception thrown if an unsupported audio sample type is encountered.

Inherits Exception.

### Public Member Functions

- [UnsupportedSampleTypeException](#) (Type t)  
Create a new [UnsupportedSampleTypeException](#).

### 3.213.1 Detailed Description

Exception thrown if an unsupported audio sample type is encountered.

PhotonVoice generally supports 32-bit floating point ("float") or 16-bit signed integer ("short") audio, but it usually won't be converted automatically due to the high CPU overhead (and potential loss of precision) involved.

### 3.213.2 Constructor & Destructor Documentation

#### 3.213.2.1 UnsupportedSampleTypeException()

```
UnsupportedSampleTypeException (
    Type t)
```

Create a new [UnsupportedSampleTypeException](#).

## Parameters

|                |                                       |
|----------------|---------------------------------------|
| <code>t</code> | The sample type actually encountered. |
|----------------|---------------------------------------|

### 3.214 OpusCodec.Util Class Reference

### 3.215 Utility Class Reference

#### Static Public Member Functions

- static bool **IsAudio** (this [Codec](#) c)
- static bool **IsVideo** (this [Codec](#) c)

### 3.216 VideoCapture Class Reference

Inherits [IDisposable](#).

Inherited by [VideoCaptureLayer](#), and [VideoCaptureUnityTexture](#).

#### Public Member Functions

- virtual void **Dispose** ()

#### Protected Member Functions

- static IntPtr **Photon\_Video\_CreateCapture** (int hostID, string deviceID, bool createPreviewLayer, IntPtr previewLayer, Action< int, IntPtr, int, int > previewTextureUpdateCallback)
- static void **Photon\_Video\_Destroy\_Capture** (IntPtr handler)
- static IntPtr **Photon\_Video\_Capture\_GetPreviewLayer** (IntPtr handler)

#### Static Protected Attributes

- const string **lib\_name** = "\_\_Internal"

#### Properties

- IntPtr **Handle** [get, protected set]
- string **Error** [get, protected set]

### 3.217 VideoCaptureLayer Class Reference

Inherits [VideoCapture](#).

#### Public Member Functions

- **VideoCaptureLayer** ([ILogger](#) logger, string deviceID, bool createPreviewLayer=true, IntPtr preview↔ Layer=default(IntPtr))

## Public Member Functions inherited from [VideoCapture](#)

- virtual void **Dispose** ()

## Properties

- IntPtr **PreviewLayer** [get]

## Properties inherited from [VideoCapture](#)

- IntPtr **Handle** [get, protected set]
- string **Error** [get, protected set]

## Additional Inherited Members

## Protected Member Functions inherited from [VideoCapture](#)

- static IntPtr **Photon\_Video\_CreateCapture** (int hostID, string deviceID, bool createPreviewLayer, IntPtr previewLayer, Action< int, IntPtr, int, int > previewTextureUpdateCallback)
- static void **Photon\_Video\_Destroy\_Capture** (IntPtr handler)
- static IntPtr **Photon\_Video\_Capture\_GetPreviewLayer** (IntPtr handler)

## Static Protected Attributes inherited from [VideoCapture](#)

- const string **lib\_name** = "\_\_Internal"

## 3.218 VideoCaptureUnityTexture Class Reference

Inherits [VideoCapture](#).

## Public Member Functions

- **VideoCaptureUnityTexture** ([ILogger](#) logger, string deviceID, Action userOnPreviewTextureReady)
- override void [Dispose](#) ()

## Properties

- UnityEngine.Texture2D **PreviewTexture** [get]

## Properties inherited from [VideoCapture](#)

- IntPtr **Handle** [get, protected set]
- string **Error** [get, protected set]

## Additional Inherited Members

## Protected Member Functions inherited from [VideoCapture](#)

- static IntPtr **Photon\_Video\_CreateCapture** (int hostID, string deviceID, bool createPreviewLayer, IntPtr previewLayer, Action< int, IntPtr, int, int > previewTextureUpdateCallback)
- static void **Photon\_Video\_Destroy\_Capture** (IntPtr handler)
- static IntPtr **Photon\_Video\_Capture\_GetPreviewLayer** (IntPtr handler)

## Static Protected Attributes inherited from [VideoCapture](#)

- const string **lib\_name** = "\_\_Internal"

## 3.218.1 Member Function Documentation

### 3.218.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoCapture](#).

## 3.219 VideoDecoder Class Reference

Inherits [IDecoder](#).

Inherited by [VideoDecoderLayer](#), and [VideoDecoderUnityTexture](#).

## Public Member Functions

- **VideoDecoder** ([ILogger](#) logger)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- virtual void **Dispose** ()

## Protected Member Functions

- static IntPtr **Photon\_Video\_CreateDecoderLayer** ()
- static IntPtr **Photon\_Video\_CreateDecoderTexture** (int instanceID, Action< int, IntPtr, int, int > previewTextureUpdateCallback)
- static int **Photon\_Video\_Decoder\_Input** (IntPtr handler, IntPtr data, int len, int flags)
- static void **Photon\_Video\_Destroy\_Decoder** (IntPtr handler)
- static IntPtr **Photon\_Video\_Decoder\_GetPreviewLayer** (IntPtr handler)

## Protected Attributes

- IntPtr **handle**
- ILogger **logger**

## Properties

- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

## 3.219.1 Member Function Documentation

### 3.219.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

### 3.219.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|             |                                          |
|-------------|------------------------------------------|
| <i>info</i> | Properties of the data stream to decode. |
|-------------|------------------------------------------|

Implements [IDecoder](#).

## 3.219.2 Property Documentation

### 3.219.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.220 VideoDecoder Class Reference

Inherits [IDecoderDirect< B >](#).

### Public Member Functions

- delegate void **OutCallbackDelegate** (int instanceID, IntPtr buf, int width, int height, int bytesPerRow)
- **VideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

### Properties

- string **Error** [get]  
*If not null, the object is in invalid state.*
- Action< [ImageBufferNative](#) > **Output** [get, set]  
*Callback to call when a new decoded data buffer is available.*

### 3.220.1 Member Function Documentation

#### 3.220.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

#### 3.220.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|                      |                                          |
|----------------------|------------------------------------------|
| <a href="#">info</a> | Properties of the data stream to decode. |
|----------------------|------------------------------------------|

Implements [IDecoder](#).



## 3.220.2 Property Documentation

### 3.220.2.1 Error

`string Error` [get]

If not null, the object is in invalid state.

Implements [IDecoder](#).

### 3.220.2.2 Output

`Action<ImageBufferNative> Output` [get], [set]

Callback to call when a new decoded data buffer is available.

Implements [IDecoderDirect< B >](#).

## 3.221 VideoDecoder Class Reference

Inherits [IDecoder](#).

Inherited by [VideoDecoderMediaPlayerElement](#), and [VideoDecoderUnityTexture](#).

### Public Member Functions

- **VideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- virtual void [Open](#) ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void [Input](#) (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- virtual void **Dispose** ()

### Protected Attributes

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- MediaPlayer **mediaPlayer** = new MediaPlayer()

### Properties

- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*

### 3.221.1 Member Function Documentation

#### 3.221.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call `buf.Retain()` to prevent the caller from disposing the buffer. In this case, the callee should call `buf.Release()` when buffer is no longer needed.

Implements [IDecoder](#).

#### 3.221.1.2 Open()

```
virtual void Open (
    VoiceInfo info) [virtual]
```

Open (initialize) the decoder.

##### Parameters

|             |                                          |
|-------------|------------------------------------------|
| <i>info</i> | Properties of the data stream to decode. |
|-------------|------------------------------------------|

Implements [IDecoder](#).

Reimplemented in [VideoDecoderMediaPlayerElement](#).

### 3.221.2 Property Documentation

#### 3.221.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.222 MFTCodec.VideoDecoder Class Reference

Inherits [IDecoderDirect< B >](#).

## Public Member Functions

- delegate void **OutCallbackDelegate** (int instanceID, IntPtr buf, int width, int height, int stride, bool keyframe)
- **VideoDecoder** (ILogger logger, VoiceInfo info)
- void **Open** (VoiceInfo info)  
*Open (initialize) the decoder.*
- void **Input** (ref FrameBuffer buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

## Properties

- string **Error** [get]  
*If not null, the object is in invalid state.*
- Action< ImageBufferNative > **Output** [get, set]  
*Callback to call when a new decoded data buffer is available.*

## 3.222.1 Member Function Documentation

### 3.222.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements [IDecoder](#).

### 3.222.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|             |                                          |
|-------------|------------------------------------------|
| <b>info</b> | Properties of the data stream to decode. |
|-------------|------------------------------------------|

Implements [IDecoder](#).

### 3.222.2 Property Documentation

#### 3.222.2.1 Error

`string Error` [get]

If not null, the object is in invalid state.

Implements [IDecoder](#).

#### 3.222.2.2 Output

`Action<ImageBufferNative> Output` [get], [set]

Callback to call when a new decoded data buffer is available.

Implements [IDecoderDirect< B >](#).

## 3.223 VideoDecoderLayer Class Reference

Inherits [VideoDecoder](#).

### Public Member Functions

- **VideoDecoderLayer** ([ILogger](#) logger)

### Public Member Functions inherited from [VideoDecoder](#)

- **VideoDecoder** ([ILogger](#) logger)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- virtual void **Dispose** ()

### Properties

- IntPtr **PreviewLayer** [get, protected set]

### Properties inherited from [VideoDecoder](#)

- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

**Additional Inherited Members****Protected Member Functions inherited from [VideoDecoder](#)**

- static IntPtr **Photon\_Video\_CreateDecoderLayer** ()
- static IntPtr **Photon\_Video\_CreateDecoderTexture** (int instanceID, Action< int, IntPtr, int, int > preview↔ TextureUpdateCallback)
- static int **Photon\_Video\_Decoder\_Input** (IntPtr handler, IntPtr data, int len, int flags)
- static void **Photon\_Video\_Destroy\_Decoder** (IntPtr handler)
- static IntPtr **Photon\_Video\_Decoder\_GetPreviewLayer** (IntPtr handler)

**Protected Attributes inherited from [VideoDecoder](#)**

- IntPtr **handle**
- [ILogger](#) **logger**

**3.224 VideoDecoderMediaPlayerElement Class Reference**

Inherits [VideoDecoder](#).

**Public Member Functions**

- **VideoDecoderMediaPlayerElement** ([ILogger](#) logger, [VoiceInfo](#) info, MediaPlayerElement custom↔ Preview=null)
- override void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- override void **Dispose** ()

**Public Member Functions inherited from [VideoDecoder](#)**

- **VideoDecoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*

**Properties**

- MediaPlayerElement **PreviewMediaPlayerElement** [get]

**Properties inherited from [VideoDecoder](#)**

- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

## Additional Inherited Members

### Protected Attributes inherited from [VideoDecoder](#)

- [ILogger](#) **logger**
- [VoiceInfo](#) **info**
- MediaPlayer **mediaPlayer** = new MediaPlayer()

## 3.224.1 Member Function Documentation

### 3.224.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoDecoder](#).

### 3.224.1.2 Open()

```
override void Open (
    VoiceInfo info) [virtual]
```

Open (initialize) the decoder.

#### Parameters

|                      |                                          |
|----------------------|------------------------------------------|
| <a href="#">info</a> | Properties of the data stream to decode. |
|----------------------|------------------------------------------|

Reimplemented from [VideoDecoder](#).

## 3.225 VideoDecoderUnityTexture Class Reference

Inherits [VideoDecoder](#).

### Public Member Functions

- **VideoDecoderUnityTexture** ([ILogger](#) logger, Action userOnPreviewTextureReady)
- override void [Dispose](#) ()

### Public Member Functions inherited from [VideoDecoder](#)

- **VideoDecoder** ([ILogger](#) logger)
- void [Open](#) ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void [Input](#) (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*

### Static Protected Member Functions

- static void **nativeTextureUpdateCallback** (int instanceID, IntPtr nativeTex, int width, int height)

### Static Protected Attributes

- static int **instanceCnt**
- static Dictionary< int, [VideoDecoderUnityTexture](#) > **instancePerHandle** = new Dictionary<int, [VideoDecoderUnityTexture](#)>()

### Properties

- UnityEngine.Texture2D **PreviewTexture** [get]
- int **Width** [get, protected set]
- int **Height** [get, protected set]

### Properties inherited from [VideoDecoder](#)

- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

### Additional Inherited Members

### Protected Member Functions inherited from [VideoDecoder](#)

- static IntPtr **Photon\_Video\_CreateDecoderLayer** ()
- static IntPtr **Photon\_Video\_CreateDecoderTexture** (int instanceID, Action< int, IntPtr, int, int > preview↔TextureUpdateCallback)
- static int **Photon\_Video\_Decoder\_Input** (IntPtr handler, IntPtr data, int len, int flags)
- static void **Photon\_Video\_Destroy\_Decoder** (IntPtr handler)
- static IntPtr **Photon\_Video\_Decoder\_GetPreviewLayer** (IntPtr handler)

### Protected Attributes inherited from [VideoDecoder](#)

- IntPtr **handle**
- [ILogger](#) **logger**

## 3.225.1 Member Function Documentation

### 3.225.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoDecoder](#).

## 3.226 VideoDecoderUnityTexture Class Reference

Inherits [VideoDecoder](#).

### Public Member Functions

- **VideoDecoderUnityTexture** ([ILogger](#) logger, [VoicelInfo](#) info)
- override void [Dispose](#) ()

### Public Member Functions inherited from [VideoDecoder](#)

- **VideoDecoder** ([ILogger](#) logger, [VoicelInfo](#) info)
- virtual void [Open](#) ([VoicelInfo](#) info)  
*Open (initialize) the decoder.*
- void [Input](#) (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*

### Properties

- Texture2D **PreviewUnityTexture** [get]

### Properties inherited from [VideoDecoder](#)

- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*

### Additional Inherited Members

### Protected Attributes inherited from [VideoDecoder](#)

- [ILogger](#) logger
- [VoicelInfo](#) info
- MediaPlayer **mediaPlayer** = new MediaPlayer()

## 3.226.1 Member Function Documentation

### 3.226.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoDecoder](#).



## 3.227 VideoEncoder Class Reference

Inherits [IEncoder](#).

Inherited by [VideoEncoderFromCapture](#), and [VideoEncoderUnityTextureInput](#).

### Public Member Functions

- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*
- virtual void **Dispose** ()

### Static Public Attributes

- static Dictionary< [Codec](#), int > [CodecCodes](#)

### Protected Member Functions

- static IntPtr **Photon\_Video\_CreateEncoder** (int instanceID, IntPtr captureHandle, Action< int, IntPtr, int, int > pushCallback, int codec, int bitrate, int width, int height, int fps, int keyFrameInt)
- static void **Photon\_Video\_Destroy\_Encoder** (IntPtr handler)
- static void **Photon\_Video\_Encoder\_GetInfo** (IntPtr handler, out int codec, out int bitrate, out int width, out int height)
- static void **Photon\_Video\_Encoder\_SetPendingSource** (IntPtr handler, IntPtr mtlTex)
- static IntPtr **Photon\_Video\_GetRenderEventFunc** ()
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, [VideoCapture](#) capture)

### Static Protected Member Functions

- static void **nativePushCallback** (int instanceID, IntPtr buf, int len, int flags)

### Protected Attributes

- IntPtr **handle**
- int **instanceID**

### Static Protected Attributes

- static int **instanceCnt**
- static Dictionary< int, [VideoEncoder](#) > **instancePerHandle** = new Dictionary<int, [VideoEncoder](#)>()

## Properties

- int **Width** [get, protected set]
- int **Height** [get, protected set]
- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

## 3.227.1 Member Function Documentation

### 3.227.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.227.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.227.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

## Type Constraints

**I : class**

## 3.227.2 Member Data Documentation

### 3.227.2.1 CodecCodes

```
Dictionary<Codec, int> CodecCodes [static]
```

#### Initial value:

```
= new Dictionary<Codec, int>()
{
    {Codec.VideoH264, fourCharCode('a', 'v', 'c', '1')},
}
```

### 3.227.3 Property Documentation

#### 3.227.3.1 Error

`string Error [get], [protected set]`

If not null, the object is in invalid state.

Implements [IEncoder](#).

#### 3.227.3.2 Output

`Action<ArraySegment<byte>, FrameFlags> Output [set]`

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.228 VideoEncoder Class Reference

Inherits [IEncoderDirectImage](#).

Inherited by [VideoEncoderUnityTextureInput](#).

### Public Member Functions

- delegate void **OutCallbackDelegate** (int instanceID, IntPtr buf, int size, int flags)
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- `ArraySegment< byte > DequeueOutput (out FrameFlags flags)`
- *Returns next encoded data frame (if such output supported).*
- void **Input** ([ImageBufferNative](#) imBuf)
- void [EndOfStream](#) ()
- *Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`
- *Returns a platform-specific interface.*
- virtual void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect< B >](#)

- void [Input](#) (B buf)
- *Consumes the given raw data.*

### Protected Member Functions

- static void **Photon\_Video\_Encoder\_SetPendingSource** (IntPtr handle, IntPtr mtlTex)
- static IntPtr **Photon\_Video\_GetRenderEventFunc** ()

## Protected Attributes

- IntPtr **handle**
- ILogger **logger**

## Static Protected Attributes

- const string **lib\_name** = "Video"

## Properties

- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- ImageFormat **ImageFormat** [get]  
*Recommended encoder input image format. Encoder may support other formats.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.228.1 Member Function Documentation

### 3.228.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.228.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.228.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

## Type Constraints

**I : class**

## 3.228.2 Property Documentation

### 3.228.2.1 Error

`string Error [get], [protected set]`

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.228.2.2 ImageFormat

`ImageFormat ImageFormat [get]`

Recommended encoder input image format. Encoder may support other formats.

Implements [IEncoderDirectImage](#).

### 3.228.2.3 Output

`Action<ArraySegment<byte>, FrameFlags> Output [get], [set]`

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.229 VideoEncoder Class Reference

Inherits [IEncoder](#).

Inherited by [VideoEncoderMediaPlayerElement](#), and [VideoEncoderUnityTexture](#).

### Public Member Functions

- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId)
- `ArraySegment< byte > DequeueOutput` (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`  
*Returns an platform-specific interface.*
- virtual void **Dispose** ()

### Protected Attributes

- [ILogger](#) **logger**
- MediaPlayer **mediaPlayer** = new MediaPlayer()

## Properties

- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*
- bool **ErrorAccess** [get]
- int **Width** [get]
- int **Height** [get]
- PlatformAPI.IMediaCapture **UWPVideoEncoderInstance** [get]
- PlatformAPI.IMixedRealityCapture **MixedRealityCaptureVideoEncoderInstance** [get]

## 3.229.1 Member Function Documentation

### 3.229.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.229.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.229.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

## Type Constraints

**I : class**

## 3.229.2 Property Documentation

### 3.229.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.229.2.2 Output

Action<ArraySegment<byte>, FrameFlags> Output [set]

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.230 MFTCodec.VideoEncoder Class Reference

Inherits [IEncoderDirectImage](#).

### Public Member Functions

- delegate void **OutCallbackDelegate** (int instanceID, IntPtr buf, int size, bool keyframe)
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **Input** ([ImageBufferNative](#) imBuf)
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*
- void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect](#)< B >

- void [Input](#) (B buf)  
*Consumes the given raw data.*

### Properties

- string [Error](#) [get]  
*If not null, the object is in invalid state.*
- ImageFormat [ImageFormat](#) [get]  
*Recommended encoder input image format. Encoder may support other formats.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.230.1 Member Function Documentation

### 3.230.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.230.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.230.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

#### Type Constraints

***I : class***

## 3.230.2 Property Documentation

### 3.230.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.230.2.2 ImageFormat

```
ImageFormat ImageFormat [get]
```

Recommended encoder input image format. Encoder may support other formats.

Implements [IEncoderDirectImage](#).

### 3.230.2.3 Output

```
Action<ArraySegment<byte>, FrameFlags> Output [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).



## 3.231 VideoEncoderFromCapture Class Reference

Inherits [VideoEncoder](#).

### Public Member Functions

- **VideoEncoderFromCapture** ([ILogger](#) logger, [VoiceInfo](#) info, [VideoCapture](#) capture)

### Public Member Functions inherited from [VideoEncoder](#)

- `ArraySegment< byte > DequeueOutput (out FrameFlags flags)`  
*Returns next encoded data frame (if such output supported).*
- `void EndOfStream ()`  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`  
*Returns an platform-specific interface.*
- `virtual void Dispose ()`

### Properties

- [VideoCapture](#) **Capture** [get]

### Properties inherited from [VideoEncoder](#)

- `int Width [get, protected set]`
- `int Height [get, protected set]`
- `Action< ArraySegment< byte >, FrameFlags > Output [set]`  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- `string Error [get, protected set]`  
*If not null, the object is in invalid state.*

### Additional Inherited Members

### Static Public Attributes inherited from [VideoEncoder](#)

- `static Dictionary< Codec, int > CodecCodes`

### Protected Member Functions inherited from [VideoEncoder](#)

- `static IntPtr Photon_Video_CreateEncoder (int instanceID, IntPtr captureHandle, Action< int, IntPtr, int, int > pushCallback, int codec, int bitrate, int width, int height, int fps, int keyFrameInt)`
- `static void Photon_Video_Destroy_Encoder (IntPtr handler)`
- `static void Photon_Video_Encoder_GetInfo (IntPtr handler, out int codec, out int bitrate, out int width, out int height)`
- `static void Photon_Video_Encoder_SetPendingSource (IntPtr handler, IntPtr mtlTex)`
- `static IntPtr Photon_Video_GetRenderEventFunc ()`
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, [VideoCapture](#) capture)

### Static Protected Member Functions inherited from [VideoEncoder](#)

- static void **nativePushCallback** (int instanceID, IntPtr buf, int len, int flags)

### Protected Attributes inherited from [VideoEncoder](#)

- IntPtr **handle**
- int **instanceID**

### Static Protected Attributes inherited from [VideoEncoder](#)

- static int **instanceCnt**
- static Dictionary< int, [VideoEncoder](#) > **instancePerHandle** = new Dictionary<int, [VideoEncoder](#)>()

## 3.232 VideoEncoderMediaPlayerElement Class Reference

Inherits [VideoEncoder](#).

### Public Member Functions

- **VideoEncoderMediaPlayerElement** (ILogger logger, VoiceInfo info, string deviceID, MediaPlayerElement customPreview=null)
- override void **Dispose** ()

### Public Member Functions inherited from [VideoEncoder](#)

- **VideoEncoder** (ILogger logger, VoiceInfo info, string deviceID)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I **GetPlatformAPI**< I > ()  
*Returns a platform-specific interface.*

### Properties

- MediaPlayerElement **PreviewMediaPlayerElement** [get]

### Properties inherited from [VideoEncoder](#)

- Action< ArraySegment< byte >, FrameFlags > **Output** [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- bool **ErrorAccess** [get]
- int **Width** [get]
- int **Height** [get]
- PlatformAPI.IMediaCapture **UWPVideoEncoderInstance** [get]
- PlatformAPI.IMixedRealityCapture **MixedRealityCaptureVideoEncoderInstance** [get]

## Additional Inherited Members

### Protected Attributes inherited from [VideoEncoder](#)

- [ILogger](#) **logger**
- MediaPlayer **mediaPlayer** = new MediaPlayer()

## 3.232.1 Member Function Documentation

### 3.232.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoEncoder](#).

## 3.233 VideoEncoderUnityTexture Class Reference

Inherits [VideoEncoder](#).

### Public Member Functions

- **VideoEncoderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId)
- override void [Dispose](#) ()

### Public Member Functions inherited from [VideoEncoder](#)

- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- [I GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*

### Properties

- Texture2D **PreviewUnityTexture** [get]

### Properties inherited from [VideoEncoder](#)

- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*
- bool **ErrorAccess** [get]
- int **Width** [get]
- int **Height** [get]
- PlatformAPI.IMediaCapture **UWPVideoEncoderInstance** [get]
- PlatformAPI.IMixedRealityCapture **MixedRealityCaptureVideoEncoderInstance** [get]

## Additional Inherited Members

### Protected Attributes inherited from [VideoEncoder](#)

- [ILogger](#) **logger**
- MediaPlayer **mediaPlayer** = new MediaPlayer()

## 3.233.1 Member Function Documentation

### 3.233.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoEncoder](#).

## 3.234 VideoEncoderUnityTextureInput Class Reference

Inherits [VideoEncoder](#).

### Public Member Functions

- **VideoEncoderUnityTextureInput** ([ILogger](#) logger, [VoiceInfo](#) info)
- int **EncodeFrame** ()
- override void [Dispose](#) ()

### Public Member Functions inherited from [VideoEncoder](#)

- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I [GetPlatformAPI](#)< I > ()  
*Returns an platform-specific interface.*

### Properties

- UnityEngine.RenderTexture **InputTexture** [get]

### Properties inherited from [VideoEncoder](#)

- int **Width** [get, protected set]
- int **Height** [get, protected set]
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*
- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*

## Additional Inherited Members

### Static Public Attributes inherited from [VideoEncoder](#)

- static Dictionary< [Codec](#), int > [CodecCodes](#)

### Protected Member Functions inherited from [VideoEncoder](#)

- static IntPtr **Photon\_Video\_CreateEncoder** (int instanceID, IntPtr captureHandle, Action< int, IntPtr, int, int > pushCallback, int codec, int bitrate, int width, int height, int fps, int keyFrameInt)
- static void **Photon\_Video\_Destroy\_Encoder** (IntPtr handler)
- static void **Photon\_Video\_Encoder\_GetInfo** (IntPtr handler, out int codec, out int bitrate, out int width, out int height)
- static void **Photon\_Video\_Encoder\_SetPendingSource** (IntPtr handler, IntPtr mtlTex)
- static IntPtr **Photon\_Video\_GetRenderEventFunc** ()
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, [VideoCapture](#) capture)

### Static Protected Member Functions inherited from [VideoEncoder](#)

- static void **nativePushCallback** (int instanceID, IntPtr buf, int len, int flags)

### Protected Attributes inherited from [VideoEncoder](#)

- IntPtr **handle**
- int **instanceID**

### Static Protected Attributes inherited from [VideoEncoder](#)

- static int **instanceCnt**
- static Dictionary< int, [VideoEncoder](#) > **instancePerHandle** = new Dictionary<int, [VideoEncoder](#)>()

## 3.234.1 Member Function Documentation

### 3.234.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoEncoder](#).

## 3.235 VideoEncoderUnityTextureInput Class Reference

Inherits [VideoEncoder](#).

## Public Member Functions

- **VideoEncoderUnityTextureInput** ([ILogger](#) logger, [VoiceInfo](#) info)
- int **EncodeFrame** ()
- override void [Dispose](#) ()

## Public Member Functions inherited from [VideoEncoder](#)

- delegate void **OutCallbackDelegate** (int instanceID, IntPtr buf, int size, int flags)
- **VideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info)
- ArraySegment< byte > [DequeueOutput](#) (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **Input** ([ImageBufferNative](#) imBuf)
- void [EndOfStream](#) ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- [I GetPlatformAPI](#)< [I](#) > ()  
*Returns an platform-specific interface.*

## Public Member Functions inherited from [IEncoderDirect](#)< [B](#) >

- void [Input](#) ([B](#) buf)  
*Consumes the given raw data.*

## Properties

- UnityEngine.RenderTexture **InputTexture** [get]

## Properties inherited from [VideoEncoder](#)

- string [Error](#) [get, protected set]  
*If not null, the object is in invalid state.*
- ImageFormat [ImageFormat](#) [get]  
*Recommended encoder input image format. Encoder may support other formats.*
- Action< ArraySegment< byte >, FrameFlags > [Output](#) [get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## Additional Inherited Members

## Protected Member Functions inherited from [VideoEncoder](#)

- static void **Photon\_Video\_Encoder\_SetPendingSource** (IntPtr handle, IntPtr mtlTex)
- static IntPtr **Photon\_Video\_GetRenderEventFunc** ()

## Protected Attributes inherited from [VideoEncoder](#)

- IntPtr **handle**
- [ILogger](#) **logger**

## Static Protected Attributes inherited from [VideoEncoder](#)

- const string **lib\_name** = "Video"

### 3.235.1 Member Function Documentation

#### 3.235.1.1 Dispose()

```
override void Dispose () [virtual]
```

Reimplemented from [VideoEncoder](#).

## 3.236 VideoInEnumerator Class Reference

Enumerates cameras available on device.

Inherits [DeviceEnumeratorBase](#).

### Public Member Functions

- **VideoInEnumerator** ([ILogger](#) logger)
- override void [Refresh](#) ()  
*Refreshes the microphones list.*
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerator< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Protected Member Functions

- static IntPtr **Photon\_Video\_Create\_CameraEnumerator** ()
- static void **Photon\_Video\_Destroy\_CameraEnumerator** (IntPtr handler)
- static int **Photon\_Video\_CameraEnumerator\_GetCount** (IntPtr handler)
- static string **Photon\_Video\_CameraEnumerator\_GetID** (IntPtr handler, int idx)
- static string **Photon\_Video\_CameraEnumerator\_GetName** (IntPtr handler, int idx)
- static int **Photon\_Video\_CameraEnumerator\_GetFacing** (IntPtr handler, int idx)

### Properties

- override bool [IsSupported](#) [get]

## Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## Additional Inherited Members

## Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### 3.236.1 Detailed Description

Enumerates cameras available on device.

### 3.236.2 Member Function Documentation

#### 3.236.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements [IDeviceEnumerator](#).

### 3.236.3 Property Documentation

#### 3.236.3.1 IsSupported

```
override bool IsSupported [get]
```

Implements [IDeviceEnumerator](#).

## 3.237 VideoInEnumerator Class Reference

Inherits [DeviceEnumeratorBase](#).

## Public Member Functions

- **VideoInEnumerator** ([ILogger](#) logger)
- override void [Refresh](#) ()
- override void **Dispose** ()



## Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- [IEnumerator](#)< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

## Properties

- override string [Error](#) [get]

## Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## Additional Inherited Members

## Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) logger

### 3.237.1 Member Function Documentation

#### 3.237.1.1 Refresh()

```
override void Refresh ()
```

Implements [IDeviceEnumerator](#).

### 3.237.2 Property Documentation

#### 3.237.2.1 Error

```
override string Error [get]
```

Implements [IDeviceEnumerator](#).

## 3.238 VideoInEnumerator Class Reference

Inherits [DeviceEnumerator](#).

### Public Member Functions

- **VideoInEnumerator** ([ILogger](#) logger)

### Public Member Functions inherited from [DeviceEnumerator](#)

- **DeviceEnumerator** ([ILogger](#) logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void **Refresh** ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerator< [DeviceInfo](#) > **GetEnumerator** ()
- void **Refresh** ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool **IsSupported** [get]
- virtual string **Error** [get, protected set]
- Action **OnReady** [protected get, set]

## 3.239 VideoPlayer Class Reference

Inherits [IVideoPlayer](#).

### Public Member Functions

- **VideoPlayer** ([IDecoder](#) decoder, object platformView, int width, int height, Action< [IVideoPlayer](#) > onReady)
- void **Dispose** ()

### Properties

- [IDecoder](#) **Decoder** [get]
- object [PlatformView](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- int [Width](#) [get]
- int [Height](#) [get]

### 3.239.1 Property Documentation

#### 3.239.1.1 Decoder

`IDecoder` Decoder [get]

Implements [IVideoPlayer](#).

#### 3.239.1.2 Flip

`Flip` Flip [get]

Implements [IVideoPreview](#).

#### 3.239.1.3 Height

`int` Height [get]

Implements [IVideoPreview](#).

#### 3.239.1.4 PlatformView

`object` PlatformView [get]

Implements [IVideoPreview](#).

#### 3.239.1.5 Rotation

`Rotation` Rotation [get]

Implements [IVideoPreview](#).

#### 3.239.1.6 Width

`int` Width [get]

Implements [IVideoPreview](#).

## 3.240 VideoPlayerUnity Class Reference

Inherits [IVideoPlayer](#).

## Public Member Functions

- **VideoPlayerUnity** ([IDecoder](#) d, Action< [IVideoPlayer](#) > onReady, GameObject gameObject=null)
- void **Draw** ([ImageBufferNative](#) b)
- void **Dispose** ()

## Properties

- [IDecoder](#) **Decoder** [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]

## Events

- Action< Texture2D > **OnTextureUpdated**

## 3.240.1 Property Documentation

### 3.240.1.1 Decoder

[IDecoder](#) **Decoder** [get]

Implements [IVideoPlayer](#).

### 3.240.1.2 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).

### 3.240.1.3 Height

int **Height** [get]

Implements [IVideoPreview](#).

### 3.240.1.4 PlatformView

object **PlatformView** [get]

Implements [IVideoPreview](#).

### 3.240.1.5 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.240.1.6 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.241 VideoPlayerUnityMB Class Reference

Inherits MonoBehaviour.

### Public Member Functions

- void **Draw** ([ImageBufferNative](#) b)

### Properties

- object **PlatformTexture** [get]
- int **Width** [get]
- int **Height** [get]

### Events

- Action< Texture2D > **OnTextureUpdated**

## 3.242 VideoPlayerUnityTexture Class Reference

Inherits [IVideoPlayer](#).

### Public Member Functions

- **VideoPlayerUnityTexture** ([ILogger](#) logger, [VoicelInfo](#) info, Action< [IVideoPlayer](#) > onReady)
- void **Dispose** ()

### Properties

- [IDecoder](#) **Decoder** [get, protected set]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- object [PlatformView](#) [get]
- int **Width** [get]
- int **Height** [get]

### 3.242.1 Property Documentation

#### 3.242.1.1 Decoder

`IDecoder` `Decoder` [get], [protected set]

Implements [IVideoPlayer](#).

#### 3.242.1.2 Flip

`Flip` `Flip` [get]

Implements [IVideoPreview](#).

#### 3.242.1.3 Height

`int` `Height` [get]

Implements [IVideoPreview](#).

#### 3.242.1.4 PlatformView

`object` `PlatformView` [get]

Implements [IVideoPreview](#).

#### 3.242.1.5 Rotation

`Rotation` `Rotation` [get]

Implements [IVideoPreview](#).

#### 3.242.1.6 Width

`int` `Width` [get]

Implements [IVideoPreview](#).

### 3.243 VideoPlayerUnityTexture Class Reference

Inherits [IVideoPlayer](#).

## Public Member Functions

- **VideoPlayerUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, Action< [IVideoPlayer](#) > onReady)
- void **Dispose** ()

## Properties

- [IDecoder](#) **Decoder** [get, protected set]
- object [PlatformView](#) [get, protected set]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- int [Width](#) [get, protected set]
- int [Height](#) [get, protected set]

## 3.243.1 Property Documentation

### 3.243.1.1 Decoder

[IDecoder](#) **Decoder** [get], [protected set]

Implements [IVideoPlayer](#).

### 3.243.1.2 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).

### 3.243.1.3 Height

int **Height** [get], [protected set]

Implements [IVideoPreview](#).

### 3.243.1.4 PlatformView

object **PlatformView** [get], [protected set]

Implements [IVideoPreview](#).

### 3.243.1.5 Rotation

Rotation **Rotation** [get]

Implements [IVideoPreview](#).

### 3.243.1.6 Width

`int Width [get], [protected set]`

Implements [IVideoPreview](#).

## 3.244 VideoRecorderLayer Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **VideoRecorderLayer** ([ILogger](#) logger, [VoicelInfo](#) info, string deviceId, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

### Properties

- [IEncoder](#) **Encoder** [get]
- object [PlatformView](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

## 3.244.1 Property Documentation

### 3.244.1.1 Encoder

[IEncoder](#) **Encoder** [get]

Implements [IVideoRecorder](#).

### 3.244.1.2 Error

string **Error** [get]

Implements [IVideoRecorder](#).

### 3.244.1.3 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).



#### 3.244.1.4 Height

int Height [get]

Implements [IVideoPreview](#).

#### 3.244.1.5 PlatformView

object PlatformView [get]

Implements [IVideoPreview](#).

#### 3.244.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

#### 3.244.1.7 Width

int Width [get]

Implements [IVideoPreview](#).

### 3.245 VideoRecorderMediaPlayerElement Class Reference

Inherits [IVideoRecorder](#).

#### Public Member Functions

- **VideoRecorderMediaPlayerElement** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

#### Properties

- [IEncoder](#) **Encoder** [get, protected set]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

### 3.245.1 Property Documentation

#### 3.245.1.1 Encoder

`IEncoder` `Encoder` [get], [protected set]

Implements [IVideoRecorder](#).

#### 3.245.1.2 Error

`string` `Error` [get]

Implements [IVideoRecorder](#).

#### 3.245.1.3 Flip

`Flip` `Flip` [get]

Implements [IVideoPreview](#).

#### 3.245.1.4 Height

`int` `Height` [get]

Implements [IVideoPreview](#).

#### 3.245.1.5 PlatformView

`object` `PlatformView` [get]

Implements [IVideoPreview](#).

#### 3.245.1.6 Rotation

`Rotation` `Rotation` [get]

Implements [IVideoPreview](#).

#### 3.245.1.7 Width

`int` `Width` [get]

Implements [IVideoPreview](#).

## 3.246 VideoRecorderUnity Class Reference

Inherits [IVideoRecorderPusher](#).

### Public Member Functions

- **VideoRecorderUnity** ([IEncoderDirectImage](#) encoder, GameObject gameObject\_not\_used, string device←Name, int width, int height, int encoderFPS, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

### Properties

- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) [Flip](#) [get]
- [IEncoder](#) [Encoder](#) [get]
- string [Error](#) [get]
- [IVideoSink](#) [VideoSink](#) [set]
- int [MemLoop](#) [get, set]

### 3.246.1 Property Documentation

#### 3.246.1.1 Encoder

[IEncoder](#) [Encoder](#) [get]

Implements [IVideoRecorder](#).

#### 3.246.1.2 Error

string [Error](#) [get]

Implements [IVideoRecorder](#).

#### 3.246.1.3 Flip

[Flip](#) [Flip](#) [get]

Implements [IVideoPreview](#).

#### 3.246.1.4 Height

int [Height](#) [get]

Implements [IVideoPreview](#).

### 3.246.1.5 PlatformView

object PlatformView [get]

Implements [IVideoPreview](#).

### 3.246.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.246.1.7 VideoSink

[IVideoSink](#) VideoSink [set]

Implements [IVideoRecorderPusher](#).

### 3.246.1.8 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.247 VideoRecorderUnityMB Class Reference

Inherits MonoBehaviour.

### Public Member Functions

- void **Init** (string deviceName, int encoderFPS, ImageFormat imageFormat, TextureFormat textureFormat)
- void **AddRecorder** ([VideoRecorderUnity](#) videoRecorder, int width, int height, Action onReady, Action< string > onError)
- void **SetSink** ([VideoRecorderUnity](#) r, [IVideoSink](#) sink)
- void **RemoveRecorder** ([VideoRecorderUnity](#) r)
- void **StartRecord** (int width, int height, int encoderFPS)
- IEnumerator **StartRecordCoroutine** ()

### Public Attributes

- string **Device**
- int **StreamCount**

### Protected Member Functions

- virtual void **Update** ()

### Properties

- object **PlatformTexture** [get]
- int **Width** [get]
- int **Height** [get]
- int **MemLoop** [get, set]

## 3.248 VideoRecorderUnityTexture Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **VideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

### Properties

- [IEncoder](#) **Encoder** [get, protected set]
- int **Width** [get]
- int **Height** [get]
- Rotation **Rotation** [get]
- [Flip](#) **Flip** [get]
- object **PlatformView** [get]
- string **Error** [get]

## 3.248.1 Property Documentation

### 3.248.1.1 Encoder

[IEncoder](#) **Encoder** [get], [protected set]

Implements [IVideoRecorder](#).

### 3.248.1.2 Error

string **Error** [get]

Implements [IVideoRecorder](#).

### 3.248.1.3 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).

### 3.248.1.4 Height

`int Height [get]`

Implements [IVideoPreview](#).

### 3.248.1.5 PlatformView

`object PlatformView [get]`

Implements [IVideoPreview](#).

### 3.248.1.6 Rotation

`Rotation Rotation [get]`

Implements [IVideoPreview](#).

### 3.248.1.7 Width

`int Width [get]`

Implements [IVideoPreview](#).

## 3.249 VideoRecorderUnityTexture Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **VideoRecorderUnityTexture** ([ILogger](#) logger, [VoicelInfo](#) info, string deviceId, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

### Properties

- [IEncoder Encoder](#) [get, protected set]
- Rotation [Rotation](#) [get]
- [Flip Flip](#) [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- string [Error](#) [get]

## 3.249.1 Property Documentation

### 3.249.1.1 Encoder

`IEncoder` Encoder [get], [protected set]

Implements [IVideoRecorder](#).

### 3.249.1.2 Error

`string` Error [get]

Implements [IVideoRecorder](#).

### 3.249.1.3 Flip

`Flip` Flip [get]

Implements [IVideoPreview](#).

### 3.249.1.4 Height

`int` Height [get]

Implements [IVideoPreview](#).

### 3.249.1.5 PlatformView

`object` PlatformView [get]

Implements [IVideoPreview](#).

### 3.249.1.6 Rotation

`Rotation` Rotation [get]

Implements [IVideoPreview](#).

### 3.249.1.7 Width

`int` Width [get]

Implements [IVideoPreview](#).

## 3.250 VideoRecorderUnityTextureInput Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **VideoRecorderUnityTextureInput** ([ILogger](#) logger, [VoiceInfo](#) info, Action< [IVideoRecorder](#) > onReady)
- int **EncodeFrame** ()
- void **Dispose** ()

### Properties

- [IEncoder Encoder](#) [get, protected set]
- UnityEngine.RenderTexture **InputTexture** [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip Flip](#) [get]
- object [PlatformView](#) [get]
- string [Error](#) [get]

### 3.250.1 Property Documentation

#### 3.250.1.1 Encoder

[IEncoder Encoder](#) [get], [protected set]

Implements [IVideoRecorder](#).

#### 3.250.1.2 Error

string [Error](#) [get]

Implements [IVideoRecorder](#).

#### 3.250.1.3 Flip

[Flip Flip](#) [get]

Implements [IVideoPreview](#).

#### 3.250.1.4 Height

int [Height](#) [get]

Implements [IVideoPreview](#).



### 3.250.1.5 PlatformView

object PlatformView [get]

Implements [IVideoPreview](#).

### 3.250.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.250.1.7 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.251 VideoRecorderUnityTextureInput Class Reference

Inherits [IVideoRecorder](#).

### Public Member Functions

- **VideoRecorderUnityTextureInput** ([ILogger](#) logger, [VoiceInfo](#) info, Action< [IVideoRecorder](#) > onReady)
- int **EncodeFrame** ()
- void **Dispose** ()

### Properties

- [IEncoder Encoder](#) [get, protected set]
- UnityEngine.RenderTexture **InputTexture** [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip Flip](#) [get]
- object [PlatformView](#) [get]
- string [Error](#) [get]

## 3.251.1 Property Documentation

### 3.251.1.1 Encoder

[IEncoder Encoder](#) [get], [protected set]

Implements [IVideoRecorder](#).

### 3.251.1.2 Error

`string Error [get]`

Implements [IVideoRecorder](#).

### 3.251.1.3 Flip

`Flip Flip [get]`

Implements [IVideoPreview](#).

### 3.251.1.4 Height

`int Height [get]`

Implements [IVideoPreview](#).

### 3.251.1.5 PlatformView

`object PlatformView [get]`

Implements [IVideoPreview](#).

### 3.251.1.6 Rotation

`Rotation Rotation [get]`

Implements [IVideoPreview](#).

### 3.251.1.7 Width

`int Width [get]`

Implements [IVideoPreview](#).

## 3.252 VideoSinkArray Class Reference

Inherits [IVideoSink](#).

### Public Member Functions

- void **Add** ([IVideoSink](#) s)
- void **Remove** ([IVideoSink](#) s)
- void **Set** (object id, [IVideoSink](#) s)
- void **PushImage** ([ImageBufferNative](#) buf)
- void **PushImageAsync** ([ImageBufferNative](#) buf)

## Properties

- int [PushImageQueueCount](#) [get]

## 3.252.1 Member Function Documentation

### 3.252.1.1 PushImage()

```
void PushImage (
    ImageBufferNative buf)
```

Implements [IVideoSink](#).

### 3.252.1.2 PushImageAsync()

```
void PushImageAsync (
    ImageBufferNative buf)
```

Implements [IVideoSink](#).

## 3.252.2 Property Documentation

### 3.252.2.1 PushImageQueueCount

```
int PushImageQueueCount [get]
```

Implements [IVideoSink](#).

## 3.253 VideoSourceSizeMode Struct Reference

Defines how the video source is resized before passing to the encoder. These options are available depending on the video source.

## Public Types

- enum [Mode](#)  
*What sizes to use.*

## Public Attributes

- [Mode](#) **Init**  
*How to initialize the sizes.*
- bool **Update**  
*If true, update the sizes when the source is resized. Ignored if Init is Fixed.*

### 3.253.1 Detailed Description

Defines how the video source is resized before passing to the encoder. These options are available depending on the video source.

### 3.253.2 Member Enumeration Documentation

#### 3.253.2.1 Mode

enum [Mode](#)

What sizes to use.

Enumerator

|  |             |                                                      |
|--|-------------|------------------------------------------------------|
|  | Fixed       | Use sizes given during the stream initialization.    |
|  | Constrained | Adjusts one of the given sizes to keep aspect ratio. |
|  | Source      | Use source sizes.                                    |

## 3.254 VideoTexture Class Reference

### Classes

- class [Shader3D](#)
- class [ShaderScreen](#)

## 3.255 VideoUtil Class Reference

### Static Public Attributes

- static Dictionary< ImageFormat, TextureFormat > [ImageFormatToUnityTextureFormat](#)

### 3.255.1 Member Data Documentation

#### 3.255.1.1 ImageFormatToUnityTextureFormat

Dictionary<ImageFormat, TextureFormat> ImageFormatToUnityTextureFormat [static]

#### Initial value:

```
= new Dictionary<ImageFormat, TextureFormat>()
{
    { ImageFormat.ABGR, TextureFormat.RGBA32 },
    { ImageFormat.BGRA, TextureFormat.ARGB32 },
    { ImageFormat.ARGB, TextureFormat.BGRA32 },
    { ImageFormat.NV12, TextureFormat.BGRA32 },
    { ImageFormat.I420, TextureFormat.BGRA32 },
}
```

## 3.256 PreviewManager.ViewState Class Reference

### Public Member Functions

- **ViewState** ([IVideoPreview](#) v)
- override string **ToString** ()

### Properties

- object **PlatformView** [get]
- int **x** [get, set]
- int **y** [get, set]
- int **w** [get, set]
- int **h** [get, set]

## 3.257 VoiceClient Class Reference

[Voice](#) client interact with other clients on network via [IVoiceTransport](#).

Inherits [IDisposable](#).

### Classes

- struct [CreateOptions](#)

### Public Member Functions

- delegate void **RemoteVoiceInfoDelegate** (int channelId, int playerId, byte voiceld, [VoiceInfo](#) voiceInfo, ref [RemoteVoiceOptions](#) options)  
*Remote voice info event delegate.*
- IEnumerable< [LocalVoice](#) > **LocalVoicesInChannel** (int channelId)  
*Iterates through copy of all local voices list of given channel.*
- void **LogSpacingProfiles** ()
- void **LogStats** ()
- void **SetRemoteVoiceDelayFrames** ([Codec](#) codec, int delayFrames)
- void **SetRemoteVoiceDelayFrames** (Func< [Codec](#), bool > predicate, int delayFrames)
- **VoiceClient** ([IVoiceTransport](#) transport, [ILogger](#) logger, [CreateOptions](#) opt=default([CreateOptions](#)))  
*Creates [VoiceClient](#) instance.*
- void **Service** ()  
*This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2..20 times a second).*
- [LocalVoice](#) **CreateLocalVoice** ([VoiceInfo](#) voiceInfo, int channelId, [VoiceCreateOptions](#) options=default([VoiceCreateOptions](#)))  
*Creates basic outgoing stream w/o data processing support. Provided encoder should generate output data stream.*
- [LocalVoiceAudio](#)< T > **CreateLocalVoiceAudio**< T > ([VoiceInfo](#) voiceInfo, [IAudioDesc](#) audioSourceDesc, int channelId, [VoiceCreateOptions](#) options=default([VoiceCreateOptions](#)))
- [LocalVoice](#) **CreateLocalVoiceAudioFromSource** ([VoiceInfo](#) voiceInfo, [IAudioDesc](#) source, [AudioSampleType](#) sampleType, int channelId, [VoiceCreateOptions](#) options=default([VoiceCreateOptions](#)))  
*Creates outgoing audio stream of type automatically assigned and adds procedures (callback or serviceable) for consuming given audio source data. Adds audio specific features (e.g. resampling, level meter) to processing pipeline and to returning stream handler.*
- [LocalVoiceVideo](#) **CreateLocalVoiceVideo** ([VoiceInfo](#) voiceInfo, [IVideoRecorder](#) recorder, int channelId, [VoiceCreateOptions](#) options=default([VoiceCreateOptions](#)))  
*Creates outgoing video stream consuming sequence of image buffers.*
- void **RemoveLocalVoice** ([LocalVoice](#) voice)  
*Removes local voice (outgoing data stream).*

### Parameters

|                          |       |                                           |
|--------------------------|-------|-------------------------------------------|
| <input type="checkbox"/> | voice | Handler of outgoing stream to be removed. |
|--------------------------|-------|-------------------------------------------|

- void **onJoinChannel** (int channelId)
- void **onJoinAllChannels** ()
- void **onLeaveChannel** (int channel)
- void **onLeaveAllChannels** ()
- void **onPlayerJoin** (int channelId, int playerId)
- void **onPlayerJoin** (int playerId)
- void **onPlayerLeave** (int channelId, int playerId)
- void **onPlayerLeave** (int playerId)
- void **onVoiceInfo** (int channelId, int playerId, byte voiceld, int eventBufferSize, [VoiceInfo](#) info)
- void **onVoiceRemove** (int playerId, byte[] voicelds)
- void **onFrame** (int playerId, byte voiceld, ushort evNumber, ref [FrameBuffer](#) receivedBytes, bool isLocal←  
Player)
- void **Dispose** ()

### Properties

- bool **ThreadingEnabled** [get, set]
- int **EventsLost** [get, set]  
*Lost events counter (the number of empty frames sent to the deocder).*
- int **FramesLost** [get, set]  
*Lost frames counter (the number of empty frames sent to the deocder).*
- int **FramesFragPart** [get, set]  
*The counter of assembled frames, fragments of which are partially missing.*
- int **FramesRecovered** [get, set]  
*Recovered frames counter.*
- int **FramesLate** [get, set]  
*Counter of late (incorrectly ordered) frames.*
- int **FramesReceived** [get]  
*Received frames counter.*
- int **FramesReceivedFEC** [get, set]  
*Received FEC events counter.*
- int **FramesTryFEC** [get, set]  
*FEC recorery attempts counter.*
- int **FramesReceivedFragments** [get, set]  
*Received events for fragmented frames counter.*
- int **FramesReceivedFragmented** [get, set]  
*Assembled fragmented frames counter.*
- int **FramesSent** [get]  
*Sent frames counter.*
- int **FramesSentBytes** [get]  
*Sent frames bytes counter.*
- int **RoundTripTime** [get]  
*Average time required voice packet to return to sender.*
- int **RoundTripTimeVariance** [get]  
*Average round trip time variation.*
- bool **SuppressInfoDuplicateWarning** [get, set]  
*Do not log warning when duplicate info received.*
- [RemoteVoiceInfoDelegate](#) **OnRemoteVoiceInfoAction** [get, set]

Register a method to be called when remote voice info arrived (after join or new new remote voice creation). Metod parameters: (int channelId, int playerId, byte voiceld, [VoiceInfo](#) voiceInfo, ref [RemoteVoiceOptions](#) options);.

- int **DebugLostPercent** [get, set]  
Lost frames simulation ratio.
- IEnumerable< [LocalVoice](#) > **LocalVoices** [get]  
Iterates through copy of all local voices list.
- IEnumerable< [RemoteVoiceInfo](#) > **RemoteVoiceInfos** [get]  
Iterates through all remote voices infos.

### 3.257.1 Detailed Description

[Voice](#) client interact with other clients on network via [IVoiceTransport](#).

### 3.257.2 Member Function Documentation

#### 3.257.2.1 CreateLocalVoice()

```
LocalVoice CreateLocalVoice (
    VoiceInfo voiceInfo,
    int channelId,
    VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates basic outgoing stream w/o data processing support. Provided encoder should generate output data stream.

##### Parameters

|                           |                                          |
|---------------------------|------------------------------------------|
| <a href="#">voiceInfo</a> | Outgoing stream parameters.              |
| <a href="#">channelId</a> | Transport channel specific to transport. |
| <a href="#">options</a>   | <a href="#">Voice</a> creation options.  |

##### Returns

Outgoing stream handler.

#### 3.257.2.2 CreateLocalVoiceAudioFromSource()

```
LocalVoice CreateLocalVoiceAudioFromSource (
    VoiceInfo voiceInfo,
    IAudioDesc source,
    AudioSampleType sampleType,
    int channelId,
    VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates outgoing audio stream of type automatically assigned and adds procedures (callback or serviceable) for consuming given audio source data. Adds audio specific features (e.g. resampling, level meter) to processing pipeline and to returning stream handler.

## Parameters

|  |                   |                                                                                                                |
|--|-------------------|----------------------------------------------------------------------------------------------------------------|
|  | <i>voiceInfo</i>  | Outgoing stream parameters.                                                                                    |
|  | <i>source</i>     | Streaming audio source.                                                                                        |
|  | <i>sampleType</i> | <a href="#">Voice</a> 's audio sample type. If does not match source audio sample type, conversion will occur. |
|  | <i>channelId</i>  | Transport channel specific to transport.                                                                       |
|  | <i>options</i>    | <a href="#">Voice</a> creation options.                                                                        |

## Returns

Outgoing stream handler.

audioSourceDesc.SamplingRate and voiceInfo.SamplingRate may do not match. Automatic resampling will occur in this case.

**3.257.2.3 CreateLocalVoiceVideo()**

```
LocalVoiceVideo CreateLocalVoiceVideo (
    VoiceInfo voiceInfo,
    IVideoRecorder recorder,
    int channelId,
    VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates outgoing video stream consuming sequence of image buffers.

## Parameters

|  |                  |                                          |
|--|------------------|------------------------------------------|
|  | <i>voiceInfo</i> | Outgoing stream parameters.              |
|  | <i>recorder</i>  | Video recorder.                          |
|  | <i>channelId</i> | Transport channel specific to transport. |
|  | <i>options</i>   | <a href="#">Voice</a> creation options.  |

## Returns

Outgoing stream handler.

**3.258 VoiceComponent Class Reference**

Inherits MonoBehaviour.

Inherited by [PhotonVoiceView](#), [AudioChangesHandler](#), [FMODRecorderSetup](#), [Recorder](#), [RecorderPreset](#), [Speaker](#), [MicAmplifier](#), [MicrophonePermission](#), [SaveIncomingStreamToFile](#), [SaveOutgoingStreamToFile](#), and [WebRtcAudioDsp](#).

**Protected Member Functions**

- virtual void **Awake** ()



**Properties**

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## 3.259 VoiceComponentImpl Class Reference

**Public Member Functions**

- void **Awake** (MonoBehaviour mb)

**Properties**

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

## 3.260 VoiceConnection Class Reference

Component that represents a [Voice](#) client.

Inherits ConnectionHandler.

Inherited by [UnityVoiceClient](#), and [VoiceFollowClient](#).

**Public Member Functions**

- virtual bool [ConnectUsingSettings](#) (AppSettings overwriteSettings=null)  
*Connect to [Photon](#) server using [Settings](#)*
- bool [AddSpeaker](#) ([Speaker](#) speaker, object userData)  
*Tries to link local [Speaker](#) with remote voice stream using UserData. Useful if [Speaker](#) created after stream is started.*
- [Speaker](#) [InstantiateSpeakerPrefab](#) (GameObject parent, bool destroyOnRemove)  
*Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.*
- bool **AddRecorder** ([Recorder](#) rec)
- void **RemoveRecorder** ([Recorder](#) rec)

**Public Attributes**

- AppSettings **Settings**  
*Settings to be used by this [Voice](#) Client.*

**Static Public Attributes**

- const int **ChannelAudio** = 1  
*Recommended [Photon](#) Transport channel for audio. Chosen not to interfere with video and default channel.*
- const int **ChannelVideo** = 2  
*Recommended [Photon](#) Transport channel for video. Chosen not to interfere with audio and default channel.*

## Protected Member Functions

- override void **Awake** ()
- virtual void **Update** ()
- virtual void **OnDestroy** ()
- virtual [Speaker](#) **InstantiateSpeakerForRemoteVoice** (int playerId, byte voiceId, object userData)
- virtual void **OnVoiceStateChanged** ([ClientState](#) fromState, [ClientState](#) toState)
- void **CalcStatistics** ()
- virtual void **OnOperationResponseReceived** (OperationResponse operationResponse)

## Properties

- virtual bool **AlwaysUsePrimaryRecorder** [get]
- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- new [LoadBalancingTransport](#) **Client** [get]
- [VoiceClient](#) **VoiceClient** [get]  
*Returns underlying [Photon Voice](#) client.*
- ClientState **ClientState** [get]  
*Returns [Photon Voice](#) client state.*
- float **FramesReceivedPerSecond** [get]  
*Number of frames received per second.*
- float **FramesLostPerSecond** [get]  
*Number of frames lost per second.*
- float **FramesLostPercent** [get]  
*Percentage of lost frames.*
- GameObject **SpeakerPrefab** [get, set]  
*Prefab that contains [Speaker](#) component to be instantiated when receiving a new remote audio source info.*
- [Recorder](#) **PrimaryRecorder** [get, set]  
*Primary [Recorder](#) to be used by [VoiceConnection](#) implementations directly or via integration objects.*
- bool **UsePrimaryRecorder** [get]  
*Use [VoiceConnection.PrimaryRecorder](#) directly.*
- string **BestRegionSummaryInPreferences** [get, set]  
*Used to store and access the "Best Region Summary" in the Player Preferences.*

## Events

- Action< [Speaker](#) > **SpeakerLinked**  
*Fires when a speaker has been linked to a remote audio stream.*
- Action< [RemoteVoiceLink](#) > **RemoteVoiceAdded**  
*Fires when a remote voice stream is added.*

### 3.260.1 Detailed Description

Component that represents a [Voice](#) client.

### 3.260.2 Member Function Documentation

#### 3.260.2.1 AddSpeaker()

```
bool AddSpeaker (
    Speaker speaker,
    object userData)
```

Tries to link local [Speaker](#) with remote voice stream using UserData. Useful if [Speaker](#) created after stream is started.

## Parameters

|                       |                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------|
| <code>speaker</code>  | <a href="#">Speaker</a> or try linking.                                              |
| <code>userData</code> | UserData object used to bind local <a href="#">Speaker</a> with remote voice stream. |

## Returns

**3.260.2.2 ConnectUsingSettings()**

```
virtual bool ConnectUsingSettings (
    AppSettings overwriteSettings = null) [virtual]
```

Connect to [Photon](#) server using [Settings](#)

## Parameters

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <code>overwriteSettings</code> | Overwrites <a href="#">Settings</a> before connecting |
|--------------------------------|-------------------------------------------------------|

## Returns

If true voice connection command was sent from client

Reimplemented in [UnityVoiceClient](#).

**3.260.2.3 InstantiateSpeakerPrefab()**

```
Speaker InstantiateSpeakerPrefab (
    GameObject parent,
    bool destroyOnRemove)
```

Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.

[VoiceConnection](#) manages the instantiated object (destroys on [OnRemoteVoiceRemoveAction](#)).

## Parameters

|                              |                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>parent</code>          | The object to attach S                                                                                    |
| <code>destroyOnRemove</code> | Automatically destroy instantiated prefab when remote voice is removed (the caller does not manages the i |

## Returns

Instantiated [Speaker](#) or null.

**3.261 VoiceConnectionMatchmaking Class Reference**

Use matchmaking to connect and join a [VoiceConnection](#).

Inherits [MonoBehaviour](#), [IConnectionCallbacks](#), and [IMatchmakingCallbacks](#).

## Public Member Functions

- void **ConnectNow** ()
- virtual void **JoinRoom** ()  
*Join the room once connected to the server. The default implementation relies on random matchmaking, using EnterRoomParams parameters.*
- virtual void **OnConnectedToMaster** ()
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)
- void **OnDisconnected** (DisconnectCause cause)

## Public Attributes

- bool **autoConnect** = true
- [StreamMatchmakingSettings](#) **matchmakingSettings** = new [StreamMatchmakingSettings](#)()
- List< [StringSessionProperty](#) > **roomCustomPropertiesPreview** = new List<[StringSessionProperty](#)>()

## Protected Member Functions

- virtual void **Start** ()

## Protected Attributes

- [VoiceConnection](#) **voiceConnection**

## Properties

- virtual AuthenticationValues **AuthValues** [get]

### 3.261.1 Detailed Description

Use matchmaking to connect and join a [VoiceConnection](#).

## 3.262 VoiceConnectionPermissionWaiter Class Reference

Wait for Microphone access permissions to be available before letting [VoiceConnection](#) connect Note: recordWhenJoined should be disabled on the recorder.

Inherits [PermissionWaiter](#).

## Public Member Functions

- override void [OnPermissionGranted](#) ()  
*Should be called manually (for instance in a [PermissionsRequester](#) callback) if `checkPermisisonGrantAutomatically` is not checked.*

## Static Public Attributes

- const int **LOWPRIORITY\_EXECUTION\_ORDER** = PermissionWaiter.EXECUTION\_ORDER + 50

## Static Public Attributes inherited from [PermissionWaiter](#)

- const int **EXECUTION\_ORDER** = PermissionsRequester.EXECUTION\_ORDER + 1

## Protected Member Functions

- override void [Awake](#) ()
- override void [OnPermissionRequired](#) ()  
*Called when the permission is required.*
- override void [Update](#) ()

## Protected Member Functions inherited from [PermissionWaiter](#)

- virtual void **OnPermissionChanged** (bool isGranted)

## Properties

- override string **PermissionName** [get]

## Properties inherited from [PermissionWaiter](#)

- bool **IsWaitingForPermission** [get]
- virtual string **PermissionName** [get]
- bool **IsPermissionGranted** [get]

## Additional Inherited Members

## Public Attributes inherited from [PermissionWaiter](#)

- bool **checkPermisisonGrantAutomatically** = true
- UnityEvent< bool > **onPermissionChanged** = new UnityEvent<bool>()
- UnityEvent **onPermissionGranted** = new UnityEvent()

## Protected Attributes inherited from [PermissionWaiter](#)

- bool **waitingForPermission** = false

### 3.262.1 Detailed Description

Wait for Microphone access permissions to be available before letting [VoiceConnection](#) connect Note: record↔  
WhenJoined should be disabled on the recorder.

Should be used alongside a [PermissionsRequester](#), to align permissions on the desired timing (to avoid conflicting requests)

### 3.262.2 Member Function Documentation

#### 3.262.2.1 Awake()

```
override void Awake () [protected], [virtual]
```

Reimplemented from [PermissionWaiter](#).

#### 3.262.2.2 OnPermissionGranted()

```
override void OnPermissionGranted () [virtual]
```

Should be called manually (for instance in a [PermissionsRequester](#) callback) if checkPermisisonGrantAutomatically is not checked.

Reimplemented from [PermissionWaiter](#).

#### 3.262.2.3 OnPermissionRequired()

```
override void OnPermissionRequired () [protected], [virtual]
```

Called when the permission is required.

Reimplemented from [PermissionWaiter](#).

#### 3.262.2.4 Update()

```
override void Update () [protected], [virtual]
```

Reimplemented from [PermissionWaiter](#).

## 3.263 VoiceCreateOptions Struct Reference

Used to initialize optional properties of the [LocalVoice](#) instance at creation time.

**Public Attributes**

- [IEncoder](#) **Encoder**  
*Encoder.*
- byte **InterestGroup**  
*See [LocalVoice.InterestGroup](#).*
- int[] **TargetPlayers**  
*See [LocalVoice.TargetPlayersSet](#) to [] to make sure that [LocalVoice](#) does not create Remote voices during creation.*
- bool **DebugEchoMode**  
*See [LocalVoice.DebugEchoMode](#).*
- bool **Reliable**  
*See [LocalVoice.Reliable](#).*
- bool **Encrypt**  
*See [LocalVoice.Encrypt](#).*
- bool **Fragment**  
*See [LocalVoice.Fragment](#).*
- int **FEC**  
*See [LocalVoice.FEC](#).*
- int **EventBufSize**  
*Receiver's ring buffer size (default is 256)*

**3.263.1 Detailed Description**

Used to initialize optional properties of the [LocalVoice](#) instance at creation time.

**3.264 VoiceDebugScript Class Reference**

[Utility](#) script to be attached next to [PhotonVoiceView](#) & PhotonView on the player prefab to be network instantiated. Call `voiceDebugScript.CantHearYou()` on the networked object of the remote (or local) player if you can't hear the corresponding player.

Inherits MonoBehaviourPun.

**Public Member Functions**

- void **CantHearYou** ()

**Public Attributes**

- bool **ForceRecordingAndTransmission**  
*Make sure recorder.TransmitEnabled and recorder.RecordingEnabled are true.*
- AudioClip **TestAudioClip**  
*Audio file to be broadcast when TestUsingAudioClip is enabled.*
- bool **TestUsingAudioClip**  
*Broadcast Audio file to make sure transmission over network works if microphone (audio input device/hardware) is not reliable. Requires setting AudioClip in TestAudioClip.*
- bool **DisableVad**  
*Disable recorder.VoiceDetection for easier testing.*
- bool **IncreaseLogLevels**  
*Set main voice component's log level to ALL (max).*
- bool **LocalDebug**  
*Debug DebugEcho mode (Can't Hear My Self?!).*

### 3.264.1 Detailed Description

[Utility](#) script to be attached next to [PhotonVoiceView](#) & PhotonView on the player prefab to be network instantiated. Call `voiceDebugScript.CantHearYou()` on the networked object of the remote (or local) player if you can't hear the corresponding player.

## 3.265 AudioUtil.VoiceDetector< T > Class Template Reference

Simple voice activity detector triggered by signal level.

Inherits [IProcessor< T >](#), and [AudioUtil.IVoiceDetector](#).

Inherited by [AudioUtil.VoiceDetectorFloat](#), and [AudioUtil.VoiceDetectorShort](#).

### Public Member Functions

- [T\[\] Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

### Protected Attributes

- float **norm**
- float **threshold**
- int **activityDelay**
- int **autoSilenceCounter** = 0
- int **valuesCountPerSec**
- int **activityDelayValuesCount**

### Properties

- bool [On](#) [get, set]  
*If true, voice detection enabled.*
- float [Threshold](#) [get, set]  
*Voice detected as soon as signal level exceeds threshold.*
- bool [Detected](#) [get, protected set]  
*If true, voice detected.*
- DateTime [DetectedTime](#) [get]  
*Last time when switched to detected state.*
- int [ActivityDelayMs](#) [get, set]  
*Keep detected state during this time after signal level dropped below threshold.*

### Events

- Action **OnDetected**  
*Called when switched to detected state.*



## Events inherited from [AudioUtil.IVoiceDetector](#)

- Action **OnDetected**

*Called when switched to detected state.*

### 3.265.1 Detailed Description

Simple voice activity detector triggered by signal level.

### 3.265.2 Member Function Documentation

#### 3.265.2.1 Process()

```
T[] Process (
    T[] buf) [abstract]
```

Process a frame of data.

##### Parameters

|            |                              |
|------------|------------------------------|
| <i>buf</i> | Buffer containing input data |
|------------|------------------------------|

##### Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements [IProcessor< T >](#).

### 3.265.3 Property Documentation

#### 3.265.3.1 ActivityDelayMs

```
int ActivityDelayMs [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implements [AudioUtil.IVoiceDetector](#).

#### 3.265.3.2 Detected

```
bool Detected [get], [protected set]
```

If true, voice detected.

Implements [AudioUtil.IVoiceDetector](#).

### 3.265.3.3 DetectedTime

`DateTime DetectedTime [get]`

Last time when switched to detected state.

Implements [AudioUtil.IVoiceDetector](#).

### 3.265.3.4 On

`bool On [get], [set]`

If true, voice detection enabled.

Implements [AudioUtil.IVoiceDetector](#).

### 3.265.3.5 Threshold

`float Threshold [get], [set]`

[Voice](#) detected as soon as signal level exceeds threshold.

Implements [AudioUtil.IVoiceDetector](#).

## 3.266 AudioUtil.VoiceDetectorCalibration< T > Class Template Reference

Calibration [Utility](#) for [Voice](#) Detector.

Inherits [IProcessor< T >](#).

### Public Member Functions

- [VoiceDetectorCalibration](#) ([IVoiceDetector](#) voiceDetector, [ILevelMeter](#) levelMeter, int samplingRate, int channels)  
*Create new VoiceDetectorCalibration instance.*
- void [Calibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Start calibration.*
- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

### Protected Attributes

- int **calibrateCount**

## Properties

- bool **IsCalibrating** [get]

### 3.266.1 Detailed Description

Calibration [Utility](#) for [Voice](#) Detector.

Using this audio processor, you can calibrate the [IVoiceDetector.Threshold](#).

### 3.266.2 Member Function Documentation

#### 3.266.2.1 Calibrate()

```
void Calibrate (
    int durationMs,
    Action< float > onCalibrated = null)
```

Start calibration.

##### Parameters

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <i>durationMs</i>   | Duration of the calibration procedure (in milliseconds).        |
| <i>onCalibrated</i> | Optional callback that is called after calibration is complete. |

This activates the Calibration process. It will reset the given LevelMeter's AccumAvgPeakAmp (accumulated average peak amplitude), and when the duration has passed, use it for the VoiceDetector's detection threshold.

#### 3.266.2.2 Process()

```
T[] Process (
    T[] buf)
```

Process a frame of data.

##### Parameters

|            |                              |
|------------|------------------------------|
| <i>buf</i> | Buffer containing input data |
|------------|------------------------------|

##### Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements [IProcessor< T >](#).

#### 3.266.2.3 VoiceDetectorCalibration()

```
VoiceDetectorCalibration (
    IVoiceDetector voiceDetector,
    ILevelMeter levelMeter,
    int samplingRate,
    int channels)
```

Create new VoiceDetectorCalibration instance.

## Parameters

|                            |                                              |
|----------------------------|----------------------------------------------|
| <code>voiceDetector</code> | <a href="#">Voice</a> Detector to calibrate. |
| <code>levelMeter</code>    | Level Meter to look at for calibration.      |
| <code>samplingRate</code>  | Sampling rate of the audio signal (in Hz).   |
| <code>channels</code>      | Number of channels in the audio signal.      |

## 3.267 AudioUtil.VoiceDetectorDummy Class Reference

Dummy VoiceDetector that doesn't actually do anything.

Inherits [AudioUtil.IVoiceDetector](#).

### Properties

- bool [On](#) [get, set]  
*If true, voice detection enabled.*
- float [Threshold](#) [get, set]  
*Voice detected as soon as signal level exceeds threshold.*
- bool [Detected](#) [get]  
*If true, voice detected.*
- int [ActivityDelayMs](#) [get, set]  
*Keep detected state during this time after signal level dropped below threshold.*
- DateTime [DetectedTime](#) [get]  
*Last time when switched to detected state.*

### Events

- Action [OnDetected](#) [add, remove]

### Events inherited from [AudioUtil.IVoiceDetector](#)

- Action [OnDetected](#)  
*Called when switched to detected state.*

### 3.267.1 Detailed Description

Dummy VoiceDetector that doesn't actually do anything.

### 3.267.2 Property Documentation

#### 3.267.2.1 ActivityDelayMs

```
int ActivityDelayMs [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implements [AudioUtil.IVoiceDetector](#).

### 3.267.2.2 Detected

bool Detected [get]

If true, voice detected.

Implements [AudioUtil.IVoiceDetector](#).

### 3.267.2.3 DetectedTime

DateTime DetectedTime [get]

Last time when switched to detected state.

Implements [AudioUtil.IVoiceDetector](#).

### 3.267.2.4 On

bool On [get], [set]

If true, voice detection enabled.

Implements [AudioUtil.IVoiceDetector](#).

### 3.267.2.5 Threshold

float Threshold [get], [set]

[Voice](#) detected as soon as signal level exceeds threshold.

Implements [AudioUtil.IVoiceDetector](#).

## 3.268 AudioUtil.VoiceDetectorFloat Class Reference

VoiceDetector specialization for float audio.

Inherits [AudioUtil.VoiceDetector< T >](#).

### Public Member Functions

- [VoiceDetectorFloat](#) (int samplingRate, int numChannels)  
*Create a new [VoiceDetectorFloat](#) instance.*
- override float[] **Process** (float[] buffer)

## Public Member Functions inherited from [AudioUtil.VoiceDetector< T >](#)

- `T[] Process (T[] buf)`  
*Process a frame of data.*
- `void Dispose ()`

## Additional Inherited Members

## Protected Attributes inherited from [AudioUtil.VoiceDetector< T >](#)

- float `norm`
- float `threshold`
- int `activityDelay`
- int `autoSilenceCounter` = 0
- int `valuesCountPerSec`
- int `activityDelayValuesCount`

## Properties inherited from [AudioUtil.VoiceDetector< T >](#)

- bool `On` [get, set]  
*If true, voice detection enabled.*
- float `Threshold` [get, set]  
*Voice detected as soon as signal level exceeds threshold.*
- bool `Detected` [get, protected set]  
*If true, voice detected.*
- `DateTime` `DetectedTime` [get]  
*Last time when switched to detected state.*
- int `ActivityDelayMs` [get, set]  
*Keep detected state during this time after signal level dropped below threshold.*

## Events inherited from [AudioUtil.VoiceDetector< T >](#)

- Action `OnDetected`  
*Called when switched to detected state.*

## Events inherited from [AudioUtil.IVoiceDetector](#)

- Action `OnDetected`  
*Called when switched to detected state.*

### 3.268.1 Detailed Description

VoiceDetector specialization for float audio.

### 3.268.2 Constructor & Destructor Documentation

#### 3.268.2.1 VoiceDetectorFloat()

```

VoiceDetectorFloat (
    int samplingRate,
    int numChannels)

```

Create a new [VoiceDetectorFloat](#) instance.

## Parameters

|                     |                                            |
|---------------------|--------------------------------------------|
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz). |
| <i>numChannels</i>  | Number of channels in the audio signal.    |

## 3.269 AudioUtil.VoiceDetectorShort Class Reference

VoiceDetector specialization for float audio.

Inherits [AudioUtil.VoiceDetector< T >](#).

### Public Member Functions

- [VoiceDetectorShort](#) (int samplingRate, int numChannels)  
*Create a new [VoiceDetectorFloat](#) instance.*
- override short[] **Process** (short[] buffer)

### Public Member Functions inherited from [AudioUtil.VoiceDetector< T >](#)

- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [AudioUtil.VoiceDetector< T >](#)

- float **norm**
- float **threshold**
- int **activityDelay**
- int **autoSilenceCounter** = 0
- int **valuesCountPerSec**
- int **activityDelayValuesCount**

### Properties inherited from [AudioUtil.VoiceDetector< T >](#)

- bool [On](#) [get, set]  
*If true, voice detection enabled.*
- float [Threshold](#) [get, set]  
*Voice detected as soon as signal level exceeds threshold.*
- bool [Detected](#) [get, protected set]  
*If true, voice detected.*
- DateTime [DetectedTime](#) [get]  
*Last time when switched to detected state.*
- int [ActivityDelayMs](#) [get, set]  
*Keep detected state during this time after signal level dropped below threshold.*

## Events inherited from [AudioUtil.VoiceDetector< T >](#)

- Action **OnDetected**

*Called when switched to detected state.*

## Events inherited from [AudioUtil.IVoiceDetector](#)

- Action **OnDetected**

*Called when switched to detected state.*

### 3.269.1 Detailed Description

VoiceDetector specialization for float audio.

### 3.269.2 Constructor & Destructor Documentation

#### 3.269.2.1 VoiceDetectorShort()

```
VoiceDetectorShort (
    int samplingRate,
    int numChannels)
```

Create a new [VoiceDetectorFloat](#) instance.

#### Parameters

|                     |                                            |
|---------------------|--------------------------------------------|
| <i>samplingRate</i> | Sampling rate of the audio signal (in Hz). |
| <i>numChannels</i>  | Number of channels in the audio signal.    |

## 3.270 VoiceEvent Class Reference

### Static Public Attributes

- const byte [Code](#) = 202  
*Single event used for voice communications.*
- const byte **FrameCode** = 203

### 3.270.1 Member Data Documentation

#### 3.270.1.1 Code

```
const byte Code = 202 [static]
```

Single event used for voice communications.

Change if it conflicts with other event codes used in the same [Photon](#) room.



## 3.271 VoiceFollowClient Class Reference

This class can be used to automatically sync client states between Leader and [Voice](#) clients.

Inherits [VoiceConnection](#).

Inherited by [FusionVoiceClient](#), and [PunVoiceClient](#).

### Public Member Functions

- bool [ConnectAndJoinRoom](#) ()  
*Connect voice client to [Photon](#) servers and join a [Voice](#) room.*
- void **Disconnect** ()  
*Disconnect voice client from all [Photon](#) servers.*

### Public Member Functions inherited from [VoiceConnection](#)

- virtual bool [ConnectUsingSettings](#) (AppSettings overwriteSettings=null)  
*Connect to [Photon](#) server using [Settings](#)*
- bool [AddSpeaker](#) ([Speaker](#) speaker, object userData)  
*Tries to link local [Speaker](#) with remote voice stream using [UserData](#). Useful if [Speaker](#) created after stream is started.*
- [Speaker](#) [InstantiateSpeakerPrefab](#) (GameObject parent, bool destroyOnRemove)  
*Instantiates [SpeakerPrefab](#), optionally attaches it to the provided parent.*
- bool **AddRecorder** ([Recorder](#) rec)
- void **RemoveRecorder** ([Recorder](#) rec)

### Public Attributes

- bool **AutoConnectAndJoin** = true  
*Auto connect voice client and join a voice room when Leader client is joined to a Leader room.*

### Public Attributes inherited from [VoiceConnection](#)

- AppSettings **Settings**  
*Settings to be used by this [Voice](#) Client.*

### Protected Member Functions

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- virtual void **Start** ()
- override void [OnDestroy](#) ()
- override void [OnOperationResponseReceived](#) (OperationResponse operationResponse)
- void **LeaderStateChanged** ([ClientState](#) toState)
- override void [OnVoiceStateChanged](#) ([ClientState](#) fromState, [ClientState](#) toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

## Protected Member Functions inherited from [VoiceConnection](#)

- override void **Awake** ()
- virtual void **Update** ()
- virtual [Speaker](#) **InstantiateSpeakerForRemoteVoice** (int playerId, byte voiceId, object userData)
- void **CalcStatistics** ()

## Properties

- bool **LeaderInRoom** [get]
- bool **LeaderOfflineMode** [get]

## Properties inherited from [VoiceConnection](#)

- virtual bool **AlwaysUsePrimaryRecorder** [get]
- [Voice.ILogger](#) **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- new [LoadBalancingTransport](#) **Client** [get]
- [VoiceClient](#) **VoiceClient** [get]  
*Returns underlying [Photon Voice](#) client.*
- ClientState **ClientState** [get]  
*Returns [Photon Voice](#) client state.*
- float **FramesReceivedPerSecond** [get]  
*Number of frames received per second.*
- float **FramesLostPerSecond** [get]  
*Number of frames lost per second.*
- float **FramesLostPercent** [get]  
*Percentage of lost frames.*
- [GameObject](#) **SpeakerPrefab** [get, set]  
*Prefab that contains [Speaker](#) component to be instantiated when receiving a new remote audio source info.*
- [Recorder](#) **PrimaryRecorder** [get, set]  
*Primary [Recorder](#) to be used by [VoiceConnection](#) implementations directly or via integration objects.*
- bool **UsePrimaryRecorder** [get]  
*Use [VoiceConnection.PrimaryRecorder](#) directly.*
- string **BestRegionSummaryInPreferences** [get, set]  
*Used to store and access the "Best Region Summary" in the Player Preferences.*

## Additional Inherited Members

## Static Public Attributes inherited from [VoiceConnection](#)

- const int **ChannelAudio** = 1  
*Recommended [Photon](#) Transport channel for audio. Chosen not to interfere with video and default channel.*
- const int **ChannelVideo** = 2  
*Recommended [Photon](#) Transport channel for video. Chosen not to interfere with audio and default channel.*

## Events inherited from [VoiceConnection](#)

- Action< [Speaker](#) > **SpeakerLinked**  
*Fires when a speaker has been linked to a remote audio stream.*
- Action< [RemoteVoiceLink](#) > **RemoteVoiceAdded**  
*Fires when a remote voice stream is added.*

### 3.271.1 Detailed Description

This class can be used to automatically sync client states between Leader and [Voice](#) clients.

### 3.271.2 Member Function Documentation

#### 3.271.2.1 ConnectAndJoinRoom()

```
bool ConnectAndJoinRoom ()
```

Connect voice client to [Photon](#) servers and join a [Voice](#) room.

##### Returns

If true, connection command send from client

#### 3.271.2.2 OnDestroy()

```
override void OnDestroy () [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.271.2.3 OnOperationResponseReceived()

```
override void OnOperationResponseReceived (  
    OperationResponse operationResponse) [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

#### 3.271.2.4 OnVoiceStateChanged()

```
override void OnVoiceStateChanged (  
    ClientState fromState,  
    ClientState toState) [protected], [virtual]
```

Reimplemented from [VoiceConnection](#).

## 3.272 VoicelInfo Struct Reference

Describes stream properties.

### Public Member Functions

- override string **ToString** ()

### Static Public Member Functions

- static [VoicelInfo CreateAudioOpus](#) (POpusCodec.Enums.SamplingRate samplingRate, int channels, OpusCodec.FrameDuration frameDurationUs, int bitrate, object userdata=null)  
*Create stream info for an Opus audio stream.*
- static [VoicelInfo CreateAudio](#) (Codec codec, int samplingRate, int channels, int frameDurationUs, object userdata=null)  
*Create stream info for an audio stream.*
- static [VoicelInfo CreateVideo](#) (Codec codec, int bitrate, int width, int height, int fps, int keyFrameInt, object userdata=null)  
*Create stream info for a video stream.*

### Properties

- [Codec](#) **Codec** [get, set]
- int **SamplingRate** [get, set]  
*Audio sampling rate (frequency, in Hz).*
- int **Channels** [get, set]  
*Number of channels.*
- int **FrameDurationUs** [get, set]  
*Uncompressed frame (audio packet) size in microseconds.*
- int **Bitrate** [get, set]  
*Target bitrate (in bits/second).*
- int **Width** [get, set]  
*Video width.*
- int **Height** [get, set]  
*Video height.*
- int **FPS** [get, set]  
*Video frames per second.*
- int **KeyFrameInt** [get, set]  
*Video keyframe interval in frames.*
- object **UserData** [get, set]  
*Optional user data. Should be serializable by [Photon](#).*
- int **FrameDurationSamples** [get]  
*Uncompressed frame (data packet) size in samples.*
- int **FrameSize** [get]  
*Uncompressed frame (data packet) array size.*

### 3.272.1 Detailed Description

Describes stream properties.

## 3.272.2 Member Function Documentation

### 3.272.2.1 CreateAudio()

```
VoiceInfo CreateAudio (
    Codec codec,
    int samplingRate,
    int channels,
    int frameDurationUs,
    object userdata = null) [static]
```

Create stream info for an audio stream.

#### Parameters

|  |                        |                                                                        |
|--|------------------------|------------------------------------------------------------------------|
|  | <i>codec</i>           | Audio codec.                                                           |
|  | <i>samplingRate</i>    | Audio sampling rate.                                                   |
|  | <i>channels</i>        | Number of channels.                                                    |
|  | <i>frameDurationUs</i> | Uncompressed frame (audio packet) size in microseconds.                |
|  | <i>userdata</i>        | Optional user data. Should be serializable by <a href="#">Photon</a> . |

#### Returns

[VoiceInfo](#) instance.

### 3.272.2.2 CreateAudioOpus()

```
VoiceInfo CreateAudioOpus (
    POpusCodec.Enums.SamplingRate samplingRate,
    int channels,
    OpusCodec.FrameDuration frameDurationUs,
    int bitrate,
    object userdata = null) [static]
```

Create stream info for an Opus audio stream.

#### Parameters

|  |                        |                                                                        |
|--|------------------------|------------------------------------------------------------------------|
|  | <i>samplingRate</i>    | Audio sampling rate.                                                   |
|  | <i>channels</i>        | Number of channels.                                                    |
|  | <i>frameDurationUs</i> | Uncompressed frame (audio packet) size in microseconds.                |
|  | <i>bitrate</i>         | Stream bitrate (in bits/second).                                       |
|  | <i>userdata</i>        | Optional user data. Should be serializable by <a href="#">Photon</a> . |

#### Returns

[VoiceInfo](#) instance.

### 3.272.2.3 CreateVideo()

```

VoiceInfo CreateVideo (
    Codec codec,
    int bitrate,
    int width,
    int height,
    int fps,
    int keyFrameInt,
    object userdata = null) [static]

```

Create stream info for a video stream.

#### Parameters

|  |                    |                                                                                   |
|--|--------------------|-----------------------------------------------------------------------------------|
|  | <i>codec</i>       | Video codec.                                                                      |
|  | <i>bitrate</i>     | Stream bitrate.                                                                   |
|  | <i>width</i>       | Streamed video width. If 0, width and height of video source used (no rescaling). |
|  | <i>height</i>      | Streamed video height. If -1, aspect ratio preserved during rescaling.            |
|  | <i>fps</i>         | Streamed video frames per second.                                                 |
|  | <i>keyFrameInt</i> | Keyframes interval in frames.                                                     |

///

#### Parameters

|  |                 |                                                                        |
|--|-----------------|------------------------------------------------------------------------|
|  | <i>userdata</i> | Optional user data. Should be serializable by <a href="#">Photon</a> . |
|--|-----------------|------------------------------------------------------------------------|

#### Returns

[VoiceInfo](#) instance.

## 3.273 AudioUtil.VoiceLevelDetectCalibrate< T > Class Template Reference

[Utility](#) Audio Processor [Voice](#) Detection Calibration.

Inherits [IProcessor< T >](#).

### Public Member Functions

- [VoiceLevelDetectCalibrate](#) (int samplingRate, int channels)  
*Create new VoiceLevelDetectCalibrate instance.*
- void [Calibrate](#) (int durationMs, Action< float > onCalibrated=null)  
*Start calibration.*
- T[] [Process](#) (T[] buf)  
*Process a frame of data.*
- void **Dispose** ()

## Properties

- [ILevelMeter](#) **LevelMeter** [get]  
*The LevelMeter in use.*
- [IVoiceDetector](#) **VoiceDetector** [get]  
*The VoiceDetector in use.*
- bool **IsCalibrating** [get]

## 3.273.1 Detailed Description

[Utility](#) Audio Processor [Voice](#) Detection Calibration.

Encapsulates level meter, voice detector and voice detector calibrator in single instance.

## 3.273.2 Member Function Documentation

### 3.273.2.1 Calibrate()

```
void Calibrate (
    int durationMs,
    Action< float > onCalibrated = null)
```

Start calibration.

#### Parameters

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <code>durationMs</code>   | Duration of the calibration procedure (in milliseconds).               |
| <code>onCalibrated</code> | Called when calibration is complete. Parameter is new threshold value. |

This activates the Calibration process. It will reset the given LevelMeter's AccumAvgPeakAmp (accumulated average peak amplitude), and when the duration has passed, use it for the VoiceDetector's detection threshold.

### 3.273.2.2 Process()

```
T[] Process (
    T[] buf)
```

Process a frame of data.

#### Parameters

|                  |                              |
|------------------|------------------------------|
| <code>buf</code> | Buffer containing input data |
|------------------|------------------------------|

#### Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements [IProcessor< T >](#).

### 3.273.2.3 VoiceLevelDetectCalibrate()

```
VoiceLevelDetectCalibrate (
    int samplingRate,
    int channels)
```

Create new VoiceLevelDetectCalibrate instance.

## Parameters

|                           |                                            |
|---------------------------|--------------------------------------------|
| <code>samplingRate</code> | Sampling rate of the audio signal (in Hz). |
| <code>channels</code>     | Number of channels in the audio signal.    |

## 3.274 VoiceLogger Class Reference

Inherits MonoBehaviour.

### Static Public Member Functions

- static [VoiceLogger](#) **FindLogger** (GameObject gameObject)
- static [VoiceLogger](#) **CreateRootLogger** ()

### Public Attributes

- LogLevel **LogLevel** = LogLevel.Warning

## 3.275 VoiceNetworkObject Class Reference

Inherits NetworkBehaviour.

### Public Member Functions

- override void **Spawned** ()
- override void **Despawned** (NetworkRunner runner, bool hasState)

### Properties

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- [Recorder](#) **RecorderInUse** [get]  
*The [Recorder](#) component currently used by this [VoiceNetworkObject](#).*
- [Speaker](#) **SpeakerInUse** [get]  
*The [Speaker](#) component currently used by this [VoiceNetworkObject](#).*
- bool **IsSpeaking** [get]  
*If true, this [VoiceNetworkObject](#) has a [Speaker](#) that is currently playing received audio frames from remote audio source.*
- bool **IsRecording** [get]  
*If true, this [VoiceNetworkObject](#) has a [Recorder](#) that is currently transmitting audio stream from local audio source.*
- bool **IsLocal** [get]



## 3.276 VPxCodec Class Reference

### Classes

- class [Decoder](#)
- class [Encoder](#)

## 3.277 AudioUtil.WaveformAudioPusher< T > Class Template Reference

IAudioPusher that provides the given waveform.

Inherits [AudioUtil.GeneratorPusher< T >](#).

### Public Member Functions

- **WaveformAudioPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)

### Public Member Functions inherited from [AudioUtil.GeneratorPusher< T >](#)

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*
- void **Dispose** ()

### Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

### Protected Member Functions inherited from [AudioUtil.GeneratorPusher< T >](#)

- int **Gen** (T[] buf, long timeSamples)

### Properties

- T[] **Waveform** [set]

### Properties inherited from [AudioUtil.GeneratorPusher< T >](#)

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

## Additional Inherited Members

### Protected Attributes inherited from [AudioUtil.GeneratorPusher< T >](#)

- long **timeSamples**

### 3.277.1 Detailed Description

IAudioPusher that provides the given waveform.

## 3.278 AudioUtil.WaveformAudioReader< T > Class Template Reference

IAudioReader that provides the given waveform.

Inherits [AudioUtil.GeneratorReader< T >](#).

### Public Member Functions

- **WaveformAudioReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)

### Public Member Functions inherited from [AudioUtil.GeneratorReader< T >](#)

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool **Read** (T[] buf)

*Fill full given frame buffer with source uncompressed data or return false if not enough such data.*

### Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

### Protected Member Functions inherited from [AudioUtil.GeneratorReader< T >](#)

- int **Gen** (T[] buf, long timeSamples)

### Properties

- T[] **Waveform** [set]

### Properties inherited from [AudioUtil.GeneratorReader< T >](#)

- int **Channels** [get]  
*Number of channels in the audio signal.*
- int **SamplingRate** [get]  
*Sampling rate of the audio signal (in Hz).*
- string **Error** [get]  
*If not null, audio object is in invalid state.*

### 3.278.1 Detailed Description

IAudioReader that provides the given waveform.

## 3.279 WaveWriter Class Reference

Inherits IDisposable.

### Public Member Functions

- **WaveWriter** (string fileName, int sampleRate, int bits, int channels)
- **WaveWriter** (Stream stream, int sampleRate, int bitsPerSample, int channels)
- void **Dispose** ()
- void **WriteSample** (float sample)
- void **WriteSamples** (float[] samples, int offset, int count)
- void **Write** (byte[] buffer, int offset, int count)
- void **Write** (byte value)
- void **Write** (short value)
- void **Write** (int value)
- void **Write** (float value)

### Protected Member Functions

- virtual void **Dispose** (bool disposing)

## 3.280 WebAudioAudioOut Class Reference

Inherits [AudioOutDelayControl< T >](#).

### Public Member Functions

- **WebAudioAudioOut** ([PlayDelayConfig](#) playDelayConfig, double spatialBlend, [ILogger](#) logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (float[] data, int offsetSamples)
- override void [Stop](#) ()
- void **SetVolume** (double x)
- void **SetSpatialBlend** (double x)
- void **SetRefDistance** (double x)
- void **SetMaxDistance** (double x)
- void **SetListenerPosition** (double x, double y, double z)
- void **SetListenerOrientation** (double fx, double fy, double fz, double ux, double uy, double uz)
- void **SetPosition** (double x, double y, double z)

## Public Member Functions inherited from [AudioOutDelayControl< T >](#)

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[] data, int offsetSamples)
- bool **IsZeroFrame** (T[] f)
- **AudioOutDelayControl** (bool processInService, [PlayDelayConfig](#) playDelayConfig, [ILogger](#) logger, string logPrefix, bool debugInfo)
- void [Start](#) (int frequency, int channels, int frameSamples)
- void [Service](#) ()
- void [Push](#) (T[] frame)
- void [Flush](#) ()

## Static Public Member Functions

- static void **ResumeAudioContext** ()

## Protected Attributes

- int **frequency**
- int **channels**
- int **bufferSamples**

## Protected Attributes inherited from [AudioOutDelayControl< T >](#)

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly [ILogger](#) **logger**
- readonly string **logPrefix**

## Properties

- string **Error** [get]
- override long **OutPos** [get]

## Properties inherited from [AudioOutDelayControl< T >](#)

- long **OutPos** [get]
- int [Lag](#) [get]
- bool **IsFlushed** [get]
- bool [IsPlaying](#) [get]

## Additional Inherited Members

## Static Public Attributes inherited from [AudioOutDelayControl< T >](#)

- const int **FRAME\_POOL\_CAPACITY** = 50

### 3.280.1 Member Function Documentation

#### 3.280.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from [AudioOutDelayControl< T >](#).

## 3.281 WebAudioInEnumerator Class Reference

Inherits [WebDeviceEnumerator](#).

### Public Member Functions

- **WebAudioInEnumerator** ([ILogger](#) logger)

### Public Member Functions inherited from [WebDeviceEnumerator](#)

- **WebDeviceEnumerator** ([ILogger](#) logger, string filter)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- [IEnumerator< DeviceInfo >](#) **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- [List< DeviceInfo >](#) **devices** = new [List< DeviceInfo >\(\)](#)
- [ILogger](#) logger

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## 3.282 WebAudioMicIn Class Reference

Inherits [IAudioPusher< T >](#).

## Public Member Functions

- **WebAudioMicIn** (string deviceId, int suggestedFrequency, int suggestedChannels, [ILogger](#) logger)
- void **SetCallback** (Action< float[]> callback, ObjectFactory< float[], int > bufferFactory, int optimalFrameSize)
- void **Dispose** ()

## Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

## Properties

- string [Error](#) [get]  
*If not null, audio object is in invalid state.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- int [Channels](#) [get]  
*Number of channels in the audio signal.*

## 3.282.1 Property Documentation

### 3.282.1.1 Channels

`int Channels [get]`

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.282.1.2 Error

`string Error [get]`

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.282.1.3 SamplingRate

`int SamplingRate [get]`

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).

## 3.283 WebCodecsCamera Class Reference

Inherits [WebCodecsVideoCapture](#).

### Public Member Functions

- **WebCodecsCamera** ([ILogger](#) logger, [VoiceInfo](#) info, string deviceId, Action onReady, Action< int, int > on↵Change)

### Public Member Functions inherited from [WebCodecsVideoCapture](#)

- **WebCodecsVideoCapture** ([ILogger](#) logger, Action onReady, Action< int, int > onChange)
- bool **SetEncoderAndPreview** ([WebCodecsVideoEncoder](#) encoder, UnityEngine.Texture2D preview)
- void **Dispose** ()

### Properties

- override string **Prefix** [get]

### Properties inherited from [WebCodecsVideoCapture](#)

- virtual string **Prefix** [get]

### Additional Inherited Members

### Public Attributes inherited from [WebCodecsVideoCapture](#)

- string **Error**

### Static Protected Member Functions inherited from [WebCodecsVideoCapture](#)

- static void **createCallbackStatic** (IntPtr managedCamObj, int err)
- static void **changeCallbackStatic** (IntPtr managedCamObj, int width, int height)

### Protected Attributes inherited from [WebCodecsVideoCapture](#)

- GCHandle **managedCamObj**

## 3.284 WebCodecsCameraRecorderUnityTexture Class Reference

Inherits [WebCodecsVideoRecorderUnityTexture](#).

**Public Member Functions**

- **WebCodecsCameraRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, string camDevice, Action<[IVideoRecorder](#)> onReady)

**Public Member Functions inherited from [WebCodecsVideoRecorderUnityTexture](#)**

- **WebCodecsVideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) voiceInfo, [VideoSourceSizeMode](#) sizeMode, Func< Action, Action< int, int >, [WebCodecsVideoCapture](#)> captureFactory, Action<[IVideoRecorder](#)> onReady)
- void **Dispose** ()

**Additional Inherited Members****Properties inherited from [WebCodecsVideoRecorderUnityTexture](#)**

- [IEncoder](#) [Encoder](#) [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) [Flip](#) [get]
- string [Error](#) [get]

**3.285 WebCodecsScreenRecorderUnityTexture Class Reference**

Inherits [WebCodecsVideoRecorderUnityTexture](#).

**Public Member Functions**

- **WebCodecsScreenRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info, [VideoSourceSizeMode](#) size↔Mode, Action<[IVideoRecorder](#)> onReady)

**Public Member Functions inherited from [WebCodecsVideoRecorderUnityTexture](#)**

- **WebCodecsVideoRecorderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) voiceInfo, [VideoSourceSizeMode](#) sizeMode, Func< Action, Action< int, int >, [WebCodecsVideoCapture](#)> captureFactory, Action<[IVideoRecorder](#)> onReady)
- void **Dispose** ()

**Additional Inherited Members****Properties inherited from [WebCodecsVideoRecorderUnityTexture](#)**

- [IEncoder](#) [Encoder](#) [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) [Flip](#) [get]
- string [Error](#) [get]



## 3.286 WebCodecsScreenShare Class Reference

Inherits [WebCodecsVideoCapture](#).

### Public Member Functions

- **WebCodecsScreenShare** ([ILogger](#) logger, [VoicelInfo](#) info, [VideoSourceSizeMode](#) sizeMode, Action onReady, Action< int, int > onChange)

### Public Member Functions inherited from [WebCodecsVideoCapture](#)

- **WebCodecsVideoCapture** ([ILogger](#) logger, Action onReady, Action< int, int > onChange)
- bool **SetEncoderAndPreview** ([WebCodecsVideoEncoder](#) encoder, UnityEngine.Texture2D preview)
- void **Dispose** ()

### Properties

- override string **Prefix** [get]

### Properties inherited from [WebCodecsVideoCapture](#)

- virtual string **Prefix** [get]

### Additional Inherited Members

### Public Attributes inherited from [WebCodecsVideoCapture](#)

- string **Error**

### Static Protected Member Functions inherited from [WebCodecsVideoCapture](#)

- static void **createCallbackStatic** (IntPtr managedCamObj, int err)
- static void **changeCallbackStatic** (IntPtr managedCamObj, int width, int height)

### Protected Attributes inherited from [WebCodecsVideoCapture](#)

- GCHandle **managedCamObj**

## 3.287 WebCodecsVideoCapture Class Reference

Inherits IDisposable.

Inherited by [WebCodecsCamera](#), and [WebCodecsScreenShare](#).

**Public Member Functions**

- **WebCodecsVideoCapture** ([ILogger](#) logger, Action onReady, Action< int, int > onChange)
- bool **SetEncoderAndPreview** ([WebCodecsVideoEncoder](#) encoder, UnityEngine.Texture2D preview)
- void **Dispose** ()

**Public Attributes**

- string **Error**

**Static Protected Member Functions**

- static void **createCallbackStatic** (IntPtr managedCamObj, int err)
- static void **changeCallbackStatic** (IntPtr managedCamObj, int width, int height)

**Protected Attributes**

- GCHandle **managedCamObj**

**Properties**

- virtual string **Prefix** [get]

## 3.288 WebCodecsVideoDecoderUnityTexture Class Reference

Inherits [IDecoder](#).

**Public Member Functions**

- **WebCodecsVideoDecoderUnityTexture** ([ILogger](#) logger, [VoiceInfo](#) info)
- void **Open** ([VoiceInfo](#) info)  
*Open (initialize) the decoder.*
- void **Input** (ref [FrameBuffer](#) buf)  
*Consumes the given encoded data.*
- void **Dispose** ()

**Protected Attributes**

- GCHandle **managedDecObj**

**Properties**

- int **nativeDecoder** [get]
- UnityEngine.Texture2D **Preview** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*

## 3.288.1 Member Function Documentation

### 3.288.1.1 Input()

```
void Input (
    ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call `buf.Retain()` to prevent the caller from disposing the buffer. In this case, the callee should call `buf.Release()` when buffer is no longer needed.

Implements [IDecoder](#).

### 3.288.1.2 Open()

```
void Open (
    VoiceInfo info)
```

Open (initialize) the decoder.

#### Parameters

|                   |                                          |
|-------------------|------------------------------------------|
| <code>info</code> | Properties of the data stream to decode. |
|-------------------|------------------------------------------|

Implements [IDecoder](#).

## 3.288.2 Property Documentation

### 3.288.2.1 Error

```
string Error [get], [protected set]
```

If not null, the object is in invalid state.

Implements [IDecoder](#).

## 3.289 WebCodecsVideoEncoder Class Reference

Inherits [IEncoder](#).

Inherited by [WebCodecsVideoEncoderDirectImage](#).

## Public Member Functions

- **WebCodecsVideoEncoder** ([ILogger](#) logger, [VoicelInfo](#) info, Action onReady)
- `ArraySegment< byte > DequeueOutput` (out `FrameFlags` flags)  
*Returns next encoded data frame (if such output supported).*
- `void EndOfStream` ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- `I GetPlatformAPI< I > ()`  
*Returns an platform-specific interface.*
- `virtual void Dispose` ()

## Protected Attributes

- `GCHandle managedEncObj`

## Properties

- `int NativeEncoder` [get]
- `string Error` [get, protected set]  
*If not null, the object is in invalid state.*
- `Action< ArraySegment< byte >, FrameFlags > Output` [protected get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## 3.289.1 Member Function Documentation

### 3.289.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
    out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements [IEncoder](#).

### 3.289.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements [IEncoder](#).

### 3.289.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements [IEncoder](#).

## Type Constraints

**I : class**

## 3.289.2 Property Documentation

### 3.289.2.1 Error

string Error [get], [protected set]

If not null, the object is in invalid state.

Implements [IEncoder](#).

### 3.289.2.2 Output

Action<ArraySegment<byte>, FrameFlags> Output [protected get], [set]

Set callback encoder calls on each encoded data frame (if such output supported).

Implements [IEncoder](#).

## 3.290 WebCodecsVideoEncoderDirectImage Class Reference

Inherits [WebCodecsVideoEncoder](#), and [IEncoderDirectImage](#).

### Public Member Functions

- **WebCodecsVideoEncoderDirectImage** ([ILogger](#) logger, [VoiceInfo](#) info, Action onReady)
- void **Input** ([ImageBufferNative](#) buf)

### Public Member Functions inherited from [WebCodecsVideoEncoder](#)

- **WebCodecsVideoEncoder** ([ILogger](#) logger, [VoiceInfo](#) info, Action onReady)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)  
*Returns next encoded data frame (if such output supported).*
- void **EndOfStream** ()  
*Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).*
- I **GetPlatformAPI**< I > ()  
*Returns an platform-specific interface.*
- virtual void **Dispose** ()

### Public Member Functions inherited from [IEncoderDirect](#)< B >

- void **Input** (B buf)  
*Consumes the given raw data.*

### Properties

- ImageFormat **ImageFormat** [get]  
*Recommended encoder input image format. Encoder may support other formats.*

## Properties inherited from [WebCodecsVideoEncoder](#)

- int **NativeEncoder** [get]
- string **Error** [get, protected set]  
*If not null, the object is in invalid state.*
- Action< ArraySegment< byte >, FrameFlags > **Output** [protected get, set]  
*Set callback encoder calls on each encoded data frame (if such output supported).*

## Additional Inherited Members

## Protected Attributes inherited from [WebCodecsVideoEncoder](#)

- GCHandle **managedEncObj**

### 3.290.1 Property Documentation

#### 3.290.1.1 ImageFormat

ImageFormat ImageFormat [get]

Recommended encoder input image format. Encoder may support other formats.

Implements [IEncoderDirectImage](#).

## 3.291 WebCodecsVideoPlayerUnityTexture Class Reference

Inherits [IVideoPlayer](#).

### Public Member Functions

- **WebCodecsVideoPlayerUnityTexture** (ILogger logger, VoiceInfo info, Action< IVideoPlayer > onReady)
- void **Dispose** ()

### Properties

- [IDecoder Decoder](#) [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip Flip](#) [get]

### 3.291.1 Property Documentation

#### 3.291.1.1 Decoder

`IDecoder` Decoder [get]

Implements [IVideoPlayer](#).

#### 3.291.1.2 Flip

`Flip` Flip [get]

Implements [IVideoPreview](#).

#### 3.291.1.3 Height

`int` Height [get]

Implements [IVideoPreview](#).

#### 3.291.1.4 PlatformView

`object` PlatformView [get]

Implements [IVideoPreview](#).

#### 3.291.1.5 Rotation

`Rotation` Rotation [get]

Implements [IVideoPreview](#).

#### 3.291.1.6 Width

`int` Width [get]

Implements [IVideoPreview](#).

## 3.292 WebCodecsVideoRecorderUnityTexture Class Reference

Inherits [IVideoRecorder](#).

Inherited by [WebCodecsCameraRecorderUnityTexture](#), and [WebCodecsScreenRecorderUnityTexture](#).

## Public Member Functions

- **WebCodecsVideoRecorderUnityTexture** ([ILogger](#) logger, [VoicelInfo](#) voicelInfo, [VideoSourceSizeMode](#) sizeMode, Func< Action, Action< int, int >, [WebCodecsVideoCapture](#) > captureFactory, Action< [IVideoRecorder](#) > onReady)
- void **Dispose** ()

## Properties

- [IEncoder](#) **Encoder** [get]
- object [PlatformView](#) [get]
- int [Width](#) [get]
- int [Height](#) [get]
- Rotation [Rotation](#) [get]
- [Flip](#) **Flip** [get]
- string [Error](#) [get]

## 3.292.1 Property Documentation

### 3.292.1.1 Encoder

[IEncoder](#) **Encoder** [get]

Implements [IVideoRecorder](#).

### 3.292.1.2 Error

string **Error** [get]

Implements [IVideoRecorder](#).

### 3.292.1.3 Flip

[Flip](#) **Flip** [get]

Implements [IVideoPreview](#).

### 3.292.1.4 Height

int **Height** [get]

Implements [IVideoPreview](#).

### 3.292.1.5 PlatformView

object **PlatformView** [get]

Implements [IVideoPreview](#).



### 3.292.1.6 Rotation

Rotation Rotation [get]

Implements [IVideoPreview](#).

### 3.292.1.7 Width

int Width [get]

Implements [IVideoPreview](#).

## 3.293 WebDeviceEnumerator Class Reference

Inherits [DeviceEnumeratorBase](#).

Inherited by [WebAudioInEnumerator](#), and [WebVideoInEnumerator](#).

### Public Member Functions

- **WebDeviceEnumerator** ([ILogger](#) logger, string filter)
- override void [Refresh](#) ()
- override void **Dispose** ()

### Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

### Additional Inherited Members

### Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

### Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

### 3.293.1 Member Function Documentation

#### 3.293.1.1 Refresh()

override void Refresh ()

Implements [IDeviceEnumerator](#).

## 3.294 WebRtcAudioDsp Class Reference

Inherits [VoiceComponent](#).

### Public Member Functions

- void **AdjustVoiceInfo** (ref [VoiceInfo](#) voiceInfo, ref [AudioSampleType](#) st)

### Protected Member Functions

- override void [Awake](#) ()

### Properties

- bool **AEC** [get, set]
- bool **AecHighPass** [get, set]
- int **ReverseStreamDelayMs** [get, set]
- [WebRTCAudioProcessor.NSLevel](#) **NoiseSuppressionLevel** [get, set]  
*Noise suppression aggressiveness. Default: Moderate.*
- bool **AgcLimiter** [get, set]  
*Hard-limit the AGC output at the target level. Default: true.*
- [WebRTCAudioProcessor.VADLikelihood](#) **VadLikelihood** [get, set]  
*How readily the voice detector calls a frame speech. VeryLow (default) yields the fewest false positives.*
- bool **NoiseSuppression** [get, set]
- bool **HighPass** [get, set]
- bool **Bypass** [get, set]
- bool **AGC** [get, set]
- int **AgcCompressionGain** [get, set]
- int **AgcTargetLevel** [get, set]
- bool **VAD** [get, set]
- bool **IsSupported** [get]

### Properties inherited from [VoiceComponent](#)

- Voice.ILogger **Logger** [get]
- [VoiceLogger](#) **VoiceLogger** [get]
- string **Name** [set]

### 3.294.1 Member Function Documentation

#### 3.294.1.1 Awake()

override void Awake () [protected], [virtual]

Reimplemented from [VoiceComponent](#).

## 3.295 WebRTCAudioLib Class Reference

Inherited by [WebRTCAudioProcessor](#).

### Public Types

- enum **Error**
- enum **Param**
- enum [NSLevel](#)  
*webrtc AudioProcessing::Config::NoiseSuppression::Level.*
- enum [VADLikelihood](#)  
*Likelihood that the voice detector reports a frame as speech. Maps onto webrtc::Vad::Aggressiveness inversely: VeryLow is the most aggressive setting and yields the fewest false positives.*

### Public Member Functions

- static IntPtr **webrtc\_audio\_processor\_create** (int samplingRate, int channels, int frameSize, int rev↵ SamplingRate, int revChannels)
- static int **webrtc\_audio\_processor\_init** (IntPtr proc)
- static int **webrtc\_audio\_processor\_set\_param** (IntPtr proc, int param, int v)
- static int **webrtc\_audio\_processor\_process** (IntPtr proc, short[] buffer, int offset, out bool voiceDetected)
- static int **webrtc\_audio\_processor\_process\_reverse** (IntPtr proc, short[] buffer, int bufferSize)
- static void **webrtc\_audio\_processor\_destroy** (IntPtr proc)

### Properties

- static string **Version** [get]  
*Which WebRTC revision the native library was built from, e.g. "WebRTC source stamp 2026-09-03T04:09:34, commit 9ab2b29b...". Returns a diagnostic string instead of throwing if the native library is missing or predates this export.*

## 3.296 WebRTCAudioProcessor Class Reference

Inherits [WebRTCAudioLib](#), and [IProcessor< T >](#).

## Public Member Functions

- **WebRTCAudioProcessor** ([ILogger](#) logger, int frameSize, int samplingRate, int channels, int reverseSamplingRate, int reverseChannels)
- short[] **Process** (short[] buf)
- void **OnAudioOutFrameFloat** (float[] data)
- void **Dispose** ()

## Public Member Functions inherited from [WebRTCAudioLib](#)

- static IntPtr **webrtc\_audio\_processor\_create** (int samplingRate, int channels, int frameSize, int reverseSamplingRate, int revChannels)
- static int **webrtc\_audio\_processor\_init** (IntPtr proc)
- static int **webrtc\_audio\_processor\_set\_param** (IntPtr proc, int param, int v)
- static int **webrtc\_audio\_processor\_process** (IntPtr proc, short[] buffer, int offset, out bool voiceDetected)
- static int **webrtc\_audio\_processor\_process\_reverse** (IntPtr proc, short[] buffer, int bufferSize)
- static void **webrtc\_audio\_processor\_destroy** (IntPtr proc)

## Public Member Functions inherited from [IProcessor< T >](#)

- T[] [Process](#) (T[] buf)  
*Process a frame of data.*

## Static Public Attributes

- static readonly int[] **SupportedSamplingRates** = { 8000, 16000, 32000, 48000 }

## Properties

- int **AECStreamDelayMs** [set]
- bool **AEC** [set]
- bool **AECHighPass** [set]
- bool **HighPass** [set]
- bool **NoiseSuppression** [set]
- [NSLevel](#) **NoiseSuppressionLevel** [set]  
*Noise suppression aggressiveness. Default: Moderate.*
- bool **AGC** [set]
- int **AGCCompressionGain** [set]
- int **AGCTargetLevel** [set]
- bool **AGCLimiter** [set]  
*Hard-limit the AGC output at the target level instead of only compressing above it. Default: true.*
- bool **AGC2** [set]
- bool **VAD** [set]
- [VADLikelihood](#) **VoiceDetectionLikelihood** [set]  
*How readily the voice detector calls a frame speech. Default: VeryLow, which reproduces the behaviour of the webrtc voice\_detection submodule this wrapper used before it was removed upstream.*
- bool **Bypass** [set]

## Properties inherited from [WebRTCAudioLib](#)

- static string **Version** [get]

*Which WebRTC revision the native library was built from, e.g. "WebRTC source stamp 2026-09-03T04:09:34, commit 9ab2b29b...". Returns a diagnostic string instead of throwing if the native library is missing or predates this export.*

## Additional Inherited Members

## Public Types inherited from [WebRTCAudioLib](#)

- enum **Error**
- enum **Param**
- enum [NSLevel](#)

*webrtc AudioProcessing::Config::NoiseSuppression::Level.*

- enum [VADLikelihood](#)

*Likelihood that the voice detector reports a frame as speech. Maps onto webrtc::Vad::Aggressiveness inversely: VeryLow is the most aggressive setting and yields the fewest false positives.*

## 3.297 WebVideoInEnumerator Class Reference

Inherits [WebDeviceEnumerator](#).

## Public Member Functions

- **WebVideoInEnumerator** ([ILogger](#) logger)

## Public Member Functions inherited from [WebDeviceEnumerator](#)

- **WebDeviceEnumerator** ([ILogger](#) logger, string filter)
- override void [Refresh](#) ()
- override void **Dispose** ()

## Public Member Functions inherited from [DeviceEnumeratorBase](#)

- **DeviceEnumeratorBase** ([ILogger](#) logger)
- IEnumerable< [DeviceInfo](#) > **GetEnumerator** ()
- void [Refresh](#) ()
- void **Dispose** ()

## Additional Inherited Members

## Protected Attributes inherited from [DeviceEnumeratorBase](#)

- List< [DeviceInfo](#) > **devices** = new List<[DeviceInfo](#)>()
- [ILogger](#) **logger**

## Properties inherited from [DeviceEnumeratorBase](#)

- virtual bool [IsSupported](#) [get]
- virtual string [Error](#) [get, protected set]
- Action [OnReady](#) [protected get, set]

## 3.298 WindowsAudioInPusher Class Reference

Inherits [IAudioPusher< T >](#).

### Public Member Functions

- **WindowsAudioInPusher** (int deviceId, [ILogger](#) logger)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimalFrameSize)
- void **Dispose** ()

### Public Member Functions inherited from [IAudioPusher< T >](#)

- void [SetCallback](#) (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)  
*Set the callback function used for pushing data.*

### Properties

- int [Channels](#) [get]  
*Number of channels in the audio signal.*
- int [SamplingRate](#) [get]  
*Sampling rate of the audio signal (in Hz).*
- string [Error](#) [get]  
*If not null, audio object is in invalid state.*

## 3.298.1 Property Documentation

### 3.298.1.1 Channels

int Channels [get]

Number of channels in the audio signal.

Implements [IAudioDesc](#).

### 3.298.1.2 Error

string Error [get]

If not null, audio object is in invalid state.

Implements [IAudioDesc](#).

### 3.298.1.3 SamplingRate

int SamplingRate [get]

Sampling rate of the audio signal (in Hz).

Implements [IAudioDesc](#).